Math 105: Calculus with Algebra
Instructor: David Zureick-Brown ("DZB")

All assignments

Gradescope code: B5J8EN

Gradescope instructions 3

1 Assignment 1, Due Friday, September 4 5

2 Assignment 1 Solutions 7

3 Assignment 2, Due Monday, September 7 10

4 Assignment 2 Solutions 11

5 Assignment 3, Due Friday, September 11 14

6 Assignment 3 Solutions 17

7 Assignment 4, Due Monday, September 14 22

8 Assignment 4 Solutions 24

9 Assignment 5, Due Friday, September 19 28

10 Assignment 5 Solutions 30

11 Assignment 6, Due Monday, September 21 35

12 Assignment 6 Solutions 37

13 Assignment 7, Due Friday, September 25 41

14 Assignment 7 Solutions 43

15 Assignment 8, Due Monday, September 28 47

16 Assignment 8 Solutions 49

17 Midterm 1 Review Packet 54

18 Midterm 1 Review Packet Solutions 61

19 Midterm 1 Review Packet Solutions 74

20 Assignment 9, Due Monday, October 5 79

21 Assignment 9 Solutions 81

22 Assignment 10, Due Friday, October 9 **84**

23 Assignment 10 Solutions **86**

24 Assignment 11, Due Thursday, October 15 **89**

25 Assignment 11 Solutions **91**

26 Assignment 12, Due Friday, October 16 **93**

27 Assignment 12 Solutions **95**

28 Assignment 13, Due Monday, October 19 **97**

29 Assignment 13 Solutions **98**

30 Assignment 14, Due Friday, October 23 **102**

31 Assignment 14 Solutions **103**

32 Assignment 15, Due Monday, October 26 **106**

33 Assignment 15 Solutions **107**

34 Midterm 2 Review Packet **110**

35 Midterm 2 Review Packet solutions **115**

36 Assignment 16, Due Wednesday, November 4 **121**

37 Assignment 16 Solutions **122**

38 Assignment 17, Due Friday, November 6 **124**

39 Assignment 17 Solutions **125**

40 Assignment 18, Due Monday, November 9 **128**

41 Assignment 18 Solutions **130**

Gradescope Instructions for submitting work in Math 105

You will be using the online Gradescope progam to submit your homework and exams. These instructions tell you how to sign up initially, and how to submit your written work.

Signing up for Gradescope the first time.

If you haven't used Gradescope for an **Amherst College** course before:

- Go to `http://www.gradescope.com`, click on "Sign up for free" (which may auto-scroll you to the bottom of the page), and select Sign up as [a] "Student".
- In the signup box:
  - Use the course entry code **B5J8EN**
  - Use your full name
  - Use your **Amherst College email** address. Or, if you are a Five-College student, use your email address from your own school.
  - Leave the "Student ID" entry blank.
- You will probably get an email asking to set a password for your account, so check your `amherst.edu` email inbox. (Or your email inbox through your own school, for Five-College students.)

Adding Math 105 to Gradescope.

If you **have** used Gradescope for an Amherst course before, and so you already have an account through your `amherst.edu` email, you still need to add Math 105, so:

- Go to `http://www.gradescope.com` and log in.
- Go to your Account Dashboard (click the Gradescope logo at upper left), and click "Add Course" at bottom right.
- Use the course code **B5J8EN**

(submission instructions on next page)

Submitting written work

First write it out on paper as you would normally. Then **scan it** to create a PDF. One method for scanning is the smartphone app **DropBox**. It makes nice clear scans, and it saves them directly into a folder so that you can have all your assignments in one place. **CamScanner** is another free scanning App, and there are others, too. **Gradescope** now has its own scanning app. You can also use a printer/scanner if you prefer.

Please be kind to our dear graders and make sure your submission is legible !

In particular, please leave some spacing between separate problems.

If you have a tablet computer, you may write your work there (instead of on paper) and save it as a PDF.

Some of you may know the math formatting package LaTeX and may want to use it in Math 105. That's fine, too; if so, you may write up your work in LaTeX and save the resulting PDF.

In short, any method is fine as long as it creates a legible **PDF** file and **NOT a photo**.

For example, if you use the DropBox app, then in your created *Math 105 Homework* Dropbox folder, you can select create (+) at the bottom of the screen and click the *Scan Document* option. Snap a shot of the first page of your homework, and then click [+] to snap shots of any subsequent pages. Do **not** use the *Take Photo* option.

After you have scanned/saved your work as a PDF, submit it on Gradescope as follows:

- Go to `http://www.gradescope.com` and log in.
- Select the course "Math 105, Fall 2026" and the appropriate assignment.
- Select "submit pdf" to submit your work in PDF format. Browse to find your PDF and upload.
- Now it is time to **tag** your problems. This is an **important step**, where you are telling Gradescope which problems are on which page(s).

  For each problem, select the pages of your submission where your written solution appears.

  I think the easiest thing to do is to click on the page of **your** homework upload where you wrote the given problem, and then click on the assigned problem listed. Repeat for each problem.

  You must tag the problems or else you will not get credit for your work.

  Gradescope will give you a warning when you go to submit your assignment if you have not selected the pages correctly. If you tag a problem incorrectly, you can fix it by clicking "More" and "Reselect Pages".
- Click Save or Submit.

After your assignment is graded, you will be able to see your score on the written problems, along with comments, on Gradescope. You should receive an email notifying you when each homework set is graded.

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 1

Due **Friday, September 4** at 11:55pm, on Gradescope

GOALS:

- Review of basic algebra.
- Review of absolute values and inequalities.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer. For a review on inequalities and absolute value, see Appendix A (Stewart Calculus book).

(1) Simplify:

$$4 - (5 - 3) + 3(4 - 7) - 4(1 + 2)$$

(2) In the following expressions, simplify and reduce to lowest terms. (Some answers may have more than one form)

(a)

$$\frac{\frac{4}{75}}{\frac{8}{25}}$$

(b)

$$\frac{7x}{3y} \cdot \frac{3y + 2}{x}$$

(c)

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}}$$

MORE PROBLEMS ON THE NEXT PAGE →

(3) Express as a single fraction and simplify:

(a)
$$\frac{1}{2}+\frac{4}{3}-\frac{2}{5}-\frac{3}{15}$$

(b)
$$\frac{\frac{1}{3}+\frac{2}{5}}{\frac{3}{2}}$$

(c)
$$\frac{1}{x}-\frac{x+1}{xy}+\frac{x-2}{xz}$$

(d)
$$\frac{\frac{1}{st}-\frac{1}{w}}{\frac{1}{tw}-\frac{2}{s}}$$

(4) Rewrite the expression without using the absolute-value symbol:

$$|x-2| \text{ if } x<2$$

(5) Solve the inequalities with the **geometric interpretation** of absolute value.

(a)
$$|x|<3$$

(b)
$$|x-4|<1$$

(c)
$$|x+5|\geq 2$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 1 SOLUTIONS

(1) Simplify:

$$4-(5-3)+3(4-7)-4(1+2)$$

Solution. We simplify the expressions inside parentheses first:

$$4-(5-3)+3(4-7)-4(1+2)=4-2+3(-3)-4(3).$$

Therefore

$$4-2-9-12=-19.$$

Hence the simplified value is $\boxed{-19}$.

(2) In the following expressions, simplify and reduce to lowest terms. (Some answers may have more than one form)

(a)

$$\frac{\frac{4}{75}}{\frac{8}{25}}$$

Solution. Dividing by a fraction is the same as multiplying by its reciprocal:

$$\frac{\frac{4}{75}}{\frac{8}{25}}=\frac{4}{75}\cdot\frac{25}{8}=\frac{1}{3}\cdot\frac{1}{2}=\boxed{\frac{1}{6}}.$$

(b)

$$\frac{7x}{3y}\cdot\frac{3y+2}{x}$$

Solution. Cancel the common factor x:

$$\frac{7x}{3y}\cdot\frac{3y+2}{x}=\boxed{\frac{7(3y+2)}{3y}}.$$

(c)

$$\frac{\frac{xy}{(x+y)}}{\frac{x^2y}{(x+y)^3}}$$

Solution. Multiply by the reciprocal of the denominator:

$$\frac{\frac{xy}{x+y}}{\frac{x^2y}{(x+y)^3}}=\frac{xy}{x+y}\cdot\frac{(x+y)^3}{x^2y}.$$

Cancelling common factors gives

$$\boxed{\frac{(x+y)^2}{x}}.$$

(3) Express as a single fraction and simplify:

(a)

$$\frac{1}{2}+\frac{4}{3}-\frac{2}{5}-\frac{3}{15}$$

Solution. A common denominator is 30:

$$\frac{1}{2}+\frac{4}{3}-\frac{2}{5}-\frac{3}{15}=\frac{15}{30}+\frac{40}{30}-\frac{12}{30}-\frac{6}{30}=\boxed{\frac{37}{30}}.$$

(b)

$$\frac{\frac{1}{3}+\frac{2}{5}}{\frac{3}{2}}$$

Solution. First simplify the numerator:

$$\frac{1}{3}+\frac{2}{5}=\frac{5}{15}+\frac{6}{15}=\frac{11}{15}.$$

Then divide by 3/2:

$$\frac{\frac{11}{15}}{\frac{3}{2}}=\frac{11}{15}\cdot\frac{2}{3}=\boxed{\frac{22}{45}}.$$

(c)

$$\frac{1}{x}-\frac{x+1}{xy}+\frac{x-2}{xz}$$

Solution. The common denominator is xyz. Thus

$$\frac{1}{x}-\frac{x+1}{xy}+\frac{x-2}{xz}=\frac{yz}{xyz}-\frac{z(x+1)}{xyz}+\frac{y(x-2)}{xyz}.$$

Combining the numerators,

$$\boxed{\frac{xy+yz-xz-2y-z}{xyz}}.$$

(d)

$$\frac{\frac{1}{st}-\frac{1}{w}}{\frac{1}{tw}-\frac{2}{s}}$$

Solution. Put the numerator and denominator over the common denominator stw:

$$\frac{1}{st}-\frac{1}{w}=\frac{w-st}{stw},\qquad \frac{1}{tw}-\frac{2}{s}=\frac{s-2tw}{stw}.$$

Therefore

$$\frac{\frac{w-st}{stw}}{\frac{s-2tw}{stw}}=\boxed{\frac{w-st}{s-2tw}}.$$

(4) Rewrite the expression without using the absolute-value symbol:

$$|x-2| \text{ if } x<2$$

Solution. If $x<2$, then $x-2<0$. The absolute value of a negative number is its opposite, so

$$|x-2|=-(x-2)=\boxed{2-x}.$$

(5) Solve the inequalities with the **geometric interpretation** of absolute value.

(a)

$$|x| < 3$$

Solution. The quantity $|x|$ is the distance from x to 0. We want that distance to be less than 3, so x must lie between -3 and 3:

$$\boxed{-3 < x < 3}.$$

(b)

$$|x - 4| < 1$$

Solution. The quantity $|x - 4|$ is the distance from x to 4. We want that distance to be less than 1, so

$$\boxed{3 < x < 5}.$$

(c)

$$|x + 5| \geq 2$$

Solution. Since $|x + 5| = |x - (-5)|$, this is the distance from x to -5. We want that distance to be at least 2. Thus x must be at least 2 units to the left or right of -5:

$$\boxed{x \leq -7 \quad \text{or} \quad x \geq -3}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 2

Due **Monday, September 7** at 11:55pm, on Gradescope

GOALS:

- Review equations of lines.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer. For a review on equations of lines, see Appendix A (Stewart Calculus book).

Find the slope of the line through P and Q:

(1) $P(1,5), Q(4,11)$

(2) $P(-1,-4), Q(6,0)$

Find the equation of the line that satisfies the given conditions.

(3) Through $(2,-3)$ with slope 6

(4) Through $(2,1)$ and $(1,6)$

(5) Slope 3, y-intercept -2

(6) x-intercept 1, y-intercept -3

(7) Through $(4,5)$, parallel to x-axis

(8) Through $(4,5)$, parallel to y-axis

(9) Through $(1,-6)$, parallel to the line $x+2y=6$

(10) Through $(-1,-2)$, perpendicular to the line $2x+5y+8=0$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 2 SOLUTIONS

Find the slope of the line through P and Q:

(1) $P(1,5), Q(4,11)$

Solution. The slope through two points (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Thus

$$m = \frac{11-5}{4-1} = \frac{6}{3} = \boxed{2}.$$

(2) $P(-1,-4), Q(6,0)$

Solution. Using the slope formula,

$$m = \frac{0-(-4)}{6-(-1)} = \frac{4}{7}.$$

Therefore the slope is

$$\boxed{\frac{4}{7}}.$$

Find the equation of the line that satisfies the given conditions.

(3) Through $(2,-3)$ with slope 6

Solution. Using point-slope form $y - y_1 = m(x - x_1)$,

$$y - (-3) = 6(x-2).$$

Hence

$$y + 3 = 6x - 12,$$

so

$$\boxed{y = 6x - 15}.$$

(4) Through $(2,1)$ and $(1,6)$

Solution. First find the slope:

$$m = \frac{6-1}{1-2} = \frac{5}{-1} = -5.$$

Using the point $(2,1)$,

$$y - 1 = -5(x-2).$$

Therefore

$$y - 1 = -5x + 10,$$

so

$$\boxed{y = -5x + 11}.$$

(5) Slope 3, y-intercept -2

Solution. A line with slope m and y-intercept b has equation $y = mx + b$. Here $m = 3$ and $b = -2$, so

$$\boxed{y = 3x - 2}.$$

(6) x-intercept 1, y-intercept -3

Solution. The x-intercept gives the point $(1, 0)$, and the y-intercept gives the point $(0, -3)$. The slope is

$$m = \frac{0 - (-3)}{1 - 0} = 3.$$

Since the y-intercept is -3, the equation is

$$\boxed{y = 3x - 3}.$$

(7) Through $(4, 5)$, parallel to x-axis

Solution. A line parallel to the x-axis is horizontal, so its y-coordinate is constant. Since the line passes through $(4, 5)$,

$$\boxed{y = 5}.$$

(8) Through $(4, 5)$, parallel to y-axis

Solution. A line parallel to the y-axis is vertical, so its x-coordinate is constant. Since the line passes through $(4, 5)$,

$$\boxed{x = 4}.$$

(9) Through $(1, -6)$, parallel to the line $x + 2y = 6$

Solution. Rewrite the given line in slope-intercept form:

$$x + 2y = 6 \quad \Longrightarrow \quad 2y = -x + 6 \quad \Longrightarrow \quad y = -\frac{1}{2}x + 3.$$

Its slope is $-\frac{1}{2}$. A parallel line has the same slope, so using point-slope form with $(1, -6)$,

$$y + 6 = -\frac{1}{2}(x - 1).$$

Therefore

$$\boxed{y = -\frac{1}{2}x - \frac{11}{2}}.$$

Equivalently, the equation can be written as $\boxed{x + 2y = -11}$.

(10) Through $(-1,-2)$, perpendicular to the line $2x+5y+8=0$

Solution. First rewrite the given line in slope-intercept form:

$$2x+5y+8=0 \quad\Longrightarrow\quad 5y=-2x-8 \quad\Longrightarrow\quad y=-\frac{2}{5}x-\frac{8}{5}.$$

Its slope is $-\frac{2}{5}$. A perpendicular line has slope equal to the negative reciprocal, so its slope is

$$\frac{5}{2}.$$

Using point-slope form with $(-1,-2)$,

$$y+2=\frac{5}{2}(x+1).$$

Thus

$$y+2=\frac{5}{2}x+\frac{5}{2},$$

and therefore

$$\boxed{y=\frac{5}{2}x+\frac{1}{2}}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 3

Due **Friday, September 11** at 11:55pm, on Gradescope

GOALS:

- Review of the essential ideas behind functions.
- Practice graphing of functions.
- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer.

(1) Let $f(x) = x^3 + 2x$. Determine:

   (a) $f(2)$

   (b) $f(2x)$

   (c) $f(-x)$

   (d) $f(2 + \Delta x)$

(2) Let $u(t) = \frac{1}{2}t^2 + t$. Determine:

   (a) $u(-1)$

   (b) $u(t + \Delta t)$

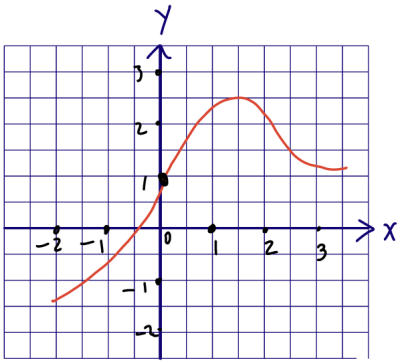
   (c) $u(-t)$

(3) Sketch the graph of the function $g(t) = |1 - 3t|$

(4) Sketch the graph of the function

$$f(x) = \begin{cases} |x| & \text{if } |x| \leq 1 \\ 1 & \text{if } |x| > 1 \end{cases}$$

(5) The graph of the function f is given.



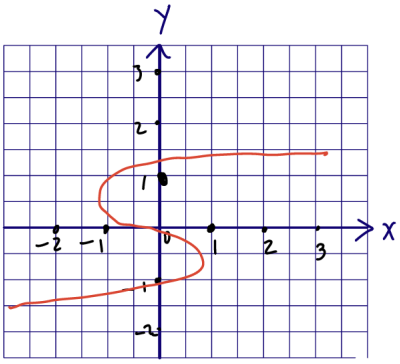
(a) State the value of $f(1.5)$.

(b) Estimate the value of $f(-1)$.

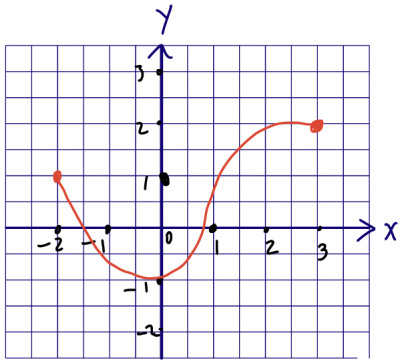
(c) For what values of x is $f(x) = 1$?

(d) Estimate the value of x such that $f(x) = 0$.

(6) Determine whether the curve is a graph of a function of x. If it is, state the domain and the range of the function.

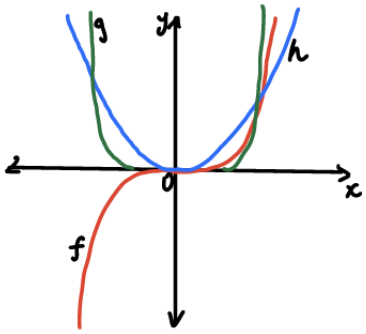


(7) State the domain and the range of the function graphed below.



(8) Find the domain of $f(x) = \frac{x+4}{x^2-9}$

(9) Match the each equation with its graph. Explain your choices. Don't use a computer or graphic calculator. (a) $y = x^2$ (b) $y = x^5$ (c) $y = x^8$



MATH 105 Calculus with Algebra – Fall 2026
Homework 3 – Solutions

1. Let $f(x) = x^3 + 2x$. Determine:

   (a) $f(2)$.

   Solution.
   $$f(2) = 2^3 + 2(2) = 8 + 4 = \boxed{12}.$$

   (b) $f(2x)$.

   Solution.
   $$f(2x) = (2x)^3 + 2(2x) = \boxed{8x^3 + 4x}.$$

   (c) $f(-x)$.

   Solution.
   $$f(-x) = (-x)^3 + 2(-x) = \boxed{-x^3 - 2x} = -f(x).$$

   (d) $f(2 + \Delta x)$.

   Solution.
   $$\begin{aligned} f(2+\Delta x) &= (2+\Delta x)^3 + 2(2+\Delta x) \\ &= 8 + 12\Delta x + 6(\Delta x)^2 + (\Delta x)^3 + 4 + 2\Delta x \\ &= \boxed{12 + 14\Delta x + 6(\Delta x)^2 + (\Delta x)^3}. \end{aligned}$$

2. Let $u(t) = \frac{1}{2}t^2 + t$. Determine:

   (a) $u(-1)$.

   Solution.
   $$u(-1) = \frac{1}{2}(-1)^2 + (-1) = \frac{1}{2} - 1 = \boxed{-\frac{1}{2}}.$$

   (b) $u(t + \Delta t)$.

   Solution.
   $$\begin{aligned} u(t+\Delta t) &= \frac{1}{2}(t+\Delta t)^2 + (t+\Delta t) \\ &= \frac{1}{2}t^2 + t\Delta t + \frac{1}{2}(\Delta t)^2 + t + \Delta t \\ &= \boxed{\frac{1}{2}t^2 + t + (t+1)\Delta t + \frac{1}{2}(\Delta t)^2}. \end{aligned}$$

   (c) $u(-t)$.

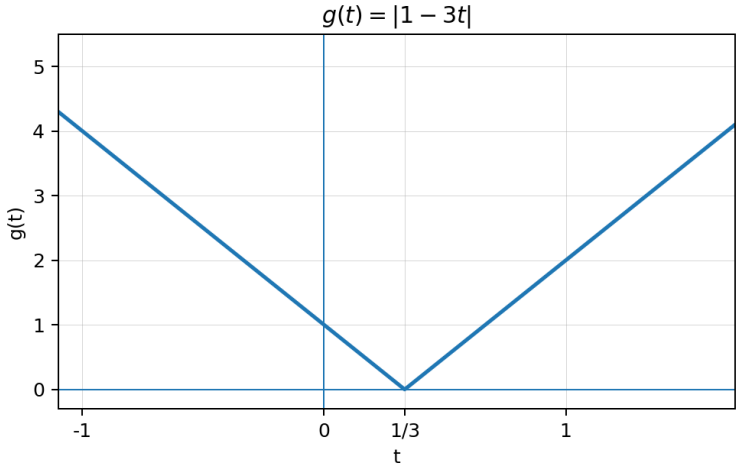
   Solution.
   $$u(-t) = \frac{1}{2}(-t)^2 - t = \boxed{\frac{1}{2}t^2 - t}.$$

3. Sketch $g(t) = |1 - 3t|$.

Solution. Since $1 - 3t = 0$ when $t = \frac{1}{3}$,

$$g(t) = \begin{cases} 1 - 3t, & t \le \frac{1}{3}, \\ 3t - 1, & t \ge \frac{1}{3}. \end{cases}$$

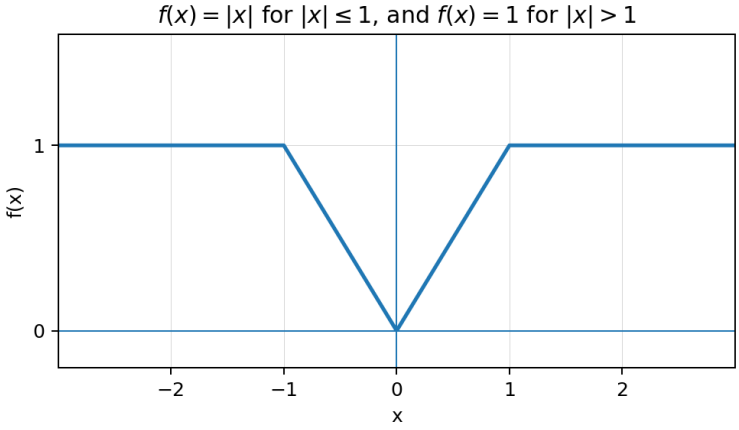
Thus the graph is a V with vertex $\left(\frac{1}{3}, 0\right)$, left slope -3, and right slope 3.



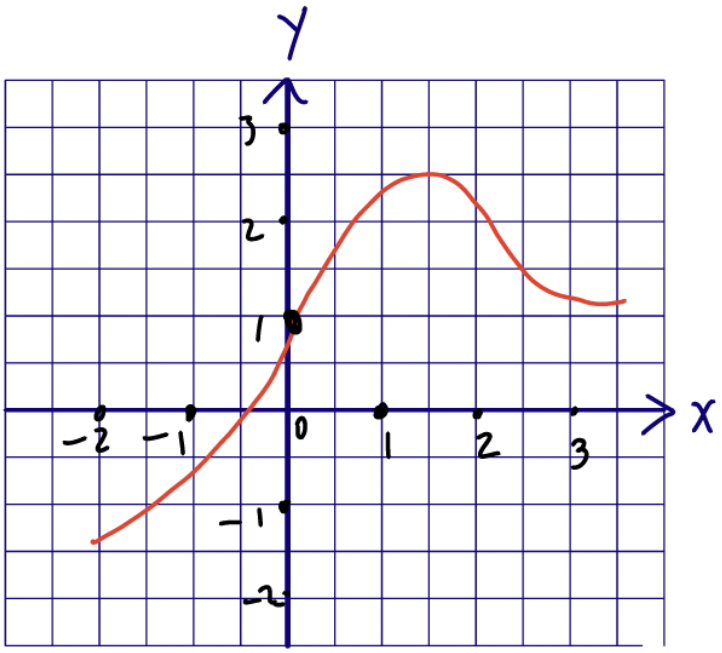
4. Sketch

$$f(x) = \begin{cases} |x|, & |x| \le 1, \\ 1, & |x| > 1. \end{cases}$$

Solution. On $[-1, 1]$ the graph is the usual V-shaped graph $y = |x|$. Outside that interval the graph is the horizontal line $y = 1$. The pieces meet at $(-1, 1)$ and $(1, 1)$.



5. The graph of f is shown below.



(a) State $f(1.5)$.

Solution. From the graph, $\boxed{f(1.5) \approx 2.5}$.

(b) Estimate $f(-1)$.

Solution. From the graph, $\boxed{f(-1) \approx -0.6}$ (an estimate near -0.5 to -0.6 is reasonable from the hand-drawn graph).

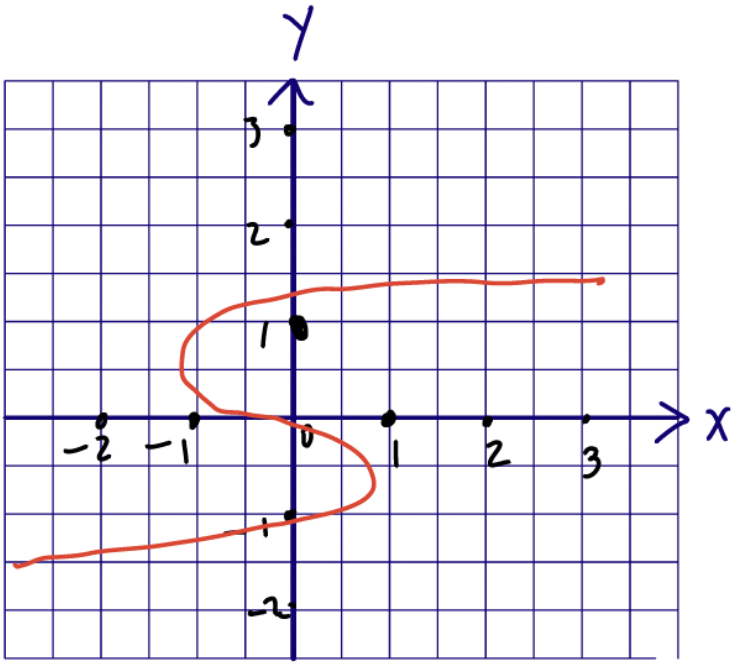
(c) For what values of x is $f(x) = 1$?

Solution. The horizontal line $y = 1$ meets the graph at $x = 0$, so $\boxed{x = 0}$.

(d) Estimate the value of x such that $f(x) = 0$.

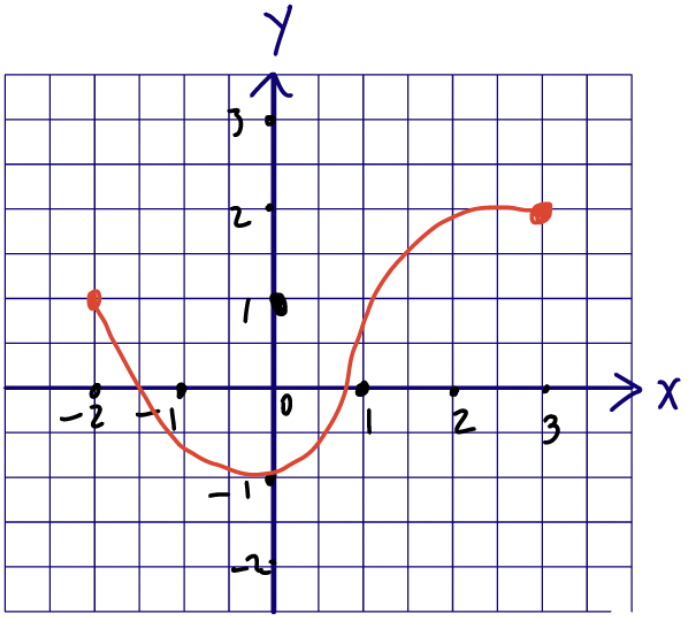
Solution. The x-intercept is approximately $\boxed{x \approx -0.4}$.

6. Determine whether the curve below is the graph of a function of x. If it is, state its domain and range.



Solution. It is not the graph of a function of x. A vertical line through some x-values intersects the red curve more than once, so it fails the vertical line test.

7. State the domain and range of the function graphed below.



Solution. The graph begins at the closed point with $x = -2$ and ends at the closed point with $x = 3$, so

$$\boxed{\text{Domain} = [-2, 3]}.$$

Its lowest value is $y = -1$ and its highest value is $y = 2$, so

$$\boxed{\text{Range} = [-1, 2]}.$$

8. Find the domain of

$$f(x) = \frac{x+4}{x^2-9}.$$

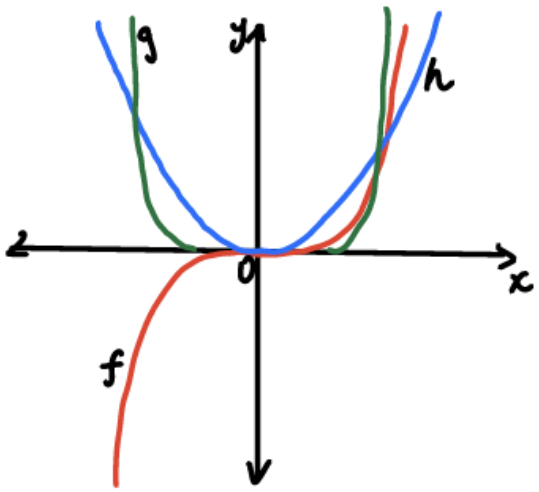
Solution. The denominator cannot equal zero:

$$x^2 - 9 = (x-3)(x+3) \neq 0.$$

Thus $x \neq -3, 3$, and

$$\boxed{\text{Domain} = (-\infty, -3) \cup (-3, 3) \cup (3, \infty)}.$$

9. Match each equation with its graph: (a) $y = x^2$, (b) $y = x^5$, (c) $y = x^8$.



Solution. The red curve f takes negative values for negative x and positive values for positive x, so it is the odd power:

$$\boxed{f(x) = x^5}.$$

The blue curve h is the wider even-power graph, while the green curve g is flatter near 0 and grows more steeply for $|x| > 1$. Therefore

$$\boxed{h(x) = x^2, \qquad g(x) = x^8}.$$

So the matching is

$$\boxed{\text{(a) } h, \qquad \text{(b) } f, \qquad \text{(c) } g}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 4

Due **Monday, September 14** at 11:55pm, on Gradescope

GOALS:

- Practice graphing of functions.
- Practice shifting and stretching graphs.
- Simplify difference quotients.
- Practice compositions of functions.

- Write complete solutions.

Justify your solution, writing down the steps that led you to the final answer.

(1) The graph of $y = f(x)$ is given. Match each equation with its graph and give reasons for your choices.

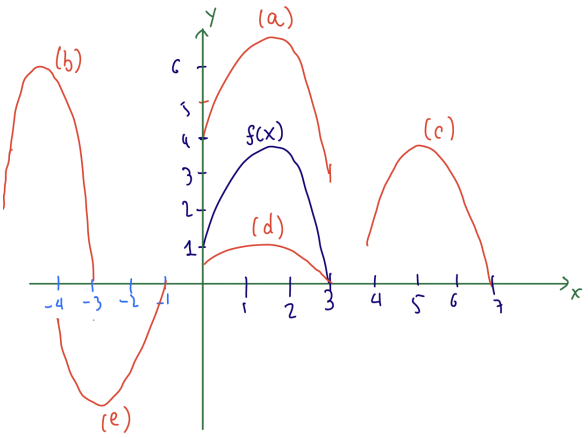
(a)
$$y = f(x-4)$$

(b)
$$y = f(x) + 3$$

(c)
$$y = \frac{1}{3} f(x)$$

(d)
$$y = -f(x+4)$$

(e)
$$y = 2f(x+4)$$



(2) Graph the following functions by hand, not by plotting points, but by starting with the graph of one of the standard functions (see Stewart, Section 1.2) and then applying the appropriate transformations.
   (a) $y = -x^2$
   (b) $1 - \frac{1}{x}$
   (c) $|x - 2|$

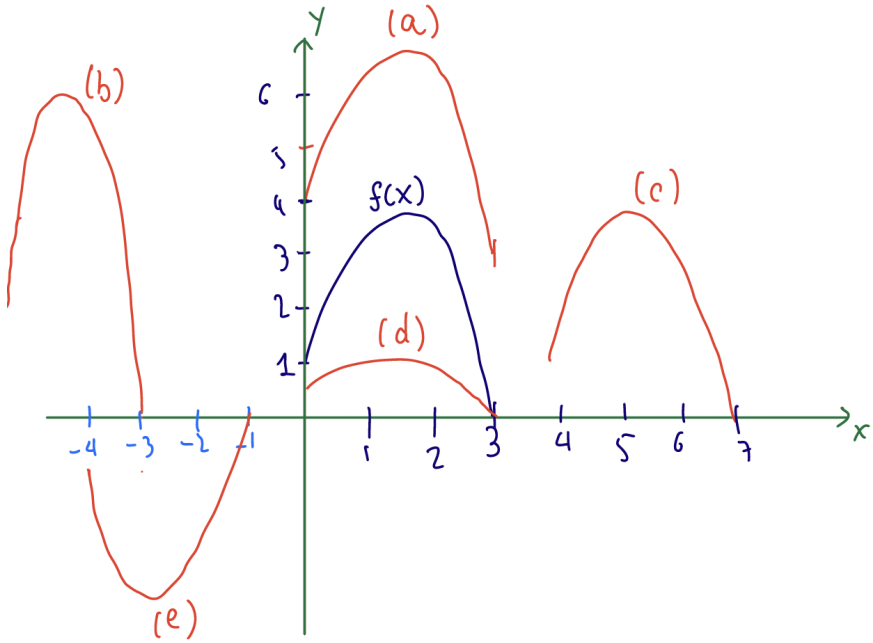
(3) Let $f(x) = \frac{x+1}{x}$.
   (a) Compute $f(a + h)$
   (b) Simplify
$$\frac{f(a+h) - f(a)}{h}$$

(4) Find $f(g(h(x)))$:
$$f(x) = \sqrt{x-3},\ g(x) = x^2,\ h(x) = x^3 + 2$$

(5) Express the function in the form $f(g(x))$:
$$F(x) = (2x + x^2)^4$$
(In other words, find functions $f(x)$ and $g(x)$ such that $F(x) = f(g(x))$).

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 4 SOLUTIONS

(1) The graph of $y = f(x)$ is given. Match each equation with its graph and give reasons for your choices.



(a) $y = f(x-4)$ is f shifted four units to the right, so it is $\boxed{\text{(c)}}$.

(b) $y = f(x)+3$ is f shifted three units up, so it is $\boxed{\text{(a)}}$.

(c) $y = \frac{1}{3}f(x)$ is a vertical compression by a factor of $\frac{1}{3}$, so it is $\boxed{\text{(d)}}$.

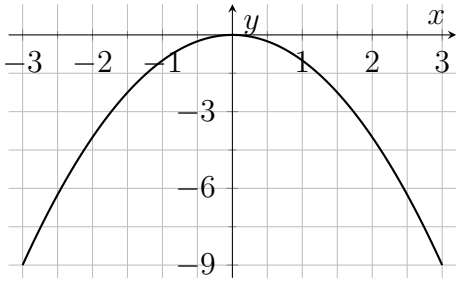
(d) $y = -f(x+4)$ shifts f four units left and reflects it across the x-axis, so it is $\boxed{\text{(e)}}$.

(e) $y = 2f(x+4)$ shifts f four units left and stretches it vertically by a factor of 2, so it is $\boxed{\text{(b)}}$.

(2) Graph the following functions by hand, starting with a standard graph and applying the appropriate transformations.

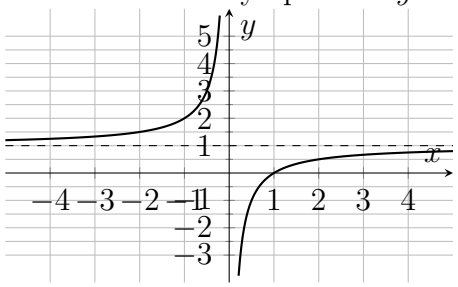
(a) $y = -x^2$

Start with $y = x^2$ and reflect it across the x-axis. The vertex remains at $(0, 0)$ and the parabola opens downward.



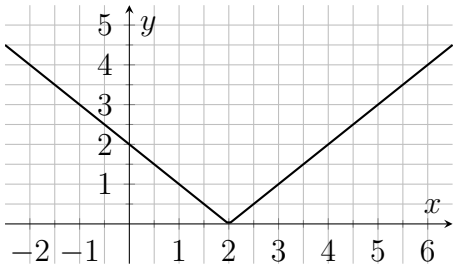
(b) $y = 1 - \frac{1}{x}$

Start with $y = \frac{1}{x}$, reflect across the x-axis, and then shift up by 1. The vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$.



(c) $y = |x - 2|$

Start with $y = |x|$ and shift it two units to the right. The vertex is $\boxed{(2, 0)}$.



(3) Let

$$f(x) = \frac{x+1}{x}.$$

(a) Replacing x by $a + h$ gives

$$\boxed{f(a+h) = \frac{a+h+1}{a+h}}.$$

(b) We simplify the difference quotient:

$$\begin{aligned}
\frac{f(a+h) - f(a)}{h} &= \frac{1}{h}\left(\frac{a+h+1}{a+h} - \frac{a+1}{a}\right) \\
&= \frac{1}{h}\left(\frac{a(a+h+1) - (a+1)(a+h)}{a(a+h)}\right) \\
&= \frac{1}{h}\left(\frac{-h}{a(a+h)}\right) \\
&= \boxed{-\frac{1}{a(a+h)}}.
\end{aligned}$$

The original expression requires $a \neq 0$, $a + h \neq 0$, and $h \neq 0$.

(4) Find $f(g(h(x)))$, where

$$f(x) = \sqrt{x-3}, \qquad g(x) = x^2, \qquad h(x) = x^3 + 2.$$

Work from the inside out:

$$h(x) = x^3 + 2,$$

so

$$g(h(x)) = (x^3+2)^2.$$

Therefore

$$f(g(h(x))) = \sqrt{(x^3+2)^2 - 3}.$$

Thus

$$\boxed{f(g(h(x))) = \sqrt{(x^3+2)^2 - 3}}.$$

(5) Express

$$F(x) = (2x + x^2)^4$$

in the form $f(g(x))$.

One natural choice is

$$\boxed{g(x) = 2x + x^2} \qquad \text{and} \qquad \boxed{f(x) = x^4}.$$

Indeed,

$$f(g(x)) = (g(x))^4 = (2x + x^2)^4 = F(x).$$

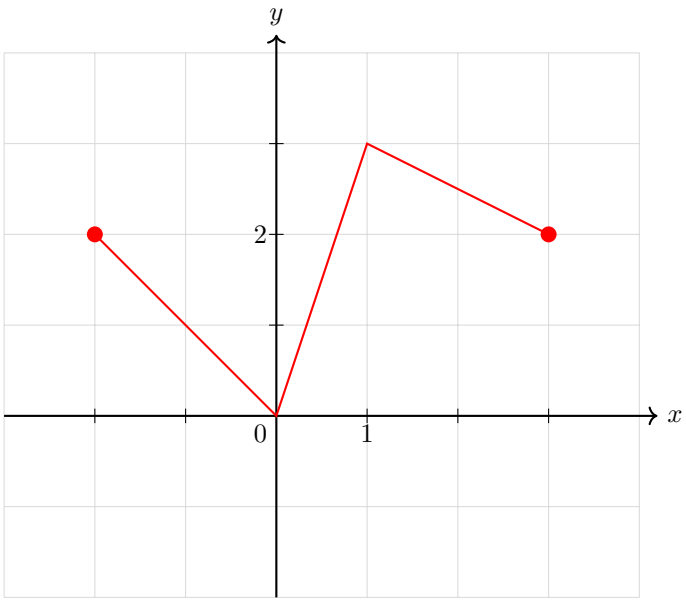
Other choices are possible.

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 5

Due **Friday, September 18** at 11:55pm, on Gradescope

Justify your solutions, writing down the steps that led you to the final answer.

(1) The graph of f is given. Draw the graphs of the following functions.
   (a) $y = f(x) - 3$
   (b) $y = f(x + 1)$
   (c) $y = \frac{1}{2}f(x)$
   (d) $y = -f(x)$
   (e) $y = f + f$.



(2) Graph the function $f(x) = -1 + \frac{1}{(2x)^2 - 1}$

(3) Let $f(x) = 1 - x^2$ and let $g(x) = \sqrt{x + 1}$. Compute (and simplify) $f + g$, fg, $f \circ g$, and $g \circ f$.

(4) Explain in your own words what is meant by the equation

$$\lim_{x \to 2} f(x) = 5.$$

Is it possible for this statement to be true and yet $f(2) = 3$? Explain.

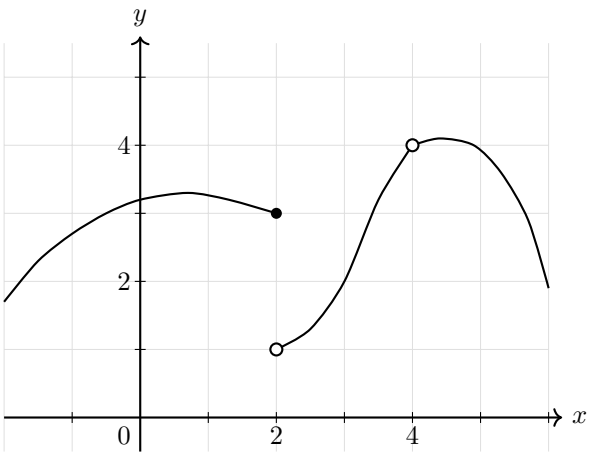
(5) Explain what it means to say that

$$\lim_{x\to 1^-} f(x) = 3 \qquad \text{and} \qquad \lim_{x\to 1^+} f(x) = 7.$$

In this situation is it possible that $\lim_{x\to 1} f(x)$ exists? Explain.

(6) For the function f whose graph is given below, state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{x\to 2^-} f(x)$
(b) $\lim_{x\to 2^+} f(x)$
(c) $\lim_{x\to 2} f(x)$
(d) $f(2)$
(e) $\lim_{x\to 4} f(x)$
(f) $f(4)$
(g) $\lim_{x\to 3} f(x)$



(7) Determine the limits:

$$\lim_{x\to 5^+} \frac{x+1}{x-5}.$$

and

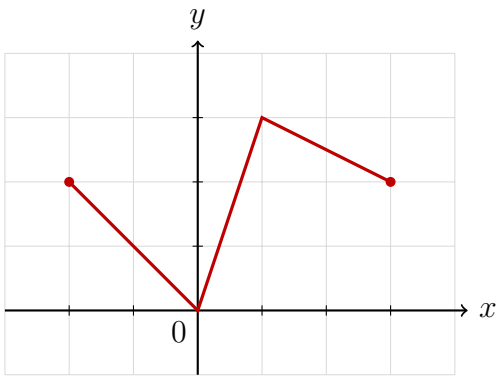
$$\lim_{x\to 5^-} \frac{x+1}{x-5}.$$

MATH 105 CALCULUS WITH ALGEBRA – FALL 2026
HOMEWORK 5 SOLUTIONS

(1) The graph of f is given by the piecewise-linear curve through

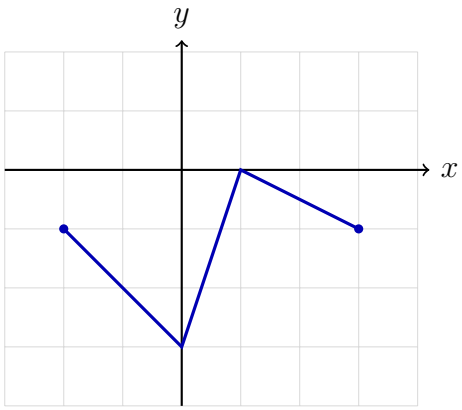
$$(-2, 2), \qquad (0, 0), \qquad (1, 3), \qquad (3, 2).$$

We transform the entire graph, rather than transforming the individual points one at a time.



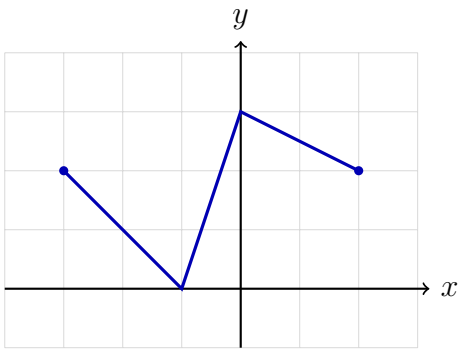
(a) $y = f(x) - 3$ shifts the graph of f **down 3 units**. The transformed vertices are

$$(-2, -1), \quad (0, -3), \quad (1, 0), \quad (3, -1).$$



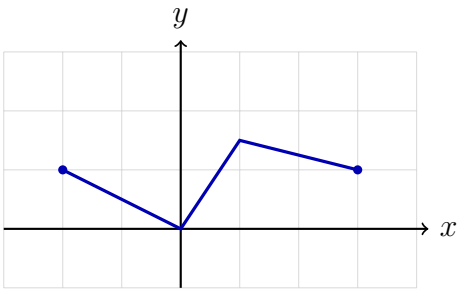
(b) $y = f(x + 1)$ shifts the graph of f **left 1 unit**. The transformed vertices are

$$(-3, 2), \quad (-1, 0), \quad (0, 3), \quad (2, 2).$$



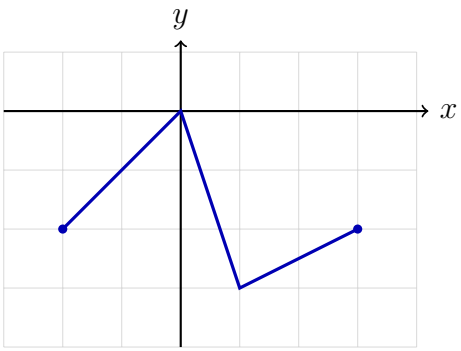
(c) $y = \frac{1}{2}f(x)$ vertically compresses the graph by a factor of $\frac{1}{2}$. The x-coordinates stay the same, while all y-coordinates are halved:

$$(-2, 1), \quad (0, 0), \quad \left(1, \frac{3}{2}\right), \quad (3, 1).$$



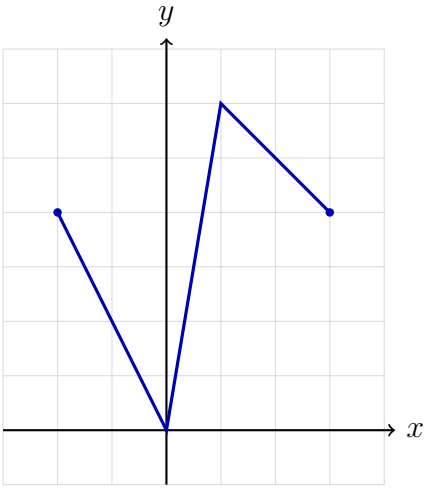
(d) $y = -f(x)$ reflects the graph across the x-axis. The transformed vertices are

$$(-2, -2), \quad (0, 0), \quad (1, -3), \quad (3, -2).$$



(e) Since $f + f = 2f$, the graph is a vertical stretch by a factor of 2. The transformed vertices are

$$(-2, 4), \quad (0, 0), \quad (1, 6), \quad (3, 4).$$



(2) Graph

$$f(x) = -1 + \frac{1}{(2x)^2 - 1} = -1 + \frac{1}{4x^2 - 1}.$$

The denominator is zero when

$$4x^2 - 1 = 0,$$

so the vertical asymptotes are

$$\boxed{x = -\frac{1}{2} \quad \text{and} \quad x = \frac{1}{2}}.$$

As $|x|$ becomes large, the fraction $1/(4x^2 - 1)$ approaches 0, so the horizontal asymptote is

$$\boxed{y = -1}.$$

The function is even, so its graph is symmetric about the y-axis. Also,

$$f(0) = -2.$$

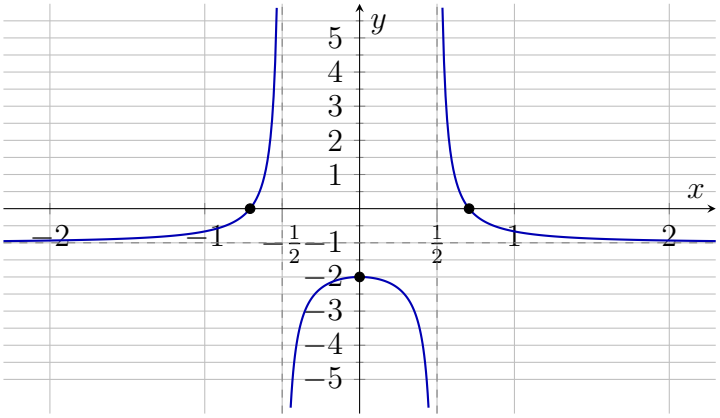
For the x-intercepts,

$$\begin{aligned} -1 + \frac{1}{4x^2 - 1} &= 0 \\ \frac{1}{4x^2 - 1} &= 1 \\ 4x^2 &= 2, \end{aligned}$$

so

$$\boxed{x = \pm\frac{1}{\sqrt{2}}}.$$

These features determine the graph:



(3) Let

$$f(x) = 1 - x^2, \qquad g(x) = \sqrt{x+1}.$$

$$\begin{aligned} (f+g)(x) &= f(x) + g(x) \\ &= \boxed{1 - x^2 + \sqrt{x+1}}. \end{aligned}$$

This is defined for $x \geq -1$.

For the product,

$$(fg)(x) = \boxed{(1 - x^2)\sqrt{x+1}},$$

again for $x \geq -1$.

For the first composition,

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= 1 - \left(\sqrt{x+1}\right)^2 \\ &= \boxed{-x}.\end{aligned}$$

Although the simplified formula is $-x$, the original composition requires $x \geq -1$ because $g(x)$ must be defined.

For the other composition,

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= \sqrt{(1-x^2)+1} \\ &= \boxed{\sqrt{2-x^2}}.\end{aligned}$$

For this to be defined we need

$$2 - x^2 \geq 0,$$

so its domain is $[-\sqrt{2}, \sqrt{2}]$.

(4) The statement

$$\lim_{x \to 2} f(x) = 5$$

means that the values of $f(x)$ can be made as close to 5 as we want by taking x sufficiently close to 2 (but with $x \neq 2$).

Yes, it is possible that $f(2) = 3$. A limit describes what happens to $f(x)$ *near* $x = 2$; it does not require the value at $x = 2$ itself to equal the limit.

(5) The statement

$$\lim_{x \to 1^-} f(x) = 3$$

means that $f(x)$ approaches 3 as x approaches 1 using values with $x < 1$. Similarly,

$$\lim_{x \to 1^+} f(x) = 7$$

means that $f(x)$ approaches 7 as x approaches 1 using values with $x > 1$.

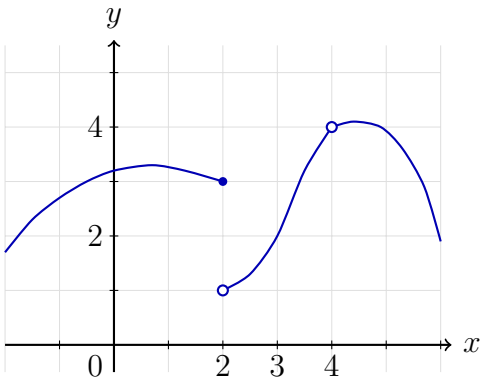
The two-sided limit **does not exist**, because a two-sided limit exists only when the left-hand and right-hand limits agree. Here

$$3 \neq 7.$$

Thus

$$\boxed{\lim_{x \to 1} f(x) \text{ does not exist.}}$$

(6) For the graph below:



(a) As x approaches 2 from the left, the graph approaches $y = 3$, so

$$\boxed{\lim_{x \to 2^-} f(x) = 3}.$$

(b) As x approaches 2 from the right, the graph approaches the open point at $(2, 1)$, so

$$\boxed{\lim_{x \to 2^+} f(x) = 1}.$$

(c) Since the one-sided limits are different,

$$\boxed{\lim_{x \to 2} f(x) \text{ does not exist.}}$$

(d) The filled point at $x = 2$ is $(2, 3)$, so

$$\boxed{f(2) = 3}.$$

(e) From both sides of $x = 4$, the graph approaches the hole at $(4, 4)$, so

$$\boxed{\lim_{x \to 4} f(x) = 4}.$$

(f) There is no filled point at $x = 4$, so

$$\boxed{f(4) \text{ is undefined.}}$$

(g) The graph is continuous near $x = 3$ and passes through $(3, 2)$, so

$$\boxed{\lim_{x \to 3} f(x) = 2}.$$

(7) Consider

$$\frac{x+1}{x-5}.$$

As $x \to 5$, the numerator approaches 6, which is positive. The sign of the denominator determines the sign of the quotient.

For $x \to 5^+$, we have $x - 5 \to 0^+$, so

$$\boxed{\lim_{x \to 5^+} \frac{x+1}{x-5} = +\infty}.$$

For $x \to 5^-$, we have $x - 5 \to 0^-$, so

$$\boxed{\lim_{x \to 5^-} \frac{x+1}{x-5} = -\infty}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 6

Due **Monday, September 21** at 11:55pm, on Gradescope

(1) Factor the following expressions:

(a)
$$xy + 4x + 2xw$$

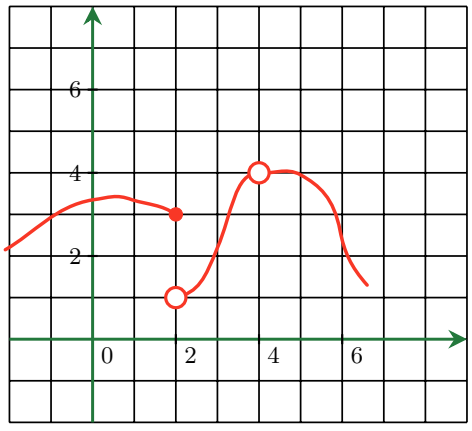
(b)
$$10x^8y^6 + 25x^2y^4 + 20x^3y^{10}$$

(2) Explain in your own words what is meant by the equation
$$\lim_{x\to 2} f(x) = 5$$
Is it possible for this statement to be true and yet $f(2) = 3$?

(3) Explain what it means to say that
$$\lim_{x\to 1^-} f(x) = 3 \text{ and } \lim_{x\to 1^+} f(x) = 7$$
In this situation is it possible than $\lim_{x\to 1} f(x)$ exists?

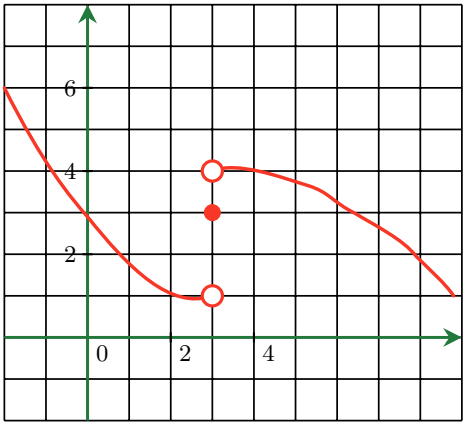
(4) For the function f whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{x\to 2^-} f(x)$
(b) $\lim_{x\to 2^+} f(x)$
(c) $\lim_{x\to 2} f(x)$
(d) $f(2)$
(e) $\lim_{x\to 4} f(x)$
(f) $f(4)$



(5) For the function g whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

(a) $\lim_{t\to 0^-} g(t)$
(b) $\lim_{t\to 0^+} g(t)$
(c) $\lim_{t\to 0} g(t)$
(d) $\lim_{t\to 3^-} g(t)$
(e) $\lim_{t\to 3^+} g(t)$
(f) $\lim_{t\to 3} g(t)$
(g) $g(3)$
(h) $\lim_{t\to 2} g(t)$



(6) Sketch the graph of the function and use it to determine the values of a for which $\lim_{x\to a} f(x)$ exists.

$$f(x) = \begin{cases} 1+x & \text{if } x < -1 \\ x^2 & \text{if } -1 \le x < 1 \\ 2-x & \text{if } x \ge 1 \end{cases}$$

(7) Guess the value of the limit (if it exists) by evaluating at the given numbers (using a calculator), correct to six decimal places.

$$\lim_{x\to 3} \frac{x^2-3x}{x^2-9},$$

where

$$x = 3.1, 3.05, 3.001, 2.9, 2.95, 2.995.$$

Note: I will not ask you to do this on quizzes/exams since calculators won't be available

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 6 SOLUTIONS

(1) Factor the following expressions.

(a) Every term has a factor of x, so

$$xy + 4x + 2xw = x(y + 4 + 2w).$$

(b) The greatest common factor is $5x^2y^4$, so

$$10x^8y^6 + 25x^2y^4 + 20x^3y^{10} = 5x^2y^4\big(2x^6y^2 + 5 + 4xy^6\big).$$

(2) Explain in your own words what is meant by the equation

$$\lim_{x\to 2} f(x) = 5.$$

Solution. This means that when x is close to 2 (but not necessarily equal to 2), the values of $f(x)$ are close to 5. The closer x gets to 2, the closer $f(x)$ gets to 5.

Yes, it is possible to have $\lim_{x\to 2} f(x) = 5$ while $f(2) = 3$. A limit describes what happens to $f(x)$ *near* $x = 2$; it does not require the value of the function at $x = 2$ itself to equal the limit.

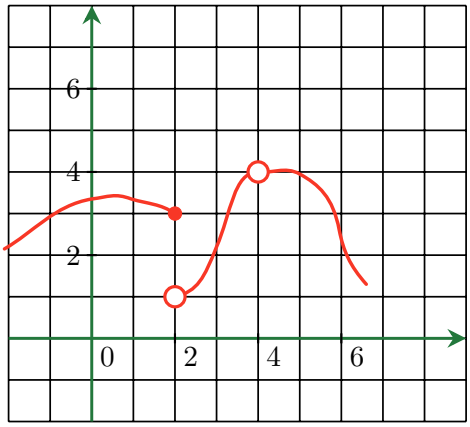
(3) Explain what it means to say that

$$\lim_{x\to 1^-} f(x) = 3 \qquad \text{and} \qquad \lim_{x\to 1^+} f(x) = 7.$$

Solution. The first statement means that as x approaches 1 through values less than 1, $f(x)$ approaches 3. The second means that as x approaches 1 through values greater than 1, $f(x)$ approaches 7.

Since the left-hand and right-hand limits are different, the two-sided limit $\lim_{x\to 1} f(x)$ does not exist.

(4) For the function f whose graph is given, state the value of each quantity, if it exists.



(a) As $x \to 2^-$, the graph approaches the filled point at height 3, so $\lim_{x\to 2^-} f(x) = 3$.

(b) As $x \to 2^+$, the graph approaches the open point at height 1, so $\lim_{x\to 2^+} f(x) = 1$.

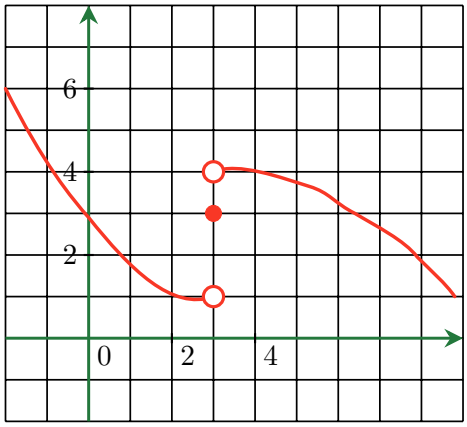
(c) Since the two one-sided limits are different, $\lim_{x\to 2} f(x)$ does not exist.

(d) The filled point at $x = 2$ is at height 3, so $f(2) = 3$.

(e) From both sides of $x = 4$, the graph approaches the open point at height 4, so $\lim_{x\to 4} f(x) = 4$.

(f) There is no filled point on the graph at $x = 4$, so $f(4)$ is undefined.

(5) For the function g whose graph is given, state the value of each quantity, if it exists.



(a) The graph is continuous near $t = 0$ and is at about height 3, so $\lim_{t\to 0^-} g(t) \approx 3$.

(b) Similarly, $\lim_{t\to 0^+} g(t) \approx 3$.

(c) Since the one-sided limits agree, $\lim_{t\to 0} g(t) \approx 3$.

(d) Approaching $t = 3$ from the left, the graph approaches the open point at height 1, so $\lim_{t\to 3^-} g(t) = 1$.

(e) Approaching $t = 3$ from the right, the graph approaches the open point at height 4, so $\lim_{t\to 3^+} g(t) = 4$.

(f) Since the two one-sided limits are different, $\lim_{t\to 3} g(t)$ does not exist.

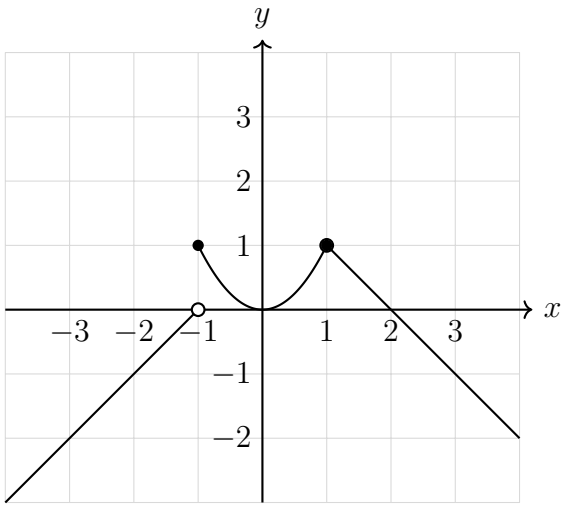
(g) The filled point at $t = 3$ is at height 3, so $g(3) = 3$.

(h) The graph is continuous near $t = 2$ and is at about height 1, so $\lim_{t\to 2} g(t) \approx 1$.

(6) Sketch the graph of

$$f(x) = \begin{cases} 1 + x & \text{if } x < -1, \\ x^2 & \text{if } -1 \le x < 1, \\ 2 - x & \text{if } x \ge 1. \end{cases}$$

Solution. For $x < -1$, the graph is the line $y = x + 1$, with an open point at $(-1, 0)$. For $-1 \le x < 1$, it is the parabola $y = x^2$, with a closed point at $(-1, 1)$ and an open point at $(1, 1)$. For $x \ge 1$, it is the line $y = 2 - x$, beginning with a closed point at $(1, 1)$.



At $x = -1$, the left-hand limit is 0 while the right-hand limit is 1, so the limit does not exist there. At $x = 1$, both sides approach 1, so the limit does exist. Everywhere else, the function is given by a polynomial or a line and has no break.

Therefore $\lim_{x\to a} f(x)$ exists for every real number a except $a = -1$; that is, for

$$\boxed{a \in \mathbb{R} \setminus \{-1\}}.$$

(7) Guess the value of

$$\lim_{x\to 3} \frac{x^2 - 3x}{x^2 - 9}$$

by evaluating the expression at the given values of x.

Solution. Using a calculator and rounding to six decimal places gives

x	$\dfrac{x^2-3x}{x^2-9}$
3.1	0.508197
3.05	0.504132
3.001	0.500083
2.9	0.491525
2.95	0.495798
2.995	0.499583

The values from both sides of 3 are getting closer to 0.5. Thus the data suggest that

$$\boxed{\lim_{x\to 3}\frac{x^2-3x}{x^2-9}=\frac{1}{2}}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026
HOMEWORK 7

Due **Friday, September 25** at 11:55pm, on Gradescope

(1) Determine the following limits.

(a) $\lim_{x\to 1} \dfrac{2-x}{(x-1)^2}$

(b) $\lim_{x\to -2^+} \dfrac{x-1}{x^2(x+2)}$

(c) $\lim_{x\to 5^+} \dfrac{x^2-25}{|x-5|}$

(d) $\lim_{x\to 1} \dfrac{1-\sqrt{x}}{x-5}$

(2) Suppose that f and g are functions such that

$$\lim_{x\to 2} f(x) = 4 \quad \lim_{x\to 2} g(x) = -2 \quad \lim_{x\to 2} h(x) = 0$$

Which of the below limits exist, and if they exist, what are their values? Explain your work.

(a) $\lim_{x\to 2} = (f(x) + 5g(x))$

(b) $\lim_{x\to 2} (g(x))^3$

(c) $\lim_{x\to 2} \sqrt{f(x)}$

(d) $\lim_{x\to 2} \dfrac{g(x)}{h(x)}$

(3) The *signum* (or sign) *function*, denoted by sgn, is defined by

$$\operatorname{sgn} = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

(a) Sketch the graph of this function.

(b) Find each of the following limits or explain why it does not exist.

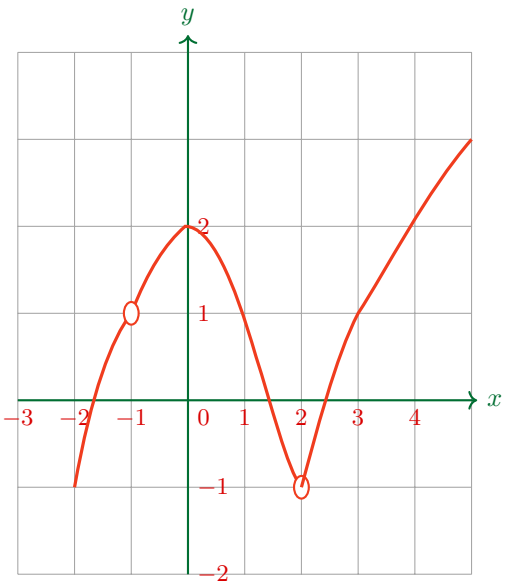
(i) $\lim_{x\to 0^+} \operatorname{sgn} x$

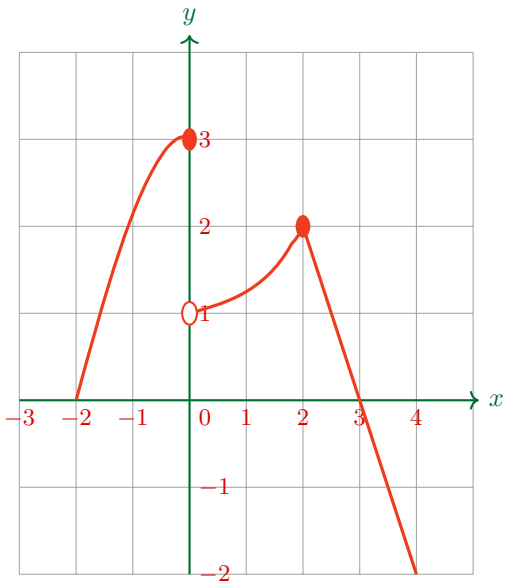
(ii) $\lim_{x\to 0^-} \operatorname{sgn} x$

(iii) $\lim_{x\to 0} \operatorname{sgn} x$

(iv) $\lim_{x\to 0} |\operatorname{sgn} x|$

(4) The graphs of f and g are given. Use them to evaluate each limit, if it exists. If the limit does not exist, explain why.





(i) $\lim_{x\to 2}[f(x)+g(x)]$

(ii) $\lim_{x\to 2}[f(x)-g(x)]$

(iii) $\lim_{x\to -1}[f(x)g(x)]$

(iv) $\lim_{x\to 2}[x^2 f(x)]$

MATH 105 Calculus with Algebra – Fall 2026
Homework 7 Solutions

1.
$$\lim_{x\to 1} \frac{2-x}{(x-1)^2}$$

As $x \to 1$, the numerator approaches 1, while $(x-1)^2$ approaches 0 through positive values. Therefore the quotient grows without bound:

$$\boxed{+\infty}.$$

2.
$$\lim_{x\to -2^+} \frac{x-1}{x^2(x+2)}$$

As $x \to -2^+$, the numerator approaches -3, while x^2 stays positive and $x+2 \to 0^+$. Thus the denominator approaches 0 through positive values, so

$$\boxed{-\infty}.$$

3.
$$\lim_{x\to 5^+} \frac{x^2-25}{|x-5|}$$

Factor the numerator:
$$\frac{x^2-25}{|x-5|} = \frac{(x-5)(x+5)}{|x-5|}.$$
Since $x \to 5^+$, we have $x-5>0$, so $|x-5| = x-5$. Therefore the expression simplifies to $x+5$, and

$$\boxed{10}.$$

4.
$$\lim_{x\to 1} \frac{1-\sqrt{x}}{x-5}$$

The denominator is nonzero at $x=1$, so direct substitution works:

$$\frac{1-\sqrt{1}}{1-5} = \frac{0}{-4} = 0.$$

Thus

$$\boxed{0}.$$

5. Suppose that
$$\lim_{x\to 2} f(x) = 4, \qquad \lim_{x\to 2} g(x) = -2, \qquad \lim_{x\to 2} h(x) = 0.$$

(a)

$$\lim_{x\to 2}\big(f(x)+5g(x)\big)$$

By the limit laws,

$$\lim_{x\to 2}\big(f(x)+5g(x)\big) = 4+5(-2) = -6.$$

Therefore

$$\boxed{-6}.$$

(b)

$$\lim_{x\to 2}(g(x))^3$$

Since the function $u \mapsto u^3$ is continuous,

$$\lim_{x\to 2}(g(x))^3 = (-2)^3 = -8.$$

Thus

$$\boxed{-8}.$$

(c)

$$\lim_{x\to 2}\sqrt{f(x)}$$

Since $f(x) \to 4$ and the square-root function is continuous at 4,

$$\lim_{x\to 2}\sqrt{f(x)} = \sqrt{4} = 2.$$

Thus

$$\boxed{2}.$$

(d)

$$\lim_{x\to 2}\frac{g(x)}{h(x)}$$

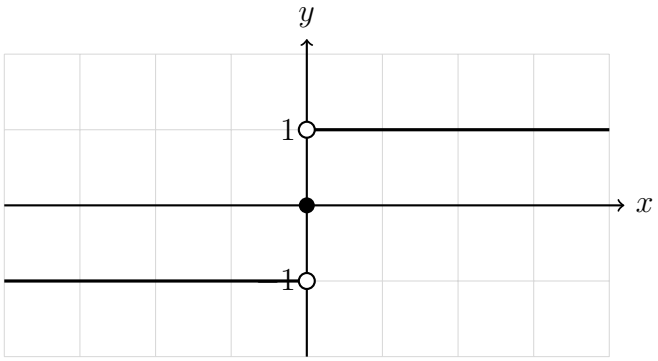
The quotient law cannot be applied because $\lim_{x\to 2} h(x) = 0$. Moreover, the given information does not tell us whether $h(x)$ approaches 0 through positive values, negative values, or with different signs from the two sides. Since the numerator approaches the nonzero number -2, there is not enough information to determine this limit.

$$\boxed{\text{The limit cannot be determined from the given information.}}$$

6. The signum function is

$$\operatorname{sgn}(x) = \begin{cases} -1 & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ 1 & \text{if } x > 0. \end{cases}$$

(a) Its graph is:



(b) (i) As $x \to 0^+$, x is positive, so $\operatorname{sgn}(x) = 1$. Hence

$$\boxed{\lim_{x \to 0^+} \operatorname{sgn}(x) = 1}.$$

(ii) As $x \to 0^-$, x is negative, so $\operatorname{sgn}(x) = -1$. Hence

$$\boxed{\lim_{x \to 0^-} \operatorname{sgn}(x) = -1}.$$

(iii) The one-sided limits are different, so

$$\boxed{\lim_{x \to 0} \operatorname{sgn}(x) \text{ does not exist.}}$$

(iv) For every $x \neq 0$, $|\operatorname{sgn}(x)| = 1$. Therefore

$$\boxed{\lim_{x \to 0} |\operatorname{sgn}(x)| = 1}.$$

7. The graphs of f and g are given below.

From the graph of f, as $x \to 2$ we have $f(x) \to -1$. From the graph of g, as $x \to 2$ we have $g(x) \to 2$.

(a)

$$\lim_{x \to 2}[f(x) + g(x)] = -1 + 2 = 1.$$

Thus

$$\boxed{1}.$$

(b)

$$\lim_{x \to 2}[f(x) - g(x)] = -1 - 2 = -3.$$

Thus

$$\boxed{-3}.$$

(c)

$$\lim_{x \to -1}[f(x)g(x)].$$

From the graph, $f(x) \to 1$ as $x \to -1$. The graph of g approaches a value a little above 2 at $x = -1$ (approximately 2.1 from the drawing). Therefore

$$\lim_{x \to -1} [f(x)g(x)] \approx 1(2.1) = 2.1.$$

Thus, from the graph,

$$\boxed{\text{approximately } 2.1}.$$

Note: The graph does not mark an exact value of the limit of $g(x)$ at $x = -1$, so only an estimate can be read from the drawing.

(d)

$$\lim_{x \to 2} [x^2 f(x)] = (2)^2(-1) = -4.$$

Thus

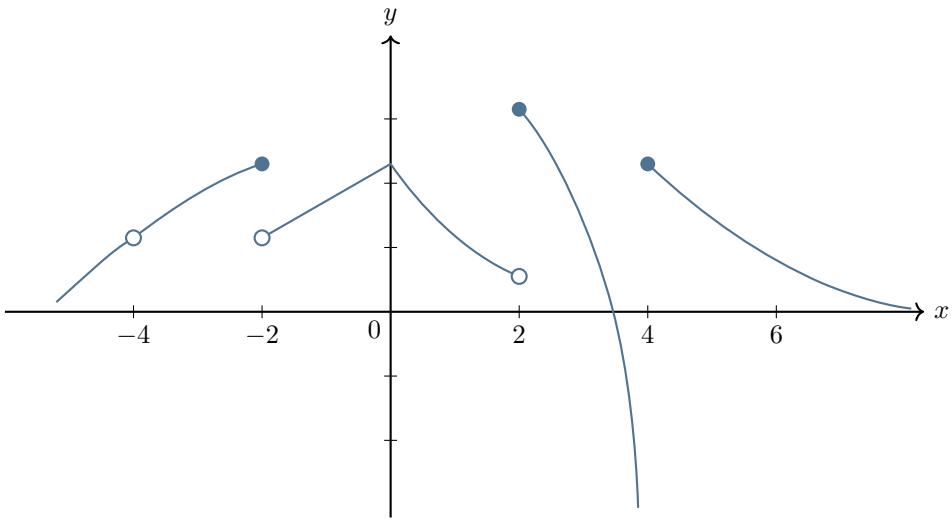
$$\boxed{-4}.$$

MATH 105 CALCULUS WITH ALGEBRA FALL 2026

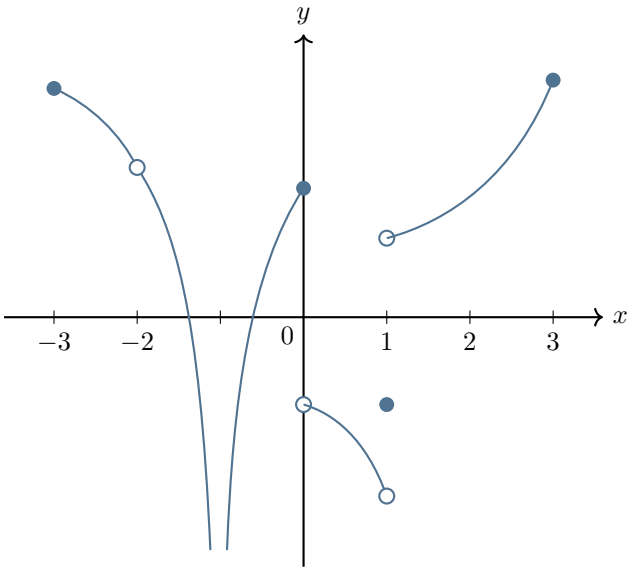
HOMEWORK 8

DUE MONDAY, SEPTEMBER 28 AT 11:55PM, ON GRADESCOPE

(1) (a) From the graph of f, state the numbers at which f is discontinuous and explain why.
(b) For each of the numbers stated in part (a), determine whether f is continuous from the right, or from the left, or neither.



(2) From the graph of g, state the intervals on which g is continuous.



(3) Explain why the following function is discontinuous at the given number a. Sketch the graph of the function.

$$f(x) = \frac{1}{x+2}, \qquad a = -2.$$

(4) Explain why the following function is discontinuous at the given number a. Sketch the graph of the function.

$$f(x) = \begin{cases} 1 - x^2, & x < 1, \\ \dfrac{1}{x}, & x \geq 1, \end{cases} \qquad a = 1.$$

(5) Is the following function continuous at 0? What about at -1? Sketch its graph.

$$f(x) = \begin{cases} \dfrac{1}{x+2}, & x < -1, \\ 1, & x \geq -1, \end{cases}$$

(6) Draw the graph of a function f whose domain is all real numbers and is continuous except for the following discontinuities:
   (a) f has a removable discontinuity at -2 and an infinite discontinuity at 2
   (b) f has a jump discontinuity at -3 and removable discontinuity at 2
   (c) f is discontinuous at 0 and 3, but continuous from the right at 0 and from the left at 3

MATH 105 Calculus with Algebra — Fall 2026
Homework 8 Solutions

1. (a) From the graph, f is discontinuous at

$$\boxed{x = -4,\ -2,\ 2,\ 4}.$$

At $x = -4$ there is a hole, so this is a removable discontinuity.

At $x = -2$, the left-hand and right-hand limits are different, so there is a jump discontinuity.

At $x = 2$, the left-hand limit is the height of the open circle, while the right-hand limit is the height of the filled point. These are different, so there is a jump discontinuity.

At $x = 4$, the graph becomes unbounded as $x \to 4^-$, while the right-hand branch begins at a finite filled point. Thus $x = 4$ is an infinite discontinuity.

(b) At $x = -4$, $f(-4)$ is not defined, so f is continuous from neither side.

At $x = -2$, the filled point lies on the branch coming from the left, so

$$\boxed{f \text{ is continuous from the left at } -2.}$$

At $x = 2$, the filled point lies on the branch coming from the right, so

$$\boxed{f \text{ is continuous from the right at } 2.}$$

At $x = 4$, the branch beginning at the filled point is continuous to the right, so

$$\boxed{f \text{ is continuous from the right at } 4.}$$

2. From the graph of g, the breaks occur at $x = -2, -1, 0$, and 1.

The continuous pieces are

$$\boxed{[-3, -2), \quad (-2, -1), \quad (-1, 0], \quad (0, 1), \quad (1, 3].}$$

The endpoints -3, 0, and 3 can be included on the indicated intervals because continuity at an endpoint of an interval only requires the appropriate one-sided limit.

3.

$$f(x) = \frac{1}{x+2}, \qquad a = -2.$$

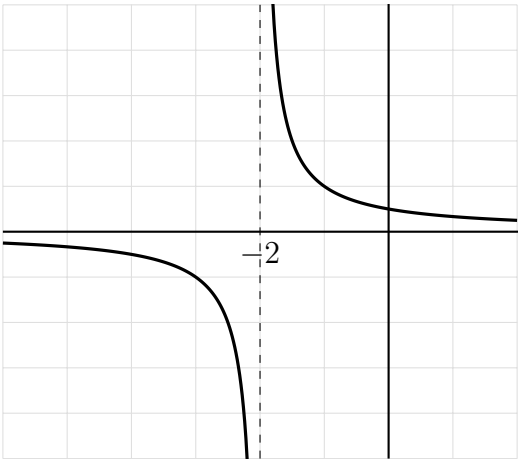
The function is not defined at $x = -2$. In fact, $x = -2$ is a vertical asymptote:

$$\lim_{x \to -2^-} \frac{1}{x+2} = -\infty, \qquad \lim_{x \to -2^+} \frac{1}{x+2} = +\infty.$$

Therefore

$$\boxed{f \text{ is discontinuous at } x = -2.}$$

A sketch is:



4.

$$f(x) = \begin{cases} 1 - x^2, & x < 1, \\ \dfrac{1}{x}, & x \geq 1. \end{cases}$$

At $x = 1$,

$$\lim_{x \to 1^-} f(x) = 1 - 1^2 = 0, \qquad \lim_{x \to 1^+} f(x) = 1, \qquad f(1) = 1.$$

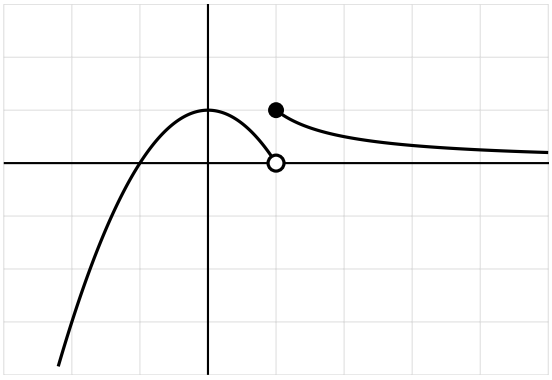
The one-sided limits are different, so

$$\boxed{f \text{ is discontinuous at } x = 1.}$$

It is continuous from the right at $x = 1$, since

$$\lim_{x \to 1^+} f(x) = f(1) = 1.$$

A sketch is:



5.

$$f(x) = \begin{cases} \dfrac{1}{x+2}, & x < -1, \\ 1, & x \geq -1. \end{cases}$$

At $x = 0$, the function is simply $f(x) = 1$ in a neighborhood of 0, so

$$\boxed{f \text{ is continuous at } 0.}$$

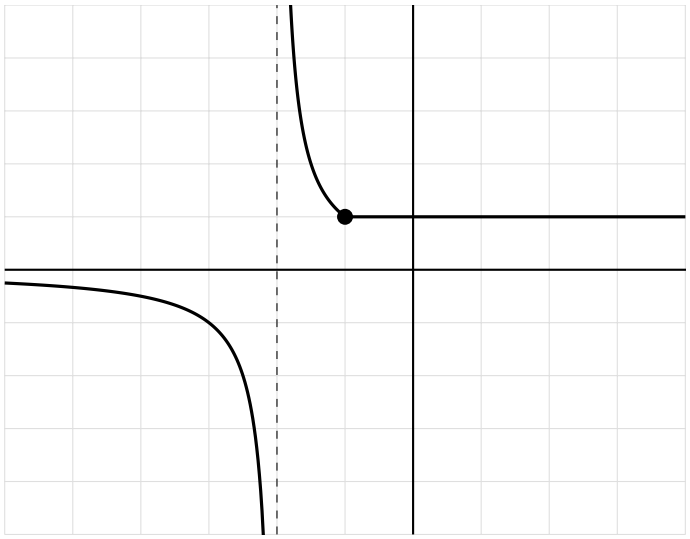
At $x = -1$,

$$\lim_{x \to -1^-} f(x) = \frac{1}{-1+2} = 1, \qquad \lim_{x \to -1^+} f(x) = 1, \qquad f(-1) = 1.$$

Therefore

$$\boxed{f \text{ is also continuous at } -1.}$$

Notice that the graph does have a vertical asymptote at $x = -2$.

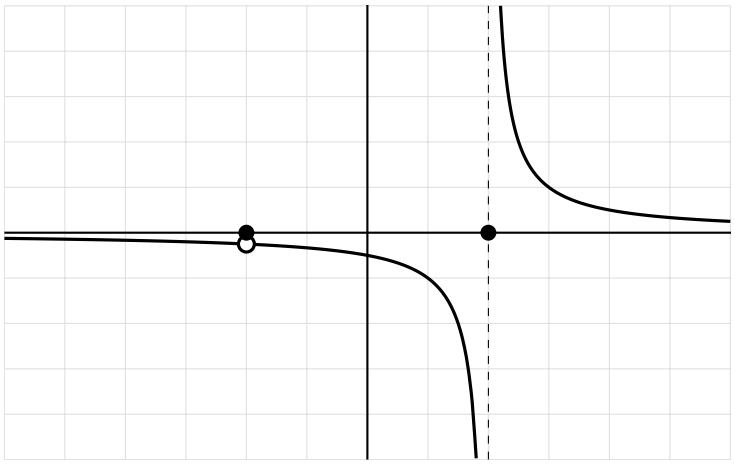


6. Many answers are possible. Here are three examples.

   (a) We want a removable discontinuity at -2 and an infinite discontinuity at 2, while keeping the domain equal to all real numbers. One example is

$$f(x) = \begin{cases} \dfrac{1}{x-2}, & x \neq -2, 2, \\ 0, & x = -2, \\ 0, & x = 2. \end{cases}$$

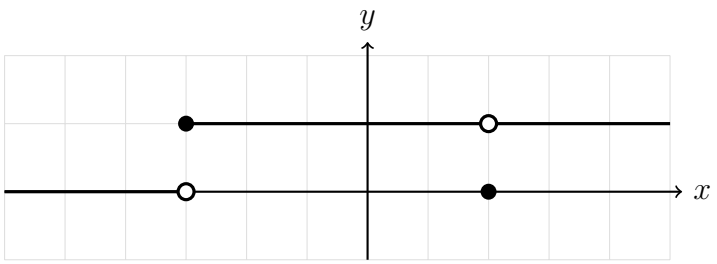
   At $x = -2$, the limit exists and equals $-1/4$, but $f(-2) = 0$, so the discontinuity is removable. At $x = 2$, the graph has a vertical asymptote, even though we have assigned a value to $f(2)$.



(b) We want a jump at -3 and a removable discontinuity at 2. For example,

$$f(x) = \begin{cases} 0, & x < -3, \\ 1, & x \geq -3,\ x \neq 2, \\ 0, & x = 2. \end{cases}$$

At $x = -3$, the left-hand limit is 0 and the right-hand limit is 1, so there is a jump. At $x = 2$, the limit is 1 but the function value is 0, so the discontinuity is removable.

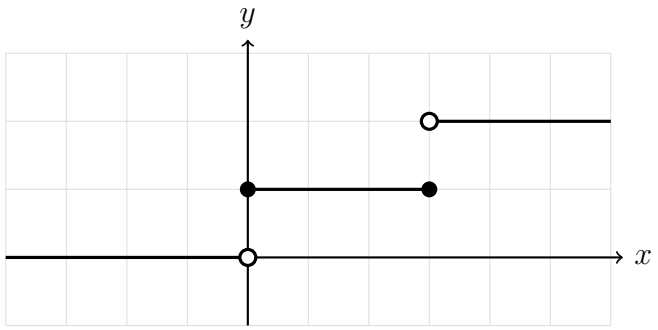


(c) We want discontinuities at 0 and 3, with right continuity at 0 and left continuity at 3. One example is

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & 0 \leq x \leq 3, \\ 2, & x > 3. \end{cases}$$

At $x = 0$, $f(0) = 1$ and the right-hand limit is 1, so the function is continuous from the right there, but the left-hand limit is 0.

At $x = 3$, $f(3) = 1$ and the left-hand limit is 1, so the function is continuous from the left there, but the right-hand limit is 2.


y
x

Instructor: David Zureick-Brown
Fall 2026

Amherst College
Dept. of Mathematics

Math 105 ∼ Calculus with Algebra: Review Packet

- These are practice problems, and are not to be turned in.
- On Wednesday in class, I will work any problems you ask about, and answer any questions.
- On Thursday, there will be a practice exam during the worksheet session.
- *Good luck!*

What you need to know for Exam 1

You should know everything from the beginning of the course up to (and including). The following is a list of most of the topics covered. **THIS IS NOT A COMPREHENSIVE LIST, BUT MERELY AN AID.** Remember, no calculators or electronics that can connect to the internet on any exams.

- Appendices A and B: Review equations of lines, and inequalities. Parallel lines have same slope. Perpendicular lines have opposite reciprocal slopes. Know the difference between *point-slope form*, and *slope-intercept form*. Know how to use *slope-intercept form* to sketch lines quickly and efficiently.
- 1.1: What is a function? (Vertical line test.) Domain and Range of a function. Know how to quickly sketch all the standard functions like x^2, x^3, $\frac{1}{x}$, $\sqrt{x}$, and $|x|$. Piecewise-defined functions, including using all these functions.
- 1.2: Standard functions: polynomials (including constant and linear), power functions (including root functions and things like $\frac{1}{x}$), rational functions.
- 1.3: Adding, subtracting, multiplying, and dividing functions. Composition of functions. Given the graph of $y = f(x)$, what do $y = f(x) \pm c$, $y = f(x \pm c)$, $y = cf(x)$, $y = f(cx)$, $y = -f(x)$, $y = f(-x)$, and $y = |f(x)|$ look like? (See boxes on pp. 36–37, and Example 5.)
- 1.5: Limits. Intuitive idea of what a limit is. Estimating limits using tables and computations. Be familiar with the various ways a limit can fail to exist. Right- and left-hand limits; and Box 3, page 55. Infinite limits. Vertical asymptotes.
- 1.6: Limit laws: sum, difference, constant multiple, product, quotient, power, root laws. Computing limits of polynomials, rational functions, and other functions. Computing harder limits, like 0/0, using algebraic tricks. What to do in each case of a DSP (Direct substitution) attempt:

  When you get

  $$\frac{\text{constant, including 0}}{\text{non-zero constant}}$$

  then it's just DSP.

  When you get

  $$\frac{\text{non-zero constant}}{0}$$

then you do *sign analysis*. Study the RHL and LHL (Right and Left Hand Limit) cases carefully. Yes it will be an infinite limit, but what sign is attached? Pay attention to your two-point argument: 1. size and 2. sign.

When you get

$$\frac{0}{0}$$

then you use an algebraic technique to cancel the zero factor. Use factoring, common denominator, or conjugate techniques. It means **MORE WORK!**

Finally, if you see absolute values, certainly for a $\frac{0}{0}$ case, then you must explore the two RHL and LHL cases.

- 1.8: Continuity: at a number a, or on an interval I. Know both the official definition

$$\lim_{x \to a} f(x) = f(a)$$

and the intuitive idea (you can draw it without lifting your pencil off the paper). The three ways continuity can fail ($f(a)$ not defined; or $\lim_{x \to a} f(x)$ not defined; or both defined, but not equal). Continuity from the right or left.

Tips

- You will be allowed to do the problems in any order, but **make sure you don't accidentally skip any** if you like to jump around.
- Make sure you know how to manipulate functions. If any of the problems from Sections 1.1–1.3 gave you serious trouble, make sure you get things cleared up.
- You will need to compute some limits carefully with full justification. Be ready. Know the algebra tricks you need for $\frac{0}{0}$ limits. Know when to use LHL and RHL sign analysis. Also know how to examine the two cases for absolute value pieces of your functions.
- Make sure you are clear when making a statement about discontinuity. You must tell me which of the three pieces of the definition fail(s).
- Do not write $\lim_{x \to a}$ by itself without the function.
- If you make a declaration about a RHL or LHL limit, then write it out formally using math notation. Look to put forward your *best presentation* for each solution.
- For your piece-wise defined graphs, make sure that when you piece together the puzzle of individual function pieces, that you pay attention to the end-point output values where those cuts are. That will help you determine whether the function pieces line up. Indeed that might change a check of continuity at a given point.
- Justify all of your work!! Don't just use random words to explain answers; I will look for clear mathematical statements.

Limit Practice Problems

Evaluate the following limits. Be clear if the limit equals a finite value, Does Not Exist, or is $+\infty$ or $-\infty$. Always justify your work:

1. $\lim_{w\to 0} \frac{16}{w} =$
2. $\lim_{t\to 2} \frac{3-t}{t-2} =$
3. $\lim_{t\to 2} \frac{3-t}{(t-2)^2} =$
4. $\lim_{x\to 4} \frac{(x+2)^2}{x^2-3x-4} =$
5. $\lim_{x\to 4} \frac{x-4}{x^2-3x-4} =$
6. $\lim_{x\to 4} \frac{x^2-2x-8}{x^2-3x-4} =$
7. $\lim_{x\to 1} \frac{x^2-4x-12}{x^2-3x-18} =$
8. $\lim_{x\to 0} \frac{x^2-4x-12}{x^2-3x-18} =$
9. $\lim_{x\to -3} \frac{x+2}{x+3} =$
10. $\lim_{x\to -2} \frac{x^2-4x-12}{x^2-3x-18} =$
11. $\lim_{x\to 0} \frac{x^2-4x}{x^2-7x} =$
12. $\lim_{x\to 3} \frac{x^2-9}{|x-3|} =$
13. $\lim_{x\to -5} \frac{x^2+6x+5}{|x+5|} =$
14. $\lim_{t\to 1} \frac{t^2-1}{t^2-11t+10} =$
15. $\lim_{t\to 1} \frac{t^2}{t^2+t-1} =$

16. $\lim_{t\to -1} \dfrac{2009(t^2+6t+5)}{t^2+t} =$

17. $\lim_{x\to 9} \dfrac{x^2-10x+9}{x^2+x-90} =$

18. $\lim_{t\to 1} \left(t^{500}+t^{400}+t^{300}+t^{200}+t^{100}\right) =$

19. $\lim_{x\to 3} \dfrac{x^2-x-6}{x^2-2x-3} =$

20. $\lim_{x\to 1} \dfrac{x-1}{\sqrt{x+3}-2} =$

21. $\lim_{x\to 9} \dfrac{9x-x^2}{3-\sqrt{x}} =$

22. $\lim_{x\to 1} \dfrac{5}{1-x} =$

23. $\lim_{x\to 5} \dfrac{6x}{5-x} =$

24. $\lim_{x\to 2} \dfrac{x^2-9x+14}{x^2-4x+4} =$

25. $\lim_{x\to 2} \dfrac{x^2-4}{|x-2|} =$

26. $\lim_{x\to 3} \dfrac{\sqrt{x+6}-3}{x^2-x-6} =$

27. $\lim_{x\to 7} \dfrac{\frac{1}{7}-\frac{1}{x}}{x-7} =$

28. $\lim_{x\to -6} \dfrac{\frac{1}{2-x}-\frac{1}{8}}{x+6} =$

29. $\lim_{x\to 3} \dfrac{\sqrt{x+1}-2}{3-x} =$

30. $\lim_{x\to 7} \dfrac{x^2-49}{2-\sqrt{x-3}} =$

31. $\lim_{x\to 5} \dfrac{\frac{1}{\sqrt{x+20}}-\frac{1}{5}}{x-5} =$ Challenge!

Functions and Limit Practice Problems Evaluate the following limits:

32. Let $g(x) = 2x + 1$. Compute $\lim_{x\to 1} \dfrac{x-1}{g(x^2)-3} =$

33. Let $G(u) = u^2 + u$. Compute $\lim_{u\to 2} \dfrac{u^2-2u}{G(u-3)} =$

34. Let $h(y) = y^2 - 3$. Compute $\lim_{x\to -2} \dfrac{x+2}{h(2x)-h(x+6)} =$

35. Let $f(t) = \dfrac{1}{t}$. Compute $\lim_{t\to 4} \dfrac{f(t-3)-4f(t)}{t-4} =$

More Functions

36. Let $f(x) = \sqrt{x}$, $g(x) = x^2 + 4$, and $h(x) = \dfrac{1}{x}$. Compute (and simplify, if possible) the following:

    (a) $f \circ g(x) =$

    (b) $g \circ f(x) =$

    (c) $h \circ g \circ f(x) =$

    (d) $g \circ g(x) =$

Piece-wise defined functions

Consider each of the following piecewise defined functions. Answer the related questions. *Justify your answers* please.

37. Let

$$f(x) = \begin{cases} x, & x < 0, \\ x^2, & 0 \le x \le 2, \\ 8 - x, & x > 2. \end{cases}$$

    Sketch the graph. Find the numbers at which f is discontinuous. Evaluate:

$$\lim_{x\to 2} f(x) = \qquad \lim_{x\to 0} f(x) = \qquad f(2) = \qquad f(4) =$$

38. Let

$$f(x) = \begin{cases} x - 1, & x < 2, \\ 1, & 2 < x < 4, \\ 3, & x = 4, \\ \sqrt{x}, & x > 4. \end{cases}$$

    Sketch the graph. Find the numbers at which f is discontinuous. Evaluate:

$$\lim_{x\to 0} f(x) = \qquad \lim_{x\to 2} f(x) = \qquad \lim_{x\to 4} f(x) = \qquad f(4) =$$

39. Let

$$h(x) = \begin{cases} \dfrac{8}{x+2}, & x < 0, \\ 2, & x = 0, \\ \dfrac{1}{2}x - 4, & 0 < x \leq 16, \\ \sqrt{x}, & x > 16. \end{cases}$$

Sketch the graph. Find the numbers at which h is discontinuous. Evaluate:

$$\lim_{x \to -2} h(x) = \qquad \lim_{x \to 0} h(x) = \qquad \lim_{x \to 16} h(x) = \qquad f(0) = \qquad f(16) =$$

40. Let

$$F(x) = \begin{cases} \dfrac{1}{x-4}, & x > 4, \\ 3 - x, & 0 < x < 4, \\ 3 - x^2, & x < 0. \end{cases}$$

Sketch the graph. Find the numbers at which F is discontinuous. Evaluate:

$$\lim_{x \to 0} F(x) = \qquad \lim_{x \to 4} F(x) = \qquad F(-3) = \qquad F(4) =$$

41. Consider the line L given by $2x + 4y = 12$.

    (a) Sketch this line L.

    (b) Find the equation of the new line M that is **parallel** to the first line L, and passes through the point $(0, 1)$.

    (c) Find the equation of the new line M' that is **perpendicular** to the first line L, and passes through the point $(1, -2)$.

    (d) Sketch this new line M found in (b).

42. Sketch the graphs of the following functions. Label a few points on the graph.

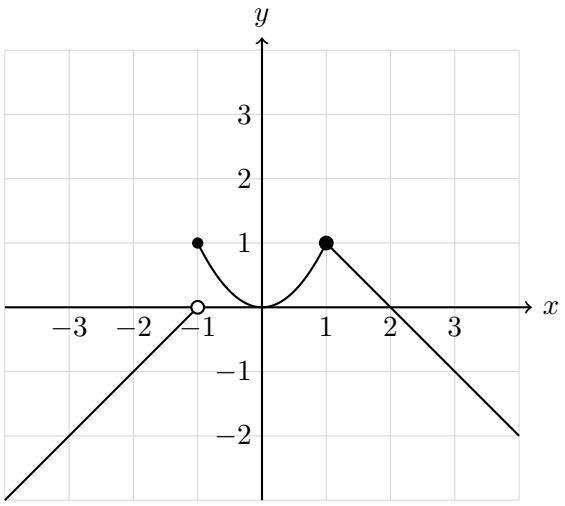
    (a) $f(x) = \sqrt{x - 3}$

    (b) $f(x) = 1 + \frac{1}{x+1}$

    (c) $f(x) = x|x|$

    (d) $f(x) = \frac{x^2-9}{|x-3|}$

43. Let $f(x)$ be the function with the following graph.



(a) Graph $-f(x)$.

(b) Graph $1 + f(2x + 2)$.

(c) Write a (piecewise) formula for $f(x)$.

44. Let $f(x) = x^2 + 3x$.

(a) Compute $f(x + h)$

(b) Simplify

$$\frac{f(x+h) - f(x)}{h}$$

45. Simplify the following expressions:

(a)

$$\frac{1}{x-1} - \frac{1}{x}$$

(b)

$$\frac{x}{x-1} - \frac{x}{x+1}$$

(c)

$$\frac{\frac{s+1}{s-1} + \frac{s-1}{s+1}}{\frac{1}{s^2-1}}$$

Math 105: Midterm 1 Review Packet Solutions

Fall 2026

These solutions are meant to model the amount of justification you should be prepared to give on an exam. In particular, for infinite limits and limits that do not exist, pay attention to the right- and left-hand limits.

Limit Practice Problems

1. $$\lim_{w\to 0} \frac{16}{w}.$$

   As $w \to 0^-$, $16/w \to -\infty$, while as $w \to 0^+$, $16/w \to +\infty$. Therefore

   $$\lim_{w\to 0} \frac{16}{w} = \boxed{\text{DNE}}.$$

2. $$\lim_{t\to 2} \frac{3-t}{t-2}.$$

   Near $t = 2$, the numerator is positive. The denominator is negative from the left and positive from the right, so the one-sided limits are $-\infty$ and $+\infty$. Thus

   $$\lim_{t\to 2} \frac{3-t}{t-2} = \boxed{\text{DNE}}.$$

3. $$\lim_{t\to 2} \frac{3-t}{(t-2)^2}.$$

   The numerator approaches $1 > 0$ and $(t-2)^2 \to 0^+$, so

   $$\lim_{t\to 2} \frac{3-t}{(t-2)^2} = \boxed{+\infty}.$$

4. $$\lim_{x\to 4} \frac{(x+2)^2}{x^2-3x-4} = \lim_{x\to 4} \frac{(x+2)^2}{(x-4)(x+1)}.$$

   Near $x = 4$, $(x+2)^2$ and $x+1$ are positive, so the sign is controlled by $x-4$. Hence the left-hand limit is $-\infty$ and the right-hand limit is $+\infty$. Therefore

   $$\lim_{x\to 4} \frac{(x+2)^2}{x^2-3x-4} = \boxed{\text{DNE}}.$$

5. $$\lim_{x\to 4} \frac{x-4}{x^2-3x-4} = \lim_{x\to 4} \frac{x-4}{(x-4)(x+1)} = \lim_{x\to 4} \frac{1}{x+1} = \boxed{\frac{1}{5}}.$$

6. $$\lim_{x\to 4} \frac{x^2-2x-8}{x^2-3x-4} = \lim_{x\to 4} \frac{(x-4)(x+2)}{(x-4)(x+1)} = \boxed{\frac{6}{5}}.$$

7. Direct substitution gives
$$\lim_{x\to 1}\frac{x^2-4x-12}{x^2-3x-18}=\frac{-15}{-20}=\boxed{\frac{3}{4}}.$$

8. Direct substitution gives
$$\lim_{x\to 0}\frac{x^2-4x-12}{x^2-3x-18}=\frac{-12}{-18}=\boxed{\frac{2}{3}}.$$

9.
$$\lim_{x\to -3}\frac{x+2}{x+3}.$$
The numerator approaches -1. From the left, $x+3<0$, so the quotient tends to $+\infty$; from the right, $x+3>0$, so the quotient tends to $-\infty$. Therefore
$$\lim_{x\to -3}\frac{x+2}{x+3}=\boxed{\text{DNE}}.$$

10. Direct substitution gives
$$\lim_{x\to -2}\frac{x^2-4x-12}{x^2-3x-18}=\frac{0}{-8}=\boxed{0}.$$

11.
$$\lim_{x\to 0}\frac{x^2-4x}{x^2-7x}=\lim_{x\to 0}\frac{x(x-4)}{x(x-7)}=\lim_{x\to 0}\frac{x-4}{x-7}=\boxed{\frac{4}{7}}.$$

12.
$$\frac{x^2-9}{|x-3|}=(x+3)\frac{x-3}{|x-3|}.$$
As $x\to 3^-$ the expression tends to -6, while as $x\to 3^+$ it tends to 6. Therefore
$$\lim_{x\to 3}\frac{x^2-9}{|x-3|}=\boxed{\text{DNE}}.$$

13.
$$\frac{x^2+6x+5}{|x+5|}=(x+1)\frac{x+5}{|x+5|}.$$
As $x\to -5^-$ the expression tends to 4, while as $x\to -5^+$ it tends to -4. Therefore
$$\lim_{x\to -5}\frac{x^2+6x+5}{|x+5|}=\boxed{\text{DNE}}.$$

14.
$$\lim_{t\to 1}\frac{t^2-1}{t^2-11t+10}=\lim_{t\to 1}\frac{(t-1)(t+1)}{(t-1)(t-10)}=\boxed{-\frac{2}{9}}.$$

15. Direct substitution gives
$$\lim_{t\to 1}\frac{t^2}{t^2+t-1}=\boxed{1}.$$

16.
$$\lim_{t\to -1}\frac{2009(t^2+6t+5)}{t^2+t}=\lim_{t\to -1}2009\frac{(t+1)(t+5)}{t(t+1)}=2009\frac{4}{-1}=\boxed{-8036}.$$

17.
$$\lim_{x\to 9}\frac{x^2-10x+9}{x^2+x-90}=\lim_{x\to 9}\frac{(x-1)(x-9)}{(x+10)(x-9)}=\boxed{\frac{8}{19}}.$$

18. By direct substitution,
$$\lim_{t\to 1}\left(t^{500}+t^{400}+t^{300}+t^{200}+t^{100}\right)=\boxed{5}.$$

19.
$$\lim_{x\to 3}\frac{x^2-x-6}{x^2-2x-3}=\lim_{x\to 3}\frac{(x-3)(x+2)}{(x-3)(x+1)}=\boxed{\frac{5}{4}}.$$

20. Rationalizing the denominator,
$$\frac{x-1}{\sqrt{x+3}-2}\cdot\frac{\sqrt{x+3}+2}{\sqrt{x+3}+2}=\frac{(x-1)(\sqrt{x+3}+2)}{x-1}=\sqrt{x+3}+2.$$
Thus
$$\lim_{x\to 1}\frac{x-1}{\sqrt{x+3}-2}=\boxed{4}.$$

21. Since $9-x=(3-\sqrt{x})(3+\sqrt{x})$,
$$\frac{9x-x^2}{3-\sqrt{x}}=\frac{x(9-x)}{3-\sqrt{x}}=x(3+\sqrt{x}),$$
so the limit is
$$\lim_{x\to 9}\frac{9x-x^2}{3-\sqrt{x}}=\boxed{54}.$$

22.
$$\lim_{x\to 1}\frac{5}{1-x}.$$
As $x\to 1^-$, $1-x\to 0^+$, so the quotient tends to $+\infty$. As $x\to 1^+$, $1-x\to 0^-$, so the quotient tends to $-\infty$. Therefore
$$\lim_{x\to 1}\frac{5}{1-x}=\boxed{\text{DNE}}.$$

23.
$$\lim_{x\to 5}\frac{6x}{5-x}.$$
The numerator approaches $30>0$. From the left the denominator is positive and from the right it is negative, so the one-sided limits are $+\infty$ and $-\infty$. Therefore
$$\lim_{x\to 5}\frac{6x}{5-x}=\boxed{\text{DNE}}.$$

24.
$$\frac{x^2-9x+14}{x^2-4x+4}=\frac{(x-7)(x-2)}{(x-2)^2}=\frac{x-7}{x-2}.$$
Near $x=2$, the numerator is negative. Thus the left-hand limit is $+\infty$ and the right-hand limit is $-\infty$, so
$$\lim_{x\to 2}\frac{x^2-9x+14}{x^2-4x+4}=\boxed{\text{DNE}}.$$

25.
$$\frac{x^2-4}{|x-2|} = (x+2)\frac{x-2}{|x-2|}.$$
The left-hand limit is -4 and the right-hand limit is 4. Hence
$$\lim_{x\to 2} \frac{x^2-4}{|x-2|} = \boxed{\text{DNE}}.$$

26. Rationalize the numerator:
$$\frac{\sqrt{x+6}-3}{x^2-x-6} = \frac{x-3}{(\sqrt{x+6}+3)(x-3)(x+2)}.$$
After canceling $x-3$,
$$\lim_{x\to 3} \frac{1}{(\sqrt{x+6}+3)(x+2)} = \boxed{\frac{1}{30}}.$$

27.
$$\frac{\frac{1}{7}-\frac{1}{x}}{x-7} = \frac{\frac{x-7}{7x}}{x-7} = \frac{1}{7x},$$
so
$$\lim_{x\to 7} \frac{\frac{1}{7}-\frac{1}{x}}{x-7} = \boxed{\frac{1}{49}}.$$

28.
$$\frac{\frac{1}{2-x}-\frac{1}{8}}{x+6} = \frac{\frac{x+6}{8(2-x)}}{x+6} = \frac{1}{8(2-x)},$$
so
$$\lim_{x\to -6} \frac{\frac{1}{2-x}-\frac{1}{8}}{x+6} = \boxed{\frac{1}{64}}.$$

29. Rationalizing the numerator,
$$\frac{\sqrt{x+1}-2}{3-x} = \frac{x-3}{(\sqrt{x+1}+2)(3-x)} = -\frac{1}{\sqrt{x+1}+2}.$$
Therefore
$$\lim_{x\to 3} \frac{\sqrt{x+1}-2}{3-x} = \boxed{-\frac{1}{4}}.$$

30. Multiply by the conjugate of the denominator:
$$\frac{x^2-49}{2-\sqrt{x-3}} \cdot \frac{2+\sqrt{x-3}}{2+\sqrt{x-3}} = \frac{(x-7)(x+7)(2+\sqrt{x-3})}{7-x}.$$
Since $7-x = -(x-7)$, this becomes
$$-(x+7)(2+\sqrt{x-3}),$$
so the limit is
$$\lim_{x\to 7} \frac{x^2-49}{2-\sqrt{x-3}} = \boxed{-56}.$$

31. Rationalize the numerator:

$$\frac{\frac{1}{\sqrt{x+20}} - \frac{1}{5}}{x-5} = \frac{5 - \sqrt{x+20}}{5\sqrt{x+20}(x-5)}.$$

Multiplying the numerator and denominator by $5 + \sqrt{x+20}$ gives

$$\frac{25 - (x+20)}{5\sqrt{x+20}(x-5)(5+\sqrt{x+20})} = -\frac{1}{5\sqrt{x+20}(5+\sqrt{x+20})}.$$

Thus

$$\lim_{x\to 5} \frac{\frac{1}{\sqrt{x+20}} - \frac{1}{5}}{x-5} = \boxed{-\frac{1}{250}}.$$

32. Since $g(x^2) = 2x^2 + 1$,

$$\frac{x-1}{g(x^2)-3} = \frac{x-1}{2x^2-2} = \frac{1}{2(x+1)},$$

so

$$\lim_{x\to 1} \frac{x-1}{g(x^2)-3} = \boxed{\frac{1}{4}}.$$

33.

$$G(u-3) = (u-3)^2 + (u-3) = u^2 - 5u + 6 = (u-2)(u-3).$$

Therefore

$$\frac{u^2-2u}{G(u-3)} = \frac{u(u-2)}{(u-2)(u-3)} = \frac{u}{u-3},$$

and the limit is

$$\lim_{u\to 2} \frac{u^2-2u}{G(u-3)} = \boxed{-2}.$$

34.

$$h(2x) - h(x+6) = \big(4x^2 - 3\big) - \big((x+6)^2 - 3\big) = 3(x-6)(x+2).$$

Hence

$$\frac{x+2}{h(2x)-h(x+6)} = \frac{1}{3(x-6)},$$

so the limit is

$$\lim_{x\to -2} \frac{x+2}{h(2x)-h(x+6)} = \boxed{-\frac{1}{24}}.$$

35. Since $f(t) = 1/t$,

$$f(t-3) - 4f(t) = \frac{1}{t-3} - \frac{4}{t} = \frac{-3(t-4)}{t(t-3)}.$$

Thus

$$\frac{f(t-3)-4f(t)}{t-4} = -\frac{3}{t(t-3)},$$

and the limit is

$$\lim_{t\to 4} \frac{f(t-3)-4f(t)}{t-4} = \boxed{-\frac{3}{4}}.$$

36. Let $f(x) = \sqrt{x}$, $g(x) = x^2 + 4$, and $h(x) = 1/x$.

(a)

$$(f \circ g)(x) = \boxed{\sqrt{x^2+4}}.$$

Its domain is all real numbers.

(b)

$$(g \circ f)(x) = (\sqrt{x})^2 + 4 = \boxed{x+4},$$

but the domain of the composition is $x \geq 0$.

(c)

$$(h \circ g \circ f)(x) = \frac{1}{(\sqrt{x})^2 + 4} = \boxed{\frac{1}{x+4}},$$

with domain $x \geq 0$.

(d)

$$(g \circ g)(x) = (x^2+4)^2 + 4 = \boxed{x^4 + 8x^2 + 20}.$$

Piecewise-defined Functions

37.

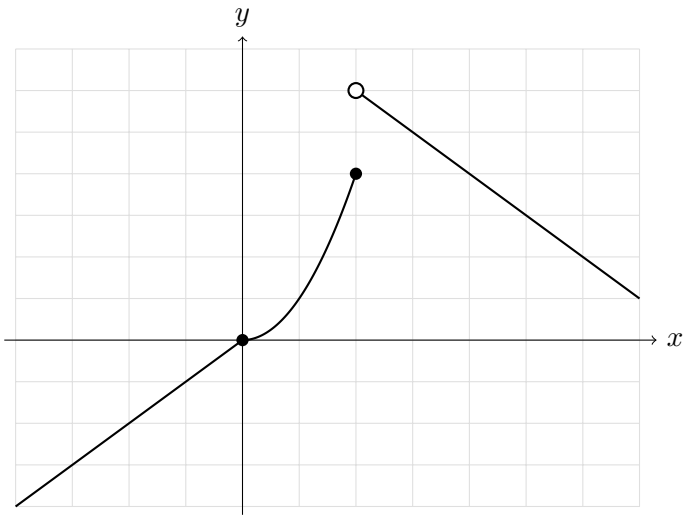
$$f(x) = \begin{cases} x, & x < 0, \\ x^2, & 0 \leq x \leq 2, \\ 8 - x, & x > 2. \end{cases}$$

At $x = 0$, both one-sided limits are 0 and $f(0) = 0$, so f is continuous there. At $x = 2$,

$$\lim_{x \to 2^-} f(x) = 4, \qquad \lim_{x \to 2^+} f(x) = 6,$$

so the two-sided limit does not exist. Therefore the only discontinuity is $x = 2$.

$$\boxed{\lim_{x \to 2} f(x) = \text{DNE}}, \qquad \boxed{\lim_{x \to 0} f(x) = 0}, \qquad \boxed{f(2) = 4}, \qquad \boxed{f(4) = 4}.$$

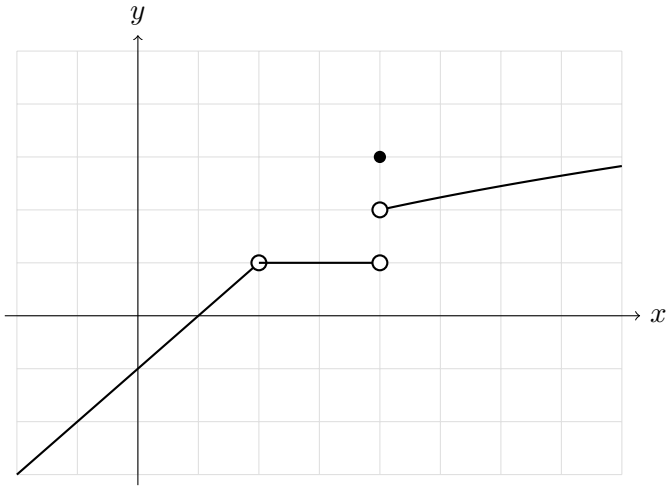


38.

$$f(x) = \begin{cases} x - 1, & x < 2, \\ 1, & 2 < x < 4, \\ 3, & x = 4, \\ \sqrt{x}, & x > 4. \end{cases}$$

At $x = 2$, the two-sided limit exists and equals 1, but $f(2)$ is undefined. At $x = 4$, the left-hand limit is 1, the right-hand limit is 2, and $f(4) = 3$. Hence the discontinuities are $x = 2$ and $x = 4$.

$$\boxed{\lim_{x \to 0} f(x) = -1}, \qquad \boxed{\lim_{x \to 2} f(x) = 1}, \qquad \boxed{\lim_{x \to 4} f(x) = \text{DNE}}, \qquad \boxed{f(4) = 3}.$$



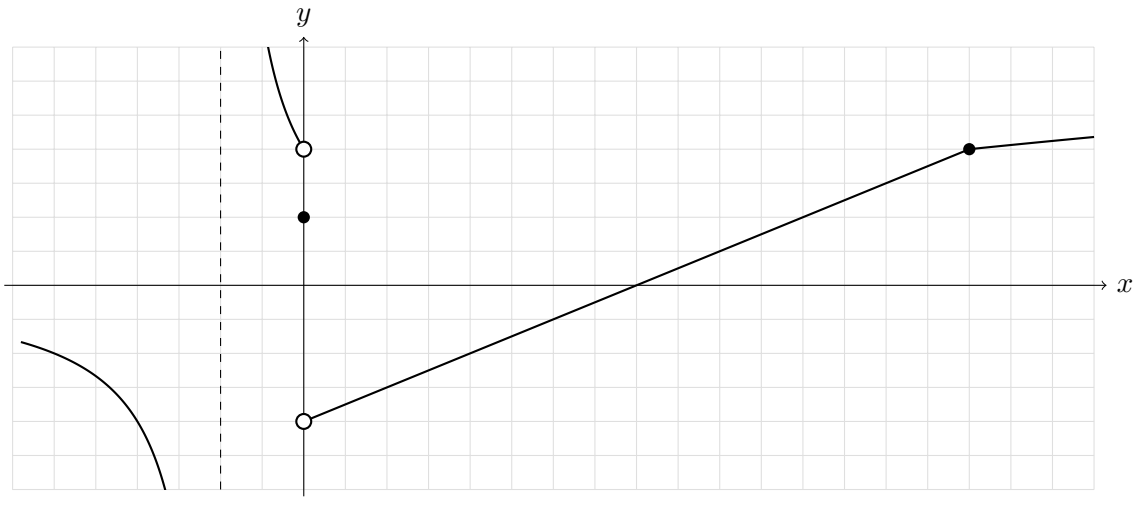
39.

$$h(x) = \begin{cases} \dfrac{8}{x+2}, & x < 0, \\ 2, & x = 0, \\ \dfrac{1}{2}x - 4, & 0 < x \le 16, \\ \sqrt{x}, & x > 16. \end{cases}$$

At $x = -2$ there is a vertical asymptote: the left-hand limit is $-\infty$ and the right-hand limit is $+\infty$. At $x = 0$, the left-hand limit is 4 and the right-hand limit is -4, while $h(0) = 2$. At $x = 16$, both one-sided limits equal 4 and $h(16) = 4$. Thus the discontinuities are $x = -2$ and $x = 0$.

$\boxed{\lim_{x\to -2} h(x) = \text{DNE}}$, $\boxed{\lim_{x\to 0} h(x) = \text{DNE}}$, $\boxed{\lim_{x\to 16} h(x) = 4}$, $\boxed{h(0) = 2}$, $\boxed{h(16) = 4}$.

Note: the review packet writes $f(0)$ and $f(16)$ in this problem; since the function here is named h, these should be $h(0)$ and $h(16)$.

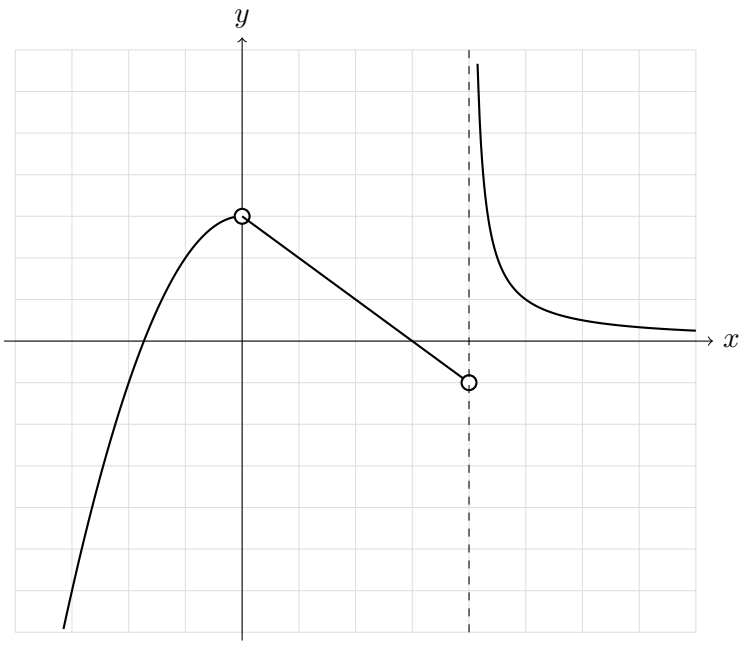


40.

$$F(x) = \begin{cases} \dfrac{1}{x-4}, & x > 4, \\ 3 - x, & 0 < x < 4, \\ 3 - x^2, & x < 0. \end{cases}$$

At $x = 0$, both one-sided limits equal 3, but $F(0)$ is undefined. At $x = 4$, the left-hand limit is -1, while the right-hand limit is $+\infty$, so the limit does not exist. Thus the discontinuities are $x = 0$ and $x = 4$.

$$\boxed{\lim_{x\to 0} F(x) = 3}, \qquad \boxed{\lim_{x\to 4} F(x) = \text{DNE}}, \qquad \boxed{F(-3) = -6}, \qquad \boxed{F(4) \text{ is undefined}}.$$



41. The line L is

$$2x + 4y = 12 \qquad \Longrightarrow \qquad \boxed{y = -\frac{1}{2}x + 3}.$$

Thus its slope is $-1/2$.

(a) Useful points for L are $(0, 3)$ and $(6, 0)$.

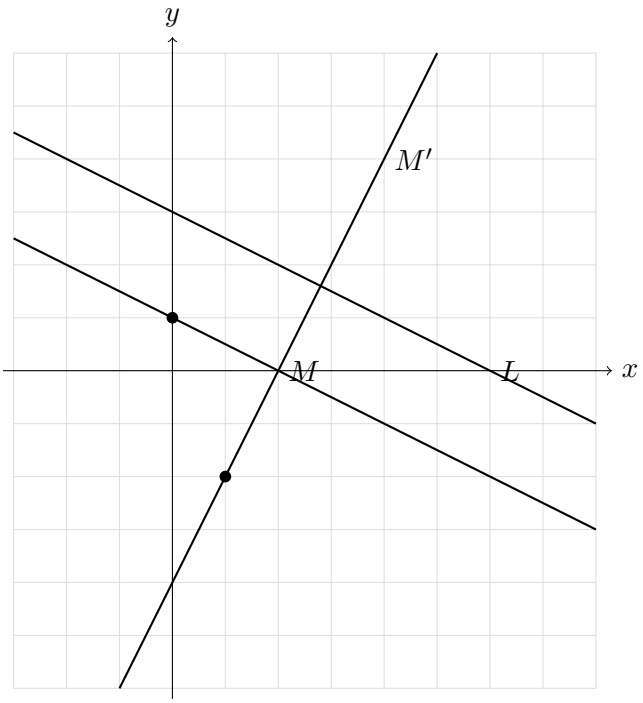
(b) A parallel line through $(0, 1)$ has the same slope:

$$\boxed{M : y = -\frac{1}{2}x + 1}.$$

(c) A perpendicular line has slope 2. Through $(1, -2)$,

$$y + 2 = 2(x - 1), \qquad \boxed{M' : y = 2x - 4}.$$

(d) The graph of M passes through $(0, 1)$ and $(2, 0)$.



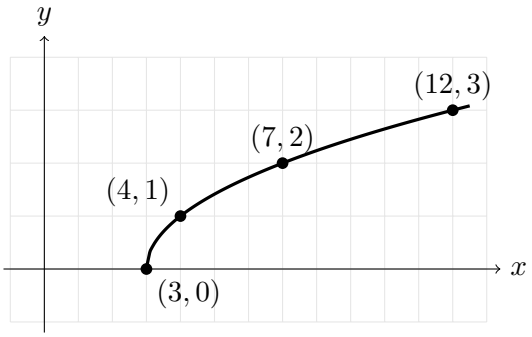
42. Sketch each graph.

(a) $f(x) = \sqrt{x-3}$ is the graph of $y = \sqrt{x}$ shifted right by 3. It begins at $(3, 0)$ and passes through $(4, 1)$, $(7, 2)$, and $(12, 3)$.

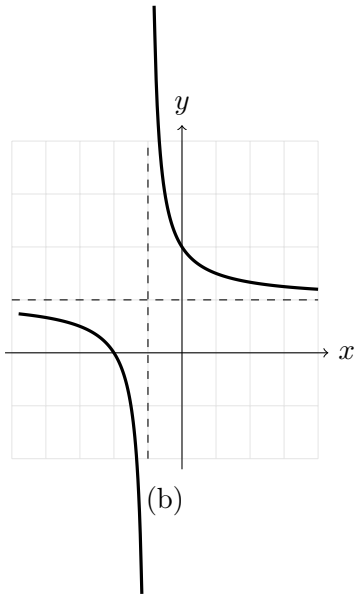
(b) $f(x) = 1 + \frac{1}{x+1}$ is the graph of $y = 1/x$ shifted left 1 and up 1. Its vertical asymptote is $x = -1$ and its horizontal asymptote is $y = 1$. For example, it passes through $(0, 2)$ and $(-2, 0)$.

(c)
$$x|x| = \begin{cases} -x^2, & x < 0, \\ x^2, & x \geq 0. \end{cases}$$
So the left half is a downward-opening parabola below the x-axis, and the right half is an upward-opening parabola. Useful points are $(-2, -4), (-1, -1), (0, 0), (1, 1), (2, 4)$.

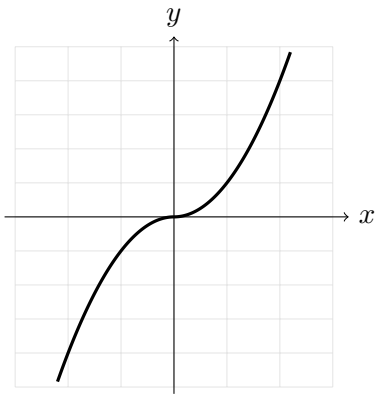
(d) For $x \neq 3$,
$$\frac{x^2 - 9}{|x-3|} = (x+3)\frac{x-3}{|x-3|} = \begin{cases} -x - 3, & x < 3, \\ x + 3, & x > 3. \end{cases}$$
There are open circles at $(3, -6)$ and $(3, 6)$.



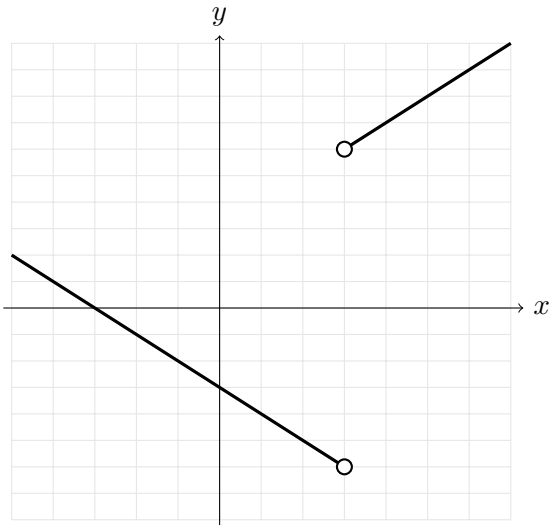
(a)



(b)



(c)



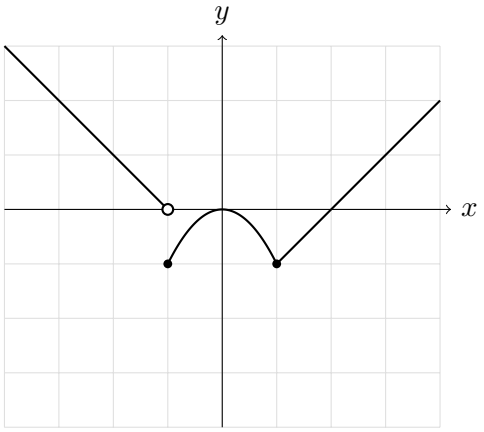
(d)

43. From the given graph,

$$f(x) = \begin{cases} x+1, & x < -1, \\ x^2, & -1 \le x < 1, \\ 2-x, & x \ge 1. \end{cases}$$

(a) Reflecting across the x-axis gives

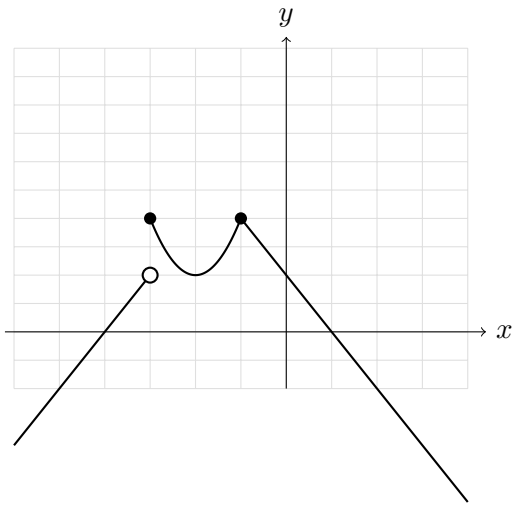
$$-f(x) = \begin{cases} -x-1, & x < -1, \\ -x^2, & -1 \le x < 1, \\ x-2, & x \ge 1. \end{cases}$$



(b) For $1+f(2x+2)$, the breakpoints occur when $2x+2=-1$ and $2x+2=1$, namely at $x=-3/2$ and $x=-1/2$. Thus

$$1+f(2x+2)=\begin{cases}2x+4, & x<-\frac{3}{2},\\ 1+(2x+2)^2, & -\frac{3}{2}\le x<-\frac{1}{2},\\ 1-2x, & x\ge -\frac{1}{2}.\end{cases}$$

A sketch is:



(c) The piecewise formula for the original graph is the formula displayed above.

44. If $f(x)=x^2+3x$, then

$$f(x+h)=(x+h)^2+3(x+h)=\boxed{x^2+2xh+h^2+3x+3h}.$$

Therefore

$$\frac{f(x+h)-f(x)}{h}=\frac{2xh+h^2+3h}{h}=\boxed{2x+h+3}.$$

45. Simplify.

(a)

$$\frac{1}{x-1}-\frac{1}{x}=\frac{x-(x-1)}{x(x-1)}=\boxed{\frac{1}{x(x-1)}}.$$

(b)

$$\frac{x}{x-1} - \frac{x}{x+1} = \frac{x[(x+1)-(x-1)]}{(x-1)(x+1)} = \boxed{\frac{2x}{x^2-1}}.$$

(c) First combine the numerator:

$$\frac{s+1}{s-1} + \frac{s-1}{s+1} = \frac{(s+1)^2 + (s-1)^2}{s^2-1} = \frac{2(s^2+1)}{s^2-1}.$$

Therefore

$$\frac{\frac{s+1}{s-1} + \frac{s-1}{s+1}}{\frac{1}{s^2-1}} = \boxed{2(s^2+1)}.$$

The original expression requires $s \neq \pm 1$.

Math 105: Midterm 1 Solution Guide

1. Consider the line L_1 given by $3x + 4y = 5$.

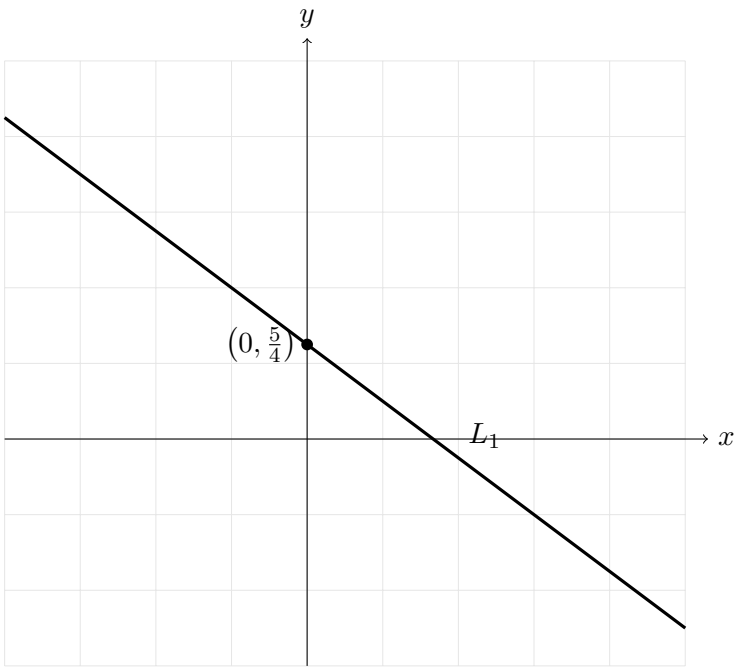
   (a) Solve for y:

   $$3x + 4y = 5 \qquad \Longrightarrow \qquad y = -\frac{3}{4}x + \frac{5}{4}.$$

   Thus L_1 has slope $-\frac{3}{4}$ and y-intercept

   $$\boxed{\left(0, \frac{5}{4}\right)}.$$

   A convenient second point is the x-intercept $\left(\frac{5}{3}, 0\right)$.



   (b) A line perpendicular to L_1 has slope equal to the opposite reciprocal of $-\frac{3}{4}$, namely $\frac{4}{3}$. Using the point $(-5, -1)$,

   $$y + 1 = \frac{4}{3}(x + 5).$$

   Solving for y gives

   $$\boxed{y = \frac{4}{3}x + \frac{17}{3}}.$$

   (c) A line parallel to L_1 must have the same slope. Therefore

   $$\boxed{m = -\frac{3}{4}}.$$

2. Let $f(x) = x^2 - x$ and $g(x) = \sqrt{x + 4}$.

(a) Substitute $2x^3$ for x in f:

$$f(2x^3) = (2x^3)^2 - 2x^3 = \boxed{4x^6 - 2x^3}.$$

(b) Substitute x^2 for x in g:

$$g(x^2) = \boxed{\sqrt{x^2+4}}.$$

(c)

$$(f \circ g)(x) = f(g(x)) = \left(\sqrt{x+4}\right)^2 - \sqrt{x+4}.$$

Therefore

$$\boxed{(f \circ g)(x) = x + 4 - \sqrt{x+4}}.$$

The expression is defined when $x + 4 \geq 0$, so its domain is $[-4, \infty)$.

3. Evaluate each limit.

(a)

$$\lim_{x \to 4} \frac{x^2 - x - 12}{x^2 - 3x - 4}.$$

Direct substitution gives $0/0$, so factor:

$$\frac{(x-4)(x+3)}{(x-4)(x+1)} = \frac{x+3}{x+1} \qquad (x \neq 4).$$

Thus

$$\boxed{\frac{7}{5}}.$$

(b)

$$\lim_{x \to -1} \frac{x^2 - 1}{2 - \sqrt{x+5}}.$$

Again direct substitution gives $0/0$. Rationalize the denominator:

$$\frac{x^2-1}{2-\sqrt{x+5}} \cdot \frac{2+\sqrt{x+5}}{2+\sqrt{x+5}} = \frac{(x-1)(x+1)(2+\sqrt{x+5})}{4-(x+5)}.$$

Since $4 - (x+5) = -(x+1)$, this simplifies to

$$-(x-1)(2+\sqrt{x+5}).$$

Now substitute $x = -1$:

$$-(-2)(2+2) = \boxed{8}.$$

(c)

$$\lim_{x \to 2} \frac{x^2 - 4}{4 - x}.$$

The denominator is nonzero at $x = 2$, so direct substitution works:

$$\frac{4-4}{4-2} = \boxed{0}.$$

(d)

$$\lim_{x\to 2} \frac{x+1}{(x-2)^2}.$$

The numerator approaches $3 > 0$, while $(x-2)^2 \to 0^+$ from both sides. Therefore

$$\boxed{+\infty}.$$

(e)

$$\lim_{x\to 1} \frac{x^2-3x+2}{|x-1|}.$$

Factor the numerator:

$$\frac{(x-1)(x-2)}{|x-1|}.$$

For $x > 1$, $|x-1| = x-1$, so

$$\lim_{x\to 1^+} \frac{x^2-3x+2}{|x-1|} = \lim_{x\to 1^+} (x-2) = -1.$$

For $x < 1$, $|x-1| = -(x-1)$, so

$$\lim_{x\to 1^-} \frac{x^2-3x+2}{|x-1|} = \lim_{x\to 1^-} -(x-2) = 1.$$

The one-sided limits are different, so

$$\boxed{\text{DNE}}.$$

4. Sketch $f(x) = 1 + \dfrac{2}{x-1}$.

The denominator is zero at $x = 1$, so the vertical asymptote is

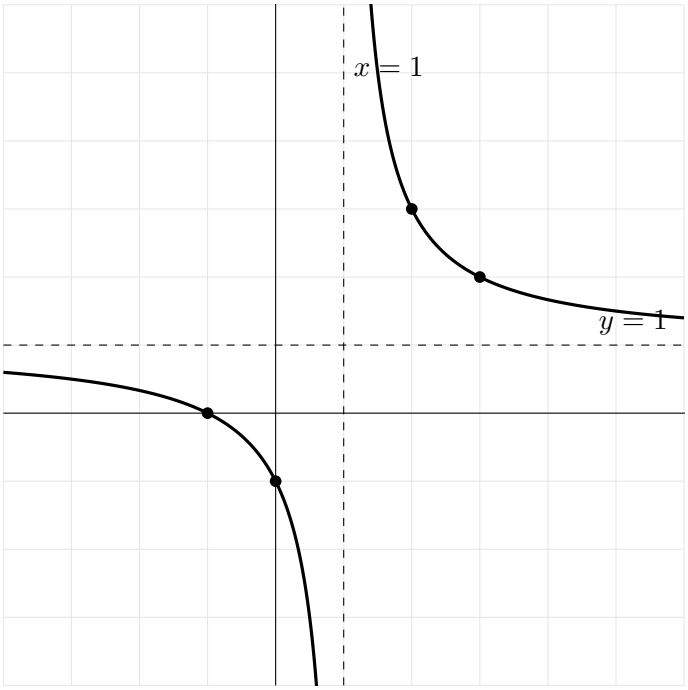
$$\boxed{x = 1}.$$

As x becomes very large in magnitude, $\frac{2}{x-1} \to 0$, so the horizontal asymptote is

$$\boxed{y = 1}.$$

Some convenient points are

$$f(-1) = 0, \qquad f(0) = -1, \qquad f(2) = 3, \qquad f(3) = 2.$$



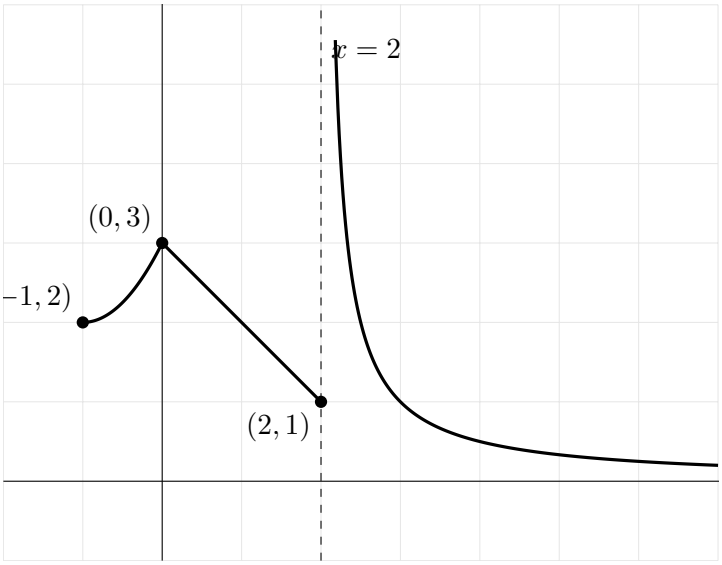
5. Let $f(x) = \dfrac{1}{x}$.

   (a)
   $$\frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h} - \frac{1}{x}}{h}.$$
   Combine the fractions in the numerator:
   $$\frac{1}{x+h} - \frac{1}{x} = \frac{x - (x+h)}{x(x+h)} = \frac{-h}{x(x+h)}.$$
   Therefore
   $$\frac{f(x+h) - f(x)}{h} = \boxed{-\frac{1}{x(x+h)}}.$$

   (b) Using the simplified expression from part (a),
   $$\lim_{h \to 0} -\frac{1}{x(x+h)} = \boxed{-\frac{1}{x^2}},$$
   for $x \neq 0$.

6. Consider
   $$f(x) = \begin{cases} \dfrac{1}{x-2}, & x > 2, \\ 3 - |x|, & 0 < x \leq 2, \\ (x+1)^2 + 2, & -1 \leq x \leq 0. \end{cases}$$

   (a) Since $0 < x \leq 2$ in the middle piece, $|x| = x$, so that piece is the line $y = 3 - x$. A sketch is shown below.



(b) Since the middle piece includes $x = 2$,

$$f(2) = 3 - |2| = \boxed{1}.$$

(c) The three pieces cover every input from -1 onward, so

$$\boxed{\text{Domain} = [-1, \infty)}.$$

For $x > 2$, the function $1/(x - 2)$ takes every positive value. Therefore

$$\boxed{\text{Range} = (0, \infty)}.$$

(d) At $x = 0$,

$$\lim_{x \to 0^-} f(x) = 3 \qquad \text{and} \qquad \lim_{x \to 0^+} f(x) = 3.$$

Hence

$$\boxed{\lim_{x \to 0} f(x) = 3}.$$

At $x = 2$,

$$\lim_{x \to 2^-} f(x) = 1, \qquad \lim_{x \to 2^+} f(x) = +\infty.$$

The one-sided limits do not agree, so

$$\boxed{\lim_{x \to 2} f(x) = \text{DNE}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 9

Due **Monday, October 5** at 11:55pm, on Gradescope

1. Consider the function defined by

$$f(x) = \begin{cases} \dfrac{1}{x-4} & \text{if } x > 4 \\ x^2 + 1 & \text{if } 0 < x < 4 \\ -3 & \text{if } x = 0 \\ x + 1 & \text{if } -2 < x < 0 \\ 3 - (x+2)^2 & \text{if } x \leq -2 \end{cases}$$

   (a) Carefully sketch the graph of $f(x)$.

   (b) State the **Domain** of the function $f(x)$.

   (c) Compute $\begin{cases} \lim\limits_{x \to -2^+} f(x) = \\ \lim\limits_{x \to -2^-} f(x) = \\ \lim\limits_{x \to -2} f(x) = \end{cases}$

   (d) Compute $\begin{cases} \lim\limits_{x \to 0^+} f(x) = \\ \lim\limits_{x \to 0^-} f(x) = \\ \lim\limits_{x \to 0} f(x) = \end{cases}$

(e) Compute $\begin{cases} \lim\limits_{x\to 4^+} f(x) = \\ \lim\limits_{x\to 4^-} f(x) = \\ \lim\limits_{x\to 4} f(x) = \end{cases}$

(f) State the value(s) at which f is **discontinuous**. Justify your answer(s) using the definition of continuity discussed in class.

2. Write an equation that expresses the fact that a function $f(x)$ is continuous at the number 4. (The equation should involve a limit.)

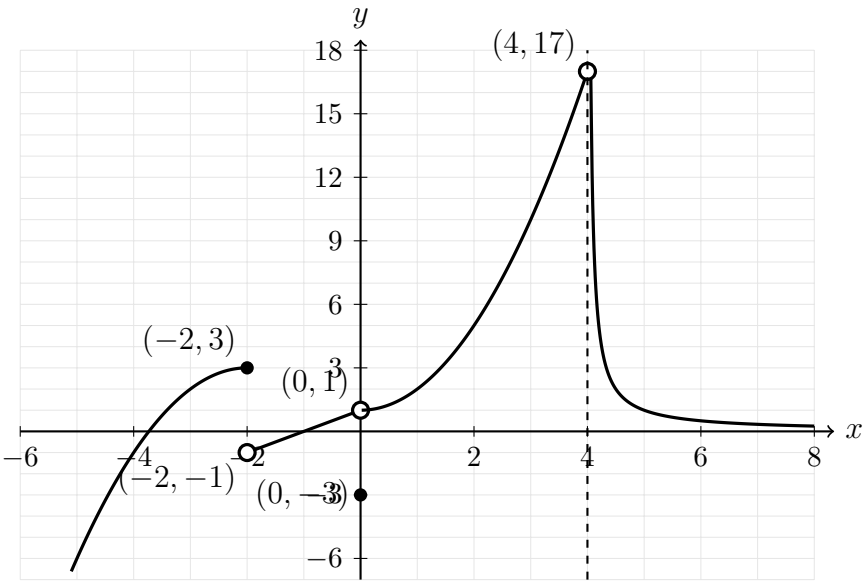
3. Sketch the graph of a function f that is continuous except for the stated discontinuity: discontinuous at 2, but continuous from the right there.

MATH 105 Calculus with Algebra – Fall 2026
Homework 9 Solutions

1. Consider

$$f(x) = \begin{cases} \dfrac{1}{x-4}, & x > 4, \\ x^2 + 1, & 0 < x < 4, \\ -3, & x = 0, \\ x + 1, & -2 < x < 0, \\ 3 - (x+2)^2, & x \le -2. \end{cases}$$

(a) A sketch of the graph is shown below. Pay particular attention to the open and closed endpoints. Note that the rightmost part of the graph has an asymptote at 4 and in particular does not "meet up" with the part to it's left. (So the limit at 4 does not exist.)



(b) The pieces cover every real number except $x = 4$. Therefore

$$\boxed{\text{Dom}(f) = (-\infty, 4) \cup (4, \infty)}.$$

(c) Near $x = -2$, the formula on the right is $x + 1$, while the formula on the left is $3 - (x+2)^2$. Hence

$$\lim_{x \to -2^+} f(x) = -1, \qquad \lim_{x \to -2^-} f(x) = 3.$$

Since the one-sided limits are different,

$$\boxed{\lim_{x \to -2} f(x) = \text{DNE}}.$$

(d) Near $x = 0$, the formula on the right is $x^2 + 1$, while the formula on the left is $x + 1$. Thus

$$\lim_{x \to 0^+} f(x) = 1, \qquad \lim_{x \to 0^-} f(x) = 1.$$

The one-sided limits agree, so

$$\boxed{\lim_{x\to 0} f(x) = 1}.$$

Notice, however, that $f(0) = -3$.

(e) As $x \to 4^+$, we use $f(x) = \frac{1}{x-4}$. The denominator is positive and approaches 0, so

$$\lim_{x\to 4^+} f(x) = +\infty.$$

From the left, $f(x) = x^2 + 1$, so

$$\lim_{x\to 4^-} f(x) = 17.$$

Since the one-sided limits do not agree,

$$\boxed{\lim_{x\to 4} f(x) = \text{DNE}}.$$

(f) The function is discontinuous at

$$\boxed{x = -2,\ 0,\ 4}.$$

To check this using the definition of continuity:
At $x = -2$, $f(-2) = 3$ is defined, but $\lim_{x\to -2} f(x)$ does not exist because the one-sided limits are different.
At $x = 0$, both $f(0)$ and $\lim_{x\to 0} f(x)$ exist, but

$$\lim_{x\to 0} f(x) = 1 \neq -3 = f(0).$$

At $x = 4$, $f(4)$ is not defined, and $\lim_{x\to 4} f(x)$ does not exist.

2. The equation expressing continuity of f at 4 is

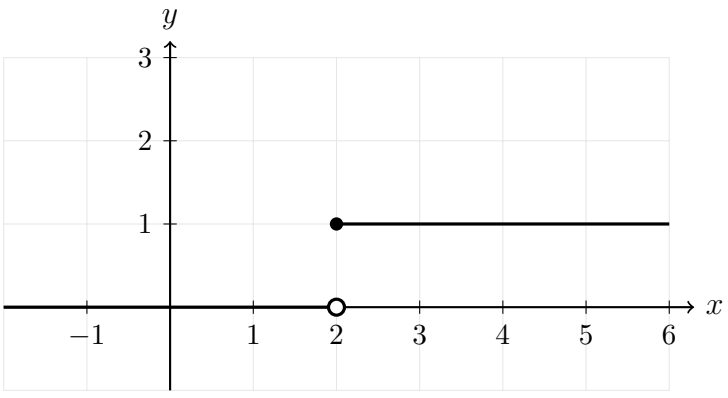
$$\boxed{\lim_{x\to 4} f(x) = f(4)}.$$

In particular, this equation requires both sides to exist.

3. One simple example is

$$f(x) = \begin{cases} 0, & x < 2, \\ 1, & x \geq 2. \end{cases}$$

Its graph is:



Here

$$\lim_{x \to 2^-} f(x) = 0, \qquad \lim_{x \to 2^+} f(x) = 1 = f(2).$$

Thus f is continuous from the right at 2, but the two-sided limit does not exist, so f is not continuous at 2.

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 10

Due **Friday, October 9** at 11:55pm, on Gradescope

1. Find an equation of the tangent line to the curve at the given point.

$$y = 4x - 3x^2, \qquad (2, -4).$$

2. Find an equation of the tangent line to the curve at the given point.

$$y = \frac{2x+1}{x+2}, \qquad (1, 1).$$

3. (a) Find the slope of the tangent to the curve

$$y = 3 + 4x^2 - 2x^3$$

at the point where $x = a$.

(b) Find equations of the tangent lines at the points $(1, 5)$ and $(2, 3)$.

4. If an equation of the tangent line to the curve $y = f(x)$ at the point where $x = 2$ is $y = 4x - 5$, find $f(2)$ and $f'(2)$.

MORE PROBLEMS ON THE NEXT PAGE →

5. For the function f graphed below:

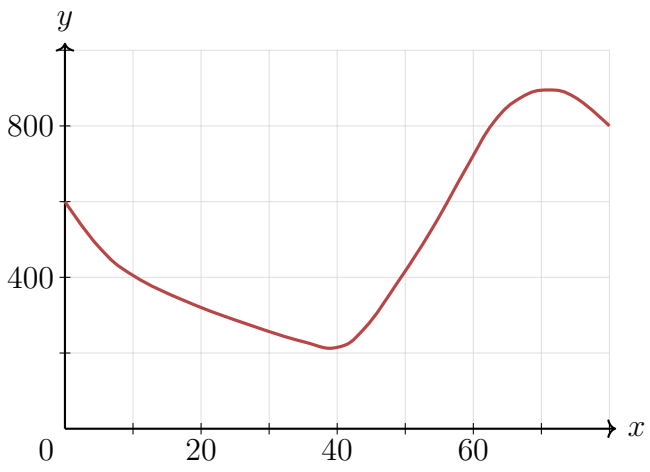
   (a) Estimate the value of $f'(50)$.

   (b) Is $f'(10) > f'(30)$?

   (c) Is

$$f'(60) > \frac{f(80) - f(40)}{80 - 40}?$$

   Explain.



MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 10 SOLUTIONS

1. We have

$$y = 4x - 3x^2, \qquad y' = 4 - 6x.$$

At $x = 2$, the slope is

$$y'(2) = 4 - 12 = -8.$$

Using the point $(2, -4)$,

$$y + 4 = -8(x - 2),$$

so the tangent line is

$$\boxed{y = -8x + 12}.$$

2. Let

$$y = \frac{2x + 1}{x + 2}.$$

By the quotient rule,

$$y' = \frac{2(x + 2) - (2x + 1)}{(x + 2)^2} = \frac{3}{(x + 2)^2}.$$

Thus

$$y'(1) = \frac{3}{9} = \frac{1}{3}.$$

Using the point $(1, 1)$,

$$y - 1 = \frac{1}{3}(x - 1),$$

so

$$\boxed{y = \frac{1}{3}x + \frac{2}{3}}.$$

3. (a) Since

$$y = 3 + 4x^2 - 2x^3,$$

we have

$$y' = 8x - 6x^2.$$

Therefore the slope at $x = a$ is

$$\boxed{8a - 6a^2}.$$

(b) At $(1, 5)$,

$$y'(1) = 8 - 6 = 2.$$

Hence

$$y - 5 = 2(x - 1),$$

so the tangent line is

$$\boxed{y = 2x + 3}.$$

At $(2, 3)$,

$$y'(2) = 16 - 24 = -8.$$

Hence

$$y - 3 = -8(x - 2),$$

so the tangent line is

$$\boxed{y = -8x + 19}.$$

4. The tangent line at $x = 2$ is

$$y = 4x - 5.$$

The point of tangency lies on this line, so

$$f(2) = 4(2) - 5 = \boxed{3}.$$

The derivative is the slope of the tangent line, so

$$\boxed{f'(2) = 4}.$$

5. From the graph:

(a) Near $x = 50$, the graph is increasing fairly steeply. Using nearby points from the graph, for example approximately $(49, 390)$ and $(54, 530)$,

$$f'(50) \approx \frac{530 - 390}{54 - 49} = 28.$$

Thus a reasonable estimate is

$$\boxed{f'(50) \approx 25\text{–}30}.$$

(b) No. At $x = 10$ the function is decreasing more steeply than it is at $x = 30$. Thus $f'(10)$ is more negative, so

$$\boxed{f'(10) < f'(30)}.$$

(c) Yes. The secant slope from $x = 40$ to $x = 80$ is approximately

$$\frac{f(80) - f(40)}{80 - 40} \approx \frac{800 - 215}{40} \approx 14.6.$$

At $x = 60$, the graph is rising much more steeply than this secant line, so

$$\boxed{f'(60) > \frac{f(80) - f(40)}{80 - 40}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 11

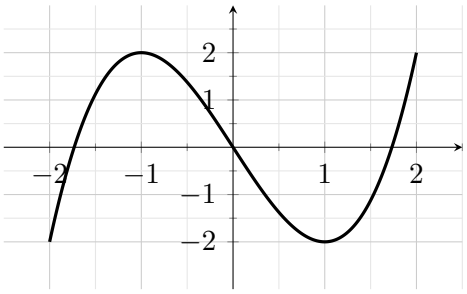
Due **Wednesday, October 14** at 11:55pm, on Gradescope

Recall that the definition of the derivative is

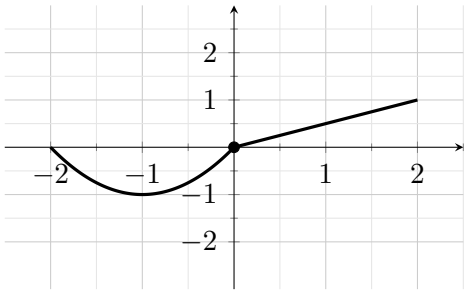
$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

1. The graphs of two functions f and g are shown below. Sketch the graphs of f' and g' on the corresponding empty coordinate planes. If a derivative does not exist at some x-value, indicate this clearly on your graph.

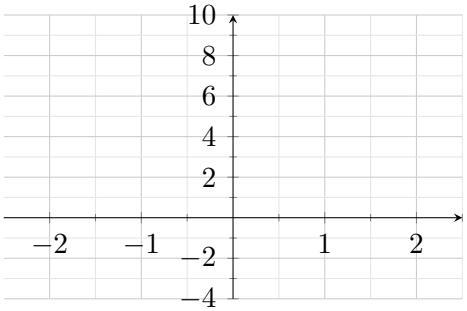
Graph of $y = f(x)$



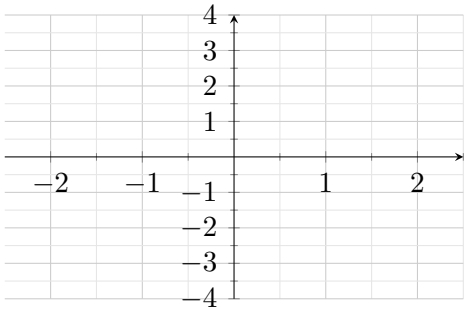
Graph of $y = g(x)$



Sketch $y = f'(x)$



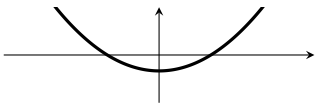
Sketch $y = g'(x)$



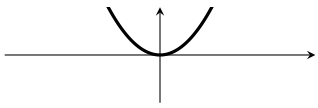
2. Below, graphs **A**–**F** are graphs of functions, and graphs **1**–**6** are graphs of their derivatives. Match each function with its derivative.

$A \to$ _____ $B \to$ _____ $C \to$ _____ $D \to$ _____ $E \to$ _____ $F \to$ _____

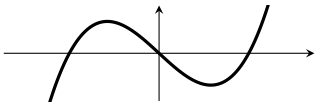
Function A



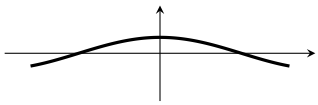
Derivative 1



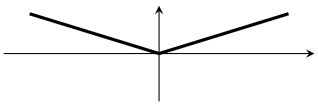
Function B



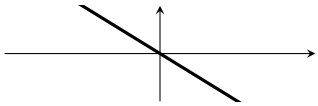
Derivative 2



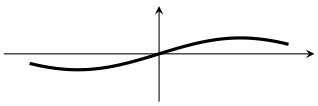
Function C



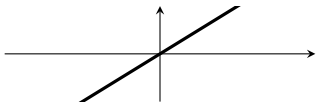
Derivative 3



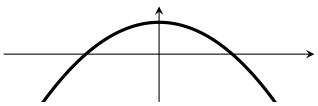
Function D



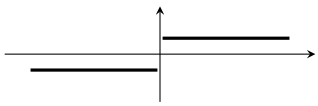
Derivative 4



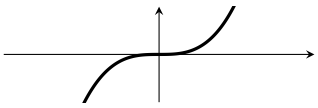
Function E



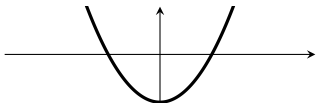
Derivative 5



Function F



Derivative 6



Recall the "difference of squares" formula $a^2 - b^2 = (a - b)(a + b)$.

3. Let $f(x) = \sqrt{1 + x^2}$.

   (a) What is the domain of $f(x)$?

   (b) Simplify the expression $(x + h)^2 - x^2$ using the difference of squares formula.

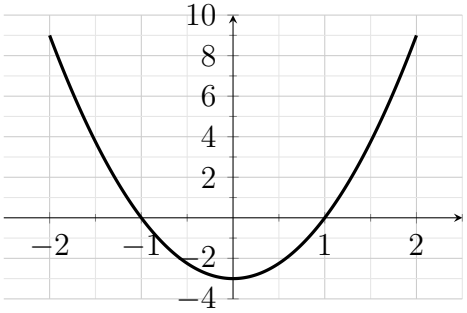
   (c) Using the limit definition of the derivative, compute $f'(x)$. (Be careful: it isn't true that $\sqrt{1 + x^2} = \sqrt{1} + \sqrt{x^2}$.)

MATH 105 Calculus with Algebra – Homework 11 Solutions

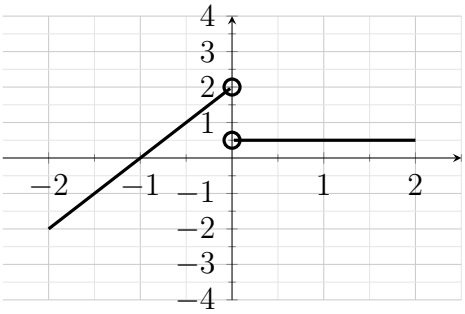
1. For g, at $x = 0$, the left-hand slope is roughly 2 and the right-hand slope is $\frac{1}{2}$, so

$$\boxed{g'(0) \text{ does not exist.}}$$

Graph of $y = f'(x)$



Graph of $y = g'(x)$



2. The matches are

$$\boxed{A \to 4, \quad B \to 6, \quad C \to 5, \quad D \to 2, \quad E \to 3, \quad F \to 1.}$$

A is an upward-opening parabola, so its derivative is an increasing line: graph 4. B is a cubic with a local maximum and minimum, so its derivative is an upward-opening parabola crossing the x-axis twice: graph 6. C is $|x|$, whose slope is -1 for $x < 0$ and 1 for $x > 0$, with no derivative at $x = 0$: graph 5. E is a downward-opening parabola, so its derivative is a decreasing line: graph 3.

3. Let

$$f(x) = \sqrt{1 + x^2}.$$

(a) Since $1 + x^2 > 0$ for every real x,

$$\boxed{\text{Domain of } f = (-\infty, \infty).}$$

(b)

$$(x + h)^2 - x^2 = ((x + h) - x)((x + h) + x) = h(2x + h),$$

so

$$\boxed{(x + h)^2 - x^2 = h(2x + h).}$$

(c) By the limit definition,

$$f'(x) = \lim_{h \to 0} \frac{\sqrt{1 + (x + h)^2} - \sqrt{1 + x^2}}{h}.$$

Multiply by the conjugate:

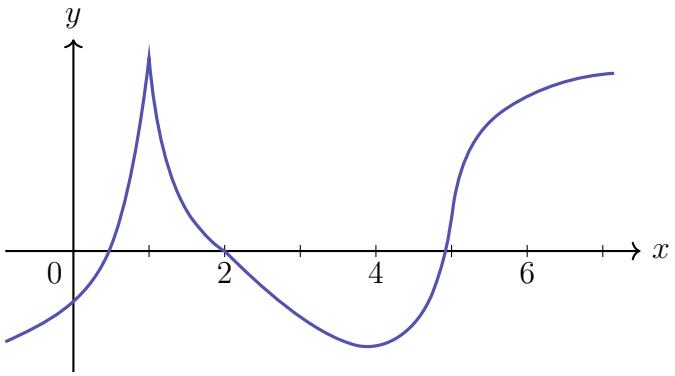
$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{1+(x+h)^2-(1+x^2)}{h\left(\sqrt{1+(x+h)^2}+\sqrt{1+x^2}\right)} \\ &= \lim_{h\to 0} \frac{(x+h)^2-x^2}{h\left(\sqrt{1+(x+h)^2}+\sqrt{1+x^2}\right)} \\ &= \lim_{h\to 0} \frac{2x+h}{\sqrt{1+(x+h)^2}+\sqrt{1+x^2}} \\ &= \boxed{\frac{x}{\sqrt{1+x^2}}}. \end{aligned}$$

MATH 105 Calculus with Algebra Fall 2026

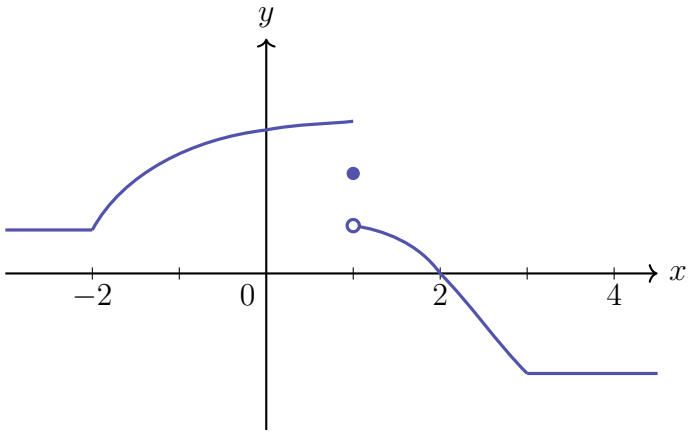
HOMEWORK 12

Due **Friday, October 16** at 11:55pm, on Gradescope

1. The graph of f is given below. State, with reasons, the numbers at which f is not differentiable.



2. The graph of f is given below. State, with reasons, the numbers at which f is not differentiable.



3. Differentiate the following functions. (Do not use the limit definition; instead use our differentiation rules.)

   (a) $g(x) = x^2 - 3x + 12$.

   (b) $f(t) = 1.4t^5 - 2.5t^2 + 6.7$.

   (c) $x^{5/3} - x^{2/3}$.

   (d) $g(x) = (1 + 2x + x^2)(1 - 2x^3)$.

   (e) $u(t) = \dfrac{1}{t} - \dfrac{1}{\sqrt{t}}$.

MATH 105 Calculus with Algebra – Homework 12 Solutions

1. The function is not differentiable at

$$\boxed{x = 1}.$$

At $x = 1$, the graph has a sharp point, so there is no single tangent slope there.

2. The function is not differentiable at

$$\boxed{x = -2, \quad x = 1, \quad x = 3}.$$

At $x = -2$, the horizontal piece meets the increasing curved piece at a corner: the slopes from the two sides do not agree.

At $x = 1$, the function is discontinuous. A differentiable function must be continuous, so f cannot be differentiable at $x = 1$.

At $x = 3$, the decreasing curved piece meets a horizontal piece at a corner, so the left- and right-hand slopes do not agree.

3. Differentiate the following functions.

(a)

$$g(x) = x^2 - 3x + 12.$$

Using the power rule,

$$\boxed{g'(x) = 2x - 3}.$$

(b)

$$f(t) = 1.4t^5 - 2.5t^2 + 6.7.$$

Therefore

$$f'(t) = 5(1.4)t^4 - 2(2.5)t,$$

so

$$\boxed{f'(t) = 7t^4 - 5t}.$$

(c)

$$f(x) = x^{5/3} - x^{2/3}.$$

Using the power rule,

$$\boxed{f'(x) = \frac{5}{3}x^{2/3} - \frac{2}{3}x^{-1/3}}.$$

(d)

$$g(x) = (1 + 2x + x^2)(1 - 2x^3).$$

Using the product rule,

$$g'(x) = (2 + 2x)(1 - 2x^3) + (1 + 2x + x^2)(-6x^2).$$

Thus

$$\boxed{g'(x) = (2 + 2x)(1 - 2x^3) - 6x^2(1 + 2x + x^2)}.$$

Equivalently, after multiplying everything out,

$$\boxed{g'(x) = 2 + 2x - 6x^2 - 16x^3 - 10x^4}.$$

(e) Rewrite using exponents:

$$u(t) = t^{-1} - t^{-1/2}.$$

Then

$$u'(t) = -t^{-2} + \frac{1}{2}t^{-3/2}.$$

Therefore

$$\boxed{u'(t) = -\frac{1}{t^2} + \frac{1}{2t^{3/2}}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 13

Due **Monday, October 19** at 11:55pm, on Gradescope

1. Let $f(x) = x^2 + |x|$. Compute $f'(x)$, and compute the equation of the tangent line for $x = 1$.

   Differentiate the following functions. (Do not use the limit definition; instead use our differentiation rules.)

2. $f(x) = (3x - 1)(x + 2)$.
3. $f(x) = 2^{40}$.
4. $y = \sqrt[3]{x}(2 + x)$.
5. $F(r) = \dfrac{5}{r^3}$.
6. $y = \dfrac{x^2 + 4x + 3}{\sqrt{x}}$.
7. $F(x) = \dfrac{x^4 - 5x^3 + \sqrt{x}}{x^2}$
8. $J(v) = (v^3 - 2v)(v^{-4} + v^{-2})$.
9. $g(x) = \dfrac{1 + 2x}{3 - 4x}$.
10. $y = \dfrac{1}{t^3 + 2t^2 - 1}$.
11. $y = \dfrac{t^3 + 3t}{t^2 - 4t + 3}$.
12. $y = \dfrac{s - \sqrt{s}}{s^2}$.

MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 13 SOLUTIONS

1. Let

$$f(x) = x^2 + |x|.$$

Since

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases}$$

we can write

$$f(x) = \begin{cases} x^2 + x, & x \geq 0, \\ x^2 - x, & x < 0. \end{cases}$$

Therefore, for $x \neq 0$,

$$f'(x) = \begin{cases} 2x + 1, & x > 0, \\ 2x - 1, & x < 0. \end{cases}$$

At $x = 0$, the slope from the left is -1 and the slope from the right is 1, so $f'(0)$ does not exist. Thus

$$\boxed{f'(x) = \begin{cases} 2x + 1, & x > 0, \\ 2x - 1, & x < 0, \end{cases} \qquad f'(0) \text{ does not exist.}}$$

At $x = 1$, we have $f(1) = 2$ and $f'(1) = 3$. The tangent line is

$$y - 2 = 3(x - 1),$$

so

$$\boxed{y = 3x - 1}.$$

2.

$$f(x) = (3x - 1)(x + 2).$$

Using the product rule,

$$\begin{aligned} f'(x) &= 3(x + 2) + (3x - 1)(1) \\ &= 6x + 5. \end{aligned}$$

Hence

$$\boxed{f'(x) = 6x + 5}.$$

3.

$$f(x) = 2^{40}.$$

This is a constant, so

$$\boxed{f'(x) = 0}.$$

4.

$$y = \sqrt[3]{x}(2+x) = x^{1/3}(2+x).$$

It is easiest to multiply first:

$$y = 2x^{1/3} + x^{4/3}.$$

Therefore

$$\boxed{y' = \frac{2}{3}x^{-2/3} + \frac{4}{3}x^{1/3}}.$$

This formula is valid for $x \neq 0$; the derivative does not exist at $x = 0$.

5.

$$F(r) = \frac{5}{r^3} = 5r^{-3}.$$

Using the power rule,

$$\boxed{F'(r) = -15r^{-4} = -\frac{15}{r^4}}.$$

6.

$$y = \frac{x^2 + 4x + 3}{\sqrt{x}}.$$

Rewrite each term using powers:

$$y = x^{3/2} + 4x^{1/2} + 3x^{-1/2}.$$

Therefore

$$\boxed{y' = \frac{3}{2}x^{1/2} + 2x^{-1/2} - \frac{3}{2}x^{-3/2}}.$$

Equivalently,

$$\boxed{y' = \frac{3}{2}\sqrt{x} + \frac{2}{\sqrt{x}} - \frac{3}{2x^{3/2}}}.$$

7.

$$F(x) = \frac{x^4 - 5x^3 + \sqrt{x}}{x^2}.$$

Simplify first:

$$F(x) = x^2 - 5x + x^{-3/2}.$$

Thus

$$\boxed{F'(x) = 2x - 5 - \frac{3}{2}x^{-5/2}}.$$

Equivalently,

$$\boxed{F'(x) = 2x - 5 - \frac{3}{2x^{5/2}}}.$$

8.

$$J(v) = (v^3 - 2v)(v^{-4} + v^{-2}).$$

Multiplying first gives

$$J(v) = v^{-1} + v - 2v^{-3} - 2v^{-1} = v - v^{-1} - 2v^{-3}.$$

Therefore

$$\boxed{J'(v) = 1 + v^{-2} + 6v^{-4}}.$$

Equivalently,

$$\boxed{J'(v) = 1 + \frac{1}{v^2} + \frac{6}{v^4}}.$$

9.

$$g(x) = \frac{1 + 2x}{3 - 4x}.$$

Using the quotient rule,

$$\begin{aligned} g'(x) &= \frac{2(3 - 4x) - (1 + 2x)(-4)}{(3 - 4x)^2} \\ &= \frac{6 - 8x + 4 + 8x}{(3 - 4x)^2}. \end{aligned}$$

Hence

$$\boxed{g'(x) = \frac{10}{(3 - 4x)^2}}.$$

10.

$$y = \frac{1}{t^3 + 2t^2 - 1} = (t^3 + 2t^2 - 1)^{-1}.$$

Using the chain rule,

$$\boxed{y' = -\frac{3t^2 + 4t}{(t^3 + 2t^2 - 1)^2}}.$$

11.

$$y = \frac{t^3 + 3t}{t^2 - 4t + 3}.$$

Using the quotient rule,

$$y' = \frac{(3t^2 + 3)(t^2 - 4t + 3) - (t^3 + 3t)(2t - 4)}{(t^2 - 4t + 3)^2}.$$

The numerator simplifies to

$$t^4 - 8t^3 + 6t^2 + 9,$$

so

$$\boxed{y' = \frac{t^4 - 8t^3 + 6t^2 + 9}{(t^2 - 4t + 3)^2}}.$$

12.

$$y = \frac{s - \sqrt{s}}{s^2}.$$

Simplify first:

$$y = s^{-1} - s^{-3/2}.$$

Therefore

$$\boxed{y' = -s^{-2} + \frac{3}{2}s^{-5/2}}.$$

Equivalently,

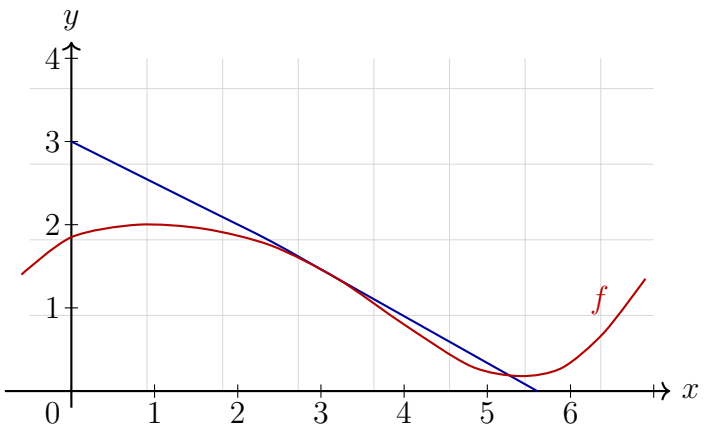
$$\boxed{y' = -\frac{1}{s^2} + \frac{3}{2s^{5/2}}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 14

Due **Friday, October 23** at 11:55pm, on Gradescope

1. Find the derivative of the function $F(x) = (1 + x + x^2)^{99}$.
2. Find the derivative of the function $g(x) = (x^2 + 1)^3(x^2 + 2)^6$.
3. Find the derivative of the function $g(u) = \left(\dfrac{u^3 - 1}{u^3 + 1}\right)^8$.
4. Find the derivative of the function $f(t) = \sqrt{\dfrac{t}{t^2 + 4}}$.
5. Find the derivative of the function $y = \sqrt{x + \sqrt{x}}$.
6. Find the derivative of the function $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$.
7. Find an equation of the tangent line to the curve at the given point $y = \sqrt{1 + x^3}, \qquad (2, 3)$.
8. If $F(x) = f(g(x))$, where $f(-2) = 8, \quad f'(-2) = 4, \quad f'(5) = 3, \quad g(5) = -2, \quad g'(5) = 6$, find $F'(5)$.
9. If $h(x) = \sqrt{4 + 3f(x)}$, where $f(1) = 7$ and $f'(1) = 4$, find $h'(1)$.
10. If $g(x) = \sqrt{f(x)}$, where the graph of f is shown below, evaluate $g'(3)$.



MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 14 SOLUTIONS

1.

$$F(x) = (1 + x + x^2)^{99}.$$

Using the chain rule,

$$F'(x) = 99(1 + x + x^2)^{98}(1 + 2x).$$

Thus

$$\boxed{F'(x) = 99(1 + 2x)(1 + x + x^2)^{98}}.$$

2.

$$g(x) = (x^2 + 1)^3(x^2 + 2)^6.$$

Using the product rule and the chain rule,

$$g'(x) = 6x(x^2 + 1)^2(x^2 + 2)^6 + 12x(x^2 + 1)^3(x^2 + 2)^5.$$

Factoring gives

$$\boxed{g'(x) = 6x(x^2 + 1)^2(x^2 + 2)^5(3x^2 + 4)}.$$

3.

$$g(u) = \left(\frac{u^3 - 1}{u^3 + 1}\right)^8.$$

By the chain rule,

$$g'(u) = 8\left(\frac{u^3 - 1}{u^3 + 1}\right)^7 \frac{d}{du}\left(\frac{u^3 - 1}{u^3 + 1}\right).$$

The quotient rule gives

$$\frac{d}{du}\left(\frac{u^3 - 1}{u^3 + 1}\right) = \frac{3u^2(u^3 + 1) - 3u^2(u^3 - 1)}{(u^3 + 1)^2} = \frac{6u^2}{(u^3 + 1)^2}.$$

Therefore

$$\boxed{g'(u) = \frac{48u^2(u^3 - 1)^7}{(u^3 + 1)^9}}.$$

4.

$$f(t) = \sqrt{\frac{t}{t^2 + 4}} = \left(\frac{t}{t^2 + 4}\right)^{1/2}.$$

First,

$$\frac{d}{dt}\left(\frac{t}{t^2 + 4}\right) = \frac{(t^2 + 4) - 2t^2}{(t^2 + 4)^2} = \frac{4 - t^2}{(t^2 + 4)^2}.$$

Thus

$$f'(t) = \frac{1}{2}\left(\frac{t}{t^2+4}\right)^{-1/2} \frac{4-t^2}{(t^2+4)^2},$$

so, for $t > 0$,

$$\boxed{f'(t) = \frac{4-t^2}{2\sqrt{t}(t^2+4)^{3/2}}}.$$

5.

$$y = \sqrt{x+\sqrt{x}}.$$

Using the chain rule twice,

$$y' = \frac{1}{2\sqrt{x+\sqrt{x}}}\left(1+\frac{1}{2\sqrt{x}}\right).$$

Therefore

$$\boxed{y' = \frac{2\sqrt{x}+1}{4\sqrt{x}\sqrt{x+\sqrt{x}}}} \qquad (x > 0).$$

6.

$$y = \sqrt{x+\sqrt{x+\sqrt{x}}}.$$

Work from the outside inward. We have

$$y' = \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}\left(1+\frac{1}{2\sqrt{x+\sqrt{x}}}\left(1+\frac{1}{2\sqrt{x}}\right)\right).$$

Thus

$$\boxed{y' = \frac{1}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}\left(1+\frac{1}{2\sqrt{x+\sqrt{x}}}\left(1+\frac{1}{2\sqrt{x}}\right)\right)} \qquad (x > 0).$$

7.

$$y = \sqrt{1+x^3}.$$

Differentiate:

$$y' = \frac{3x^2}{2\sqrt{1+x^3}}.$$

At $x = 2$,

$$y'(2) = \frac{3(2)^2}{2\sqrt{1+2^3}} = \frac{12}{6} = 2.$$

The tangent line passes through $(2, 3)$, so

$$y - 3 = 2(x-2).$$

Hence

$$\boxed{y = 2x - 1}.$$

8. If

$$F(x) = f(g(x)),$$

then the chain rule gives

$$F'(x) = f'(g(x))g'(x).$$

Since $g(5) = -2$, $f'(-2) = 4$, and $g'(5) = 6$,

$$F'(5) = f'(g(5))g'(5) = f'(-2)(6) = 4 \cdot 6.$$

Therefore

$$\boxed{F'(5) = 24}.$$

9.

$$h(x) = \sqrt{4 + 3f(x)}.$$

Using the chain rule,

$$h'(x) = \frac{3f'(x)}{2\sqrt{4 + 3f(x)}}.$$

At $x = 1$, we have $f(1) = 7$ and $f'(1) = 4$, so

$$h'(1) = \frac{3(4)}{2\sqrt{4 + 3(7)}} = \frac{12}{2\sqrt{25}} = \frac{12}{10}.$$

Thus

$$\boxed{h'(1) = \frac{6}{5}}.$$

10.

$$g(x) = \sqrt{f(x)}.$$

By the chain rule,

$$g'(x) = \frac{f'(x)}{2\sqrt{f(x)}}.$$

From the graph, at $x = 3$ we read

$$f(3) = \frac{3}{2}, \qquad f'(3) = -\frac{1}{2},$$

where $f'(3)$ is the slope of the tangent line shown on the graph. Therefore

$$g'(3) = \frac{-1/2}{2\sqrt{3/2}} = -\frac{1}{4\sqrt{3/2}}.$$

Equivalently,

$$\boxed{g'(3) = -\frac{\sqrt{6}}{12}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 15

Due **Monday, October 26** at 11:55pm, on Gradescope

In problems 1-6, find dy/dx by implicit differentiation.

1. $\sqrt{x} + \sqrt{y} = 1$.
2. $x^2 - 4xy + y^2 = 4$.
3. $x^3 - xy^2 + y^3 = 1$.
4. $\frac{x^2}{x+y} = y^2 + 1$.
5. $y^5 + x^2y^3 = 1 + x^4y$.
6. $\sqrt{xy} = 1 + x^2y$.
7. Find an equation of the tangent line to the curve at the given point.

$$x^2 - xy - y^2 = 1, \qquad (2, 1)$$

MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 15 SOLUTIONS

In Problems 1–6, find dy/dx by implicit differentiation.

1.

$$\sqrt{x} + \sqrt{y} = 1.$$

Differentiate both sides:

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\, y' = 0.$$

Therefore

$$\boxed{\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}}.$$

2.

$$x^2 - 4xy + y^2 = 4.$$

Differentiating gives

$$2x - 4\big(y + xy'\big) + 2yy' = 0.$$

Collecting the y' terms,

$$(-4x + 2y)y' = -2x + 4y.$$

Hence

$$\boxed{\frac{dy}{dx} = \frac{x - 2y}{2x - y}}.$$

3.

$$x^3 - xy^2 + y^3 = 1.$$

Differentiating,

$$3x^2 - \big(y^2 + 2xyy'\big) + 3y^2y' = 0.$$

Thus

$$(-2xy + 3y^2)y' = -3x^2 + y^2,$$

so

$$\boxed{\frac{dy}{dx} = \frac{3x^2 - y^2}{2xy - 3y^2}}.$$

4.

$$\frac{x^2}{x + y} = y^2 + 1.$$

It is convenient to clear the denominator first:

$$x^2 = (y^2+1)(x+y).$$

Differentiate:

$$2x = 2yy'(x+y) + (y^2+1)(1+y').$$

Collecting the y' terms gives

$$2x - (y^2+1) = y'\big(2y(x+y) + y^2 + 1\big).$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{2x - y^2 - 1}{2y(x+y) + y^2 + 1}}.$$

5.

$$y^5 + x^2y^3 = 1 + x^4y.$$

Differentiating,

$$5y^4y' + 2xy^3 + 3x^2y^2y' = 4x^3y + x^4y'.$$

Collecting terms containing y',

$$\big(5y^4 + 3x^2y^2 - x^4\big)y' = 4x^3y - 2xy^3.$$

Hence

$$\boxed{\frac{dy}{dx} = \frac{4x^3y - 2xy^3}{5y^4 + 3x^2y^2 - x^4}}.$$

6.

$$\sqrt{xy} = 1 + x^2y.$$

Write the left side as $(xy)^{1/2}$. Then

$$\frac{1}{2\sqrt{xy}}\big(y + xy'\big) = 2xy + x^2y'.$$

Multiply by $2\sqrt{xy}$:

$$y + xy' = 4xy\sqrt{xy} + 2x^2\sqrt{xy}\,y'.$$

Thus

$$\big(x - 2x^2\sqrt{xy}\big)y' = 4xy\sqrt{xy} - y,$$

so

$$\boxed{\frac{dy}{dx} = \frac{4xy\sqrt{xy} - y}{x - 2x^2\sqrt{xy}}}.$$

7. Find an equation of the tangent line to

$$x^2 - xy - y^2 = 1$$

at $(2, 1)$.

Differentiate implicitly:

$$2x - \big(y + xy'\big) - 2yy' = 0.$$

Therefore

$$(x + 2y)y' = 2x - y,$$

so

$$\frac{dy}{dx} = \frac{2x - y}{x + 2y}.$$

At $(2, 1)$,

$$y' = \frac{4 - 1}{2 + 2} = \frac{3}{4}.$$

Using point-slope form,

$$y - 1 = \frac{3}{4}(x - 2).$$

Thus the tangent line is

$$\boxed{y = \frac{3}{4}x - \frac{1}{2}}.$$

Instructor: David Zureick-Brown
Fall 2026

Amherst College
Dept. of Mathematics

Math 105 ∼ Calculus with Algebra: Review Packet

- These are practice problems, and are not to be turned in.
- On Wednesday in class, I will work any problems you ask about, and answer any questions.
- On Thursday, there will be a practice exam during the worksheet session.
- *Good luck!*

What you need to know for Exam 2

You should know the first parts of Chapter 2; specifically Sections 2.1–2.6, excluding 2.4 on Trig. functions). The test will not explicitly cover material from Chapter 1, but of course it will be assumed that you understand that material. (For example, you still need to know how to work with limits.) The following is a list of most of the topics covered. **THIS IS NOT A COMPREHENSIVE LIST, BUT MERELY AN AID.** Remember, no calculators in any exams.

- **2.1: Limit definition of derivatives.** (We've stuck to box 4 on page 107; just don't mix-and-match parts of one with the other.) Computing derivatives with the limit definition. Derivatives as slopes of tangent lines or instantaneous velocities or rates of change. Know how to compute derivatives using the **limit definition of the derivative**. What is it?
- **2.2: The derivative as a function.** Differentiability. How can differentiability fail? Differentiable implies continuous, but not necessarily the other way around. The graph of f' in relation to the graph of f. Higher derivatives.
- **2.3: Differentiation Rules:** Sum, Difference, Constant Multiple, Product, Quotient, Power. Memorize them and Know how to use them.
- **2.5: Chain rule.** Know it well!! Be able to compute derivatives involving both chain rule and quotient/product rules, paying attention to the order. Place parentheses carefully!
- **2.6: Implicit differentiation.** To find the tangent line to a curve described by some equation involving x and y, we think of y as a function of x, without knowing the actual formula for $y = f(x)$ (hence, an implicit, rather than explicit, function). So if we differentiate with respect to x, we must remember that y is a function of x and use the chain rule (or product rule, or quotient rule) when appropriate.

Tips

- Know how to compute derivatives both ways, using the limit definition of the derivative and the differentiation rules.
- Know how to compute the equation of a tangent line. you need two things: a point and the specific slope at that point. You can use point-slope form or slope-intercept form.
- Read the directions twice. Follow carefully if the question asks you to simplify your answer.
- $\frac{d}{dx}$ is an instruction to "take the derivative", whereas $\frac{dy}{dx}$ is the derivative of y with respect to x.
- Know your differentiation rules quickly! Efficiency is key here.
- If any of the problems in Sections 2.3, or 2.5 gave you trouble, make sure you have them cleared up; those problems will be the *easier* ones on the exam.
- Implicit differentiation can also be tricky. Besides the fact that it comes up in future related rates problems, you also need to be able to recognize when you need to use it. Anytime you run into a y chunk in an implicitly defined equation, you need to use the Chain Rule on that term.
- Most importantly, be clear and neat on your answer work. People usually score higher when the work is easy to see and grade. Don't leave any guessing, mind-reading, or searching to the grader.
- Any time you can double check your work, please do that!

Derivatives: Compute the derivative of each of the following functions **two** different ways:

- First use the **limit definition of the derivative**.
- Second use the Differentiation Rules.

1. $f(x) = \dfrac{1}{x^2}$
2. $f(x) = \sqrt{3x - 7}$
3. $f(x) = \dfrac{3x + 2}{5 - 6x}$

Horizontal Tangent Lines: Find all x-coordinates at which the graphs of the following functions have horizontal tangent lines. Please **simplify** your derivatives first by factoring out common factors.

4. $f(x) = \dfrac{1}{(x^2 - 3x)^7}$
5. $f(x) = (4x - 1)^3(2x + 7)^5$

Differentiation Rules: Differentiate the following functions. Please do **not** simplify your derivatives here.

6. $H(x) = \left(1 - \dfrac{2}{x^2}\right)^5$
7. $g(t) = \dfrac{t^3 + \frac{1}{t}}{1 + t^2}$
8. $p(x) = \dfrac{1}{(-2x + 3)^5}$
9. $r(x) = \dfrac{(2x + 1)^3}{(3x + 1)^4}$
10. $S(x) = \left(\dfrac{1 + 2x}{1 + 3x}\right)^4$
11. $y = \left((x^2 + 3x)^4 + x\right)^{-5/7}$

Tangent Lines:

12. Find an equation for the tangent line to the graph of $y = x^2 - 4x + 2$ when $x = 1$.

13. Find an equation for the tangent line to the graph of

$$f(x) = \frac{1}{x-1}$$

at the point $(0, -1)$.

14. Find an equation for the tangent line to the graph of $f(x) = \sqrt{x}$ at $x = 4$.

15. Find an equation for the tangent line to the graph of

$$f(x) = \frac{1+x^2}{\sqrt{2x+1}}$$

at $x = 0$.

16. At which point(s) of the graph of $f(x) = -x^3 + 13$ is the slope of the tangent line equal to -27? What's the picture representing this problem?

More derivatives:

17. Let f and g be two differentiable functions, and suppose that their values and the values of their derivatives at $x = 1, 2, 3$ are given by the following table:

x	1	2	3
$f(x)$	3	2	5
$f'(x)$	-2	1	3
$g(x)$	3	1	4
$g'(x)$	-3	2	7

Let $h(x) = f \circ g(x)$ and $k(x) = f(x) \cdot g(f(x))$. Compute $h'(2)$ and $k'(1)$.

18. Let f and g be two differentiable functions, and suppose that their values and the values of their derivatives at $x = 2, 3$ are given by the following table:

x	2	3
$f(x)$	4	0
$f'(x)$	1	-7
$g(x)$	3	-1
$g'(x)$	-5	4

Let $h(x) = f(x)g(x)$, $k(x) = \dfrac{f(x)}{g(x)}$, and $W(x) = f \circ g(x)$. Compute $h'(2)$, $k'(2)$, and $W'(2)$.

19. Let $f(x) = \sqrt{x+1} \cdot g(x)$ where $g(0) = -7$ and $g'(0) = 4$. Compute $f'(0)$.

20. Let $f(x) = \dfrac{\sqrt{x^2+1}}{g(x)}$ where $g(0) = -7$ and $g'(0) = 4$. Compute $f'(0)$.

Implicit Differentiation: Use implicit differentiation to find the derivative of y with respect to x, for each of the following equations.

21. $x^2 + x^2y^2 + y^3 = 3x + 7$

22. $x^3 + x^2y^{3/2} = y^3 + 7$

Math 105: Exam 2 Practice Problems — Solutions

Derivatives: Compute the derivative in two different ways.

1. $f(x) = \dfrac{1}{x^2}$.

   Using the limit definition,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} \\ &= \lim_{h\to 0} \frac{x^2 - (x+h)^2}{hx^2(x+h)^2} \\ &= \lim_{h\to 0} \frac{-2xh - h^2}{hx^2(x+h)^2} \\ &= \lim_{h\to 0} \frac{-(2x+h)}{x^2(x+h)^2} \\ &= \boxed{-\frac{2}{x^3}}. \end{aligned}$$

   Using the differentiation rules,

$$f(x) = x^{-2}, \qquad f'(x) = -2x^{-3} = \boxed{-\frac{2}{x^3}}.$$

2. $f(x) = \sqrt{3x-7}$.

   Using the limit definition,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\sqrt{3(x+h)-7} - \sqrt{3x-7}}{h} \\ &= \lim_{h\to 0} \frac{3h}{h\left(\sqrt{3x+3h-7} + \sqrt{3x-7}\right)} \\ &= \boxed{\frac{3}{2\sqrt{3x-7}}}. \end{aligned}$$

   Using the chain rule,

$$f'(x) = \frac{1}{2}(3x-7)^{-1/2}(3) = \boxed{\frac{3}{2\sqrt{3x-7}}}.$$

3. $f(x) = \dfrac{3x+2}{5-6x}$.

   Using the limit definition,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{1}{h}\left(\frac{3x+3h+2}{5-6x-6h} - \frac{3x+2}{5-6x}\right) \\ &= \lim_{h\to 0} \frac{(3x+3h+2)(5-6x) - (3x+2)(5-6x-6h)}{h(5-6x-6h)(5-6x)} \\ &= \lim_{h\to 0} \frac{27h}{h(5-6x-6h)(5-6x)} \\ &= \boxed{\frac{27}{(5-6x)^2}}. \end{aligned}$$

Using the quotient rule,

$$f'(x) = \frac{3(5-6x) - (3x+2)(-6)}{(5-6x)^2}$$
$$= \boxed{\frac{27}{(5-6x)^2}}.$$

Horizontal Tangent Lines

4. $f(x) = \dfrac{1}{(x^2-3x)^7}$.

Rewrite as

$$f(x) = (x^2-3x)^{-7}.$$

Then

$$f'(x) = -7(x^2-3x)^{-8}(2x-3) = \frac{-7(2x-3)}{(x^2-3x)^8}.$$

A horizontal tangent occurs when $f'(x) = 0$, so

$$2x - 3 = 0.$$

Thus

$$\boxed{x = \frac{3}{2}}.$$

5. $f(x) = (4x-1)^3(2x+7)^5$.

Using the product and chain rules,

$$\begin{aligned} f'(x) &= 12(4x-1)^2(2x+7)^5 + 10(4x-1)^3(2x+7)^4 \\ &= (4x-1)^2(2x+7)^4\left(12(2x+7) + 10(4x-1)\right) \\ &= 2(4x-1)^2(2x+7)^4(32x+37). \end{aligned}$$

Therefore $f'(x) = 0$ at

$$\boxed{x = \frac{1}{4}, \quad x = -\frac{7}{2}, \quad x = -\frac{37}{32}}.$$

Differentiation Rules

6. $H(x) = \left(1 - \dfrac{2}{x^2}\right)^5$.

By the chain rule,

$$\boxed{H'(x) = 5\left(1 - \frac{2}{x^2}\right)^4\left(\frac{4}{x^3}\right)}.$$

7. $g(t) = \dfrac{t^3 + \frac{1}{t}}{1+t^2}$.

Using the quotient rule,

$$\boxed{g'(t) = \frac{\left(3t^2 - \frac{1}{t^2}\right)(1+t^2) - \left(t^3 + \frac{1}{t}\right)(2t)}{(1+t^2)^2}}.$$

8. $p(x) = \dfrac{1}{(-2x+3)^5}$.

   Write $p(x) = (-2x+3)^{-5}$. Then

$$\boxed{p'(x) = -5(-2x+3)^{-6}(-2)}.$$

9. $r(x) = \dfrac{(2x+1)^3}{(3x+1)^4}$.

   Using the quotient rule,

$$\boxed{r'(x) = \frac{6(2x+1)^2(3x+1)^4 - (2x+1)^3\,12(3x+1)^3}{(3x+1)^8}}.$$

10. $S(x) = \left(\dfrac{1+2x}{1+3x}\right)^4$.

    Using the chain rule followed by the quotient rule,

$$\boxed{S'(x) = 4\left(\frac{1+2x}{1+3x}\right)^3 \frac{2(1+3x) - 3(1+2x)}{(1+3x)^2}}.$$

11. $y = \left((x^2+3x)^4 + x\right)^{-5/7}$.

    Using the chain rule twice,

$$\boxed{y' = -\frac{5}{7}\left((x^2+3x)^4 + x\right)^{-12/7}\left(4(x^2+3x)^3(2x+3)+1\right)}.$$

Tangent Lines

12. Find the tangent line to $y = x^2 - 4x + 2$ at $x = 1$.

    We have

$$y(1) = 1 - 4 + 2 = -1, \qquad y' = 2x - 4, \qquad y'(1) = -2.$$

    Thus

$$y + 1 = -2(x-1),$$

    so

$$\boxed{y = -2x + 1}.$$

13. Find the tangent line to $f(x) = \dfrac{1}{x-1}$ at $(0,-1)$.

    Since

$$f'(x) = -\frac{1}{(x-1)^2},$$

    we have $f'(0) = -1$. Therefore

$$y + 1 = -1(x-0),$$

    so

$$\boxed{y = -x - 1}.$$

14. Find the tangent line to $f(x) = \sqrt{x}$ at $x = 4$.

The point is $(4, 2)$ and

$$f'(x) = \frac{1}{2\sqrt{x}}, \qquad f'(4) = \frac{1}{4}.$$

Hence

$$y - 2 = \frac{1}{4}(x - 4),$$

so

$$\boxed{y = \frac{1}{4}x + 1}.$$

15. Find the tangent line to

$$f(x) = \frac{1 + x^2}{\sqrt{2x + 1}}$$

at $x = 0$.

First,

$$f(0) = 1.$$

Write

$$f(x) = (1 + x^2)(2x + 1)^{-1/2}.$$

Then

$$f'(x) = 2x(2x + 1)^{-1/2} - (1 + x^2)(2x + 1)^{-3/2}.$$

Thus

$$f'(0) = -1.$$

The tangent line is

$$y - 1 = -1(x - 0),$$

so

$$\boxed{y = 1 - x}.$$

16. Let $f(x) = -x^3 + 13$. Find the points where the tangent line has slope -27.

We need

$$f'(x) = -3x^2 = -27.$$

Thus

$$x^2 = 9, \qquad x = \pm 3.$$

The corresponding points are

$$f(3) = -14, \qquad f(-3) = 40.$$

Therefore the points are

$$\boxed{(3, -14) \quad \text{and} \quad (-3, 40)}.$$

Geometrically, the graph has two tangent lines of slope -27 at these two points, so the two tangent lines are parallel.

More Derivatives

17. For the first table, let $h(x) = f(g(x))$ and $k(x) = f(x)g(f(x))$.

For h,

$$h'(x) = f'(g(x))g'(x).$$

At $x = 2$,

$$g(2) = 1, \qquad f'(1) = -2, \qquad g'(2) = 2,$$

so

$$\boxed{h'(2) = (-2)(2) = -4}.$$

For k,

$$k'(x) = f'(x)g(f(x)) + f(x)g'(f(x))f'(x).$$

At $x = 1$,

$$f(1) = 3, \quad f'(1) = -2, \quad g(3) = 4, \quad g'(3) = 7.$$

Hence

$$\boxed{k'(1) = (-2)(4) + 3(7)(-2) = -50}.$$

18. For the second table, let

$$h(x) = f(x)g(x), \qquad k(x) = \frac{f(x)}{g(x)}, \qquad W(x) = f(g(x)).$$

Using the product rule,

$$h'(2) = f'(2)g(2) + f(2)g'(2) = (1)(3) + (4)(-5),$$

so

$$\boxed{h'(2) = -17}.$$

Using the quotient rule,

$$k'(2) = \frac{f'(2)g(2) - f(2)g'(2)}{g(2)^2} = \frac{(1)(3) - 4(-5)}{3^2},$$

so

$$\boxed{k'(2) = \frac{23}{9}}.$$

Finally,

$$W'(2) = f'(g(2))g'(2) = f'(3)(-5) = (-7)(-5),$$

so

$$\boxed{W'(2) = 35}.$$

19. Let $f(x) = \sqrt{x+1}\, g(x)$, where $g(0) = -7$ and $g'(0) = 4$.

Using the product rule,

$$f'(x) = \frac{1}{2\sqrt{x+1}} g(x) + \sqrt{x+1}\, g'(x).$$

Therefore

$$f'(0) = \frac{1}{2}(-7) + (1)(4) = \boxed{\frac{1}{2}}.$$

20. Let $f(x) = \dfrac{\sqrt{x^2+1}}{g(x)}$, where $g(0) = -7$ and $g'(0) = 4$.

The derivative of the numerator is

$$\frac{x}{\sqrt{x^2+1}}.$$

Thus, by the quotient rule,

$$f'(x) = \frac{\frac{x}{\sqrt{x^2+1}}g(x) - \sqrt{x^2+1}\,g'(x)}{g(x)^2}.$$

At $x = 0$,

$$f'(0) = \frac{0(-7) - 1(4)}{(-7)^2} = \boxed{-\frac{4}{49}}.$$

Implicit Differentiation

21. $x^2 + x^2y^2 + y^3 = 3x + 7$.

Differentiate both sides with respect to x:

$$2x + 2xy^2 + 2x^2y\,y' + 3y^2y' = 3.$$

Collect the y' terms:

$$(2x^2y + 3y^2)y' = 3 - 2x - 2xy^2.$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{3 - 2x - 2xy^2}{2x^2y + 3y^2}}.$$

22. $x^3 + x^2y^{3/2} = y^3 + 7$.

Differentiate both sides:

$$3x^2 + 2xy^{3/2} + x^2\left(\frac{3}{2}y^{1/2}y'\right) = 3y^2y'.$$

Move the y' terms to one side:

$$3x^2 + 2xy^{3/2} = \left(3y^2 - \frac{3}{2}x^2\sqrt{y}\right)y'.$$

Hence

$$\boxed{\frac{dy}{dx} = \frac{3x^2 + 2xy^{3/2}}{3y^2 - \frac{3}{2}x^2\sqrt{y}}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 16

Due **Wednesday, November 4** at 11:55pm, on Gradescope

1. If
$$x^2 + y^2 + z^2 = 9, \qquad \frac{dx}{dt} = 5, \qquad \frac{dy}{dt} = 4,$$
find dz/dt when $(x, y, z) = (2, 2, 1)$.

2. Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm^2?

3. The radius of a sphere is increasing at a rate of 4 mm/s. How fast is the volume increasing when the diameter is 80 mm?

4. At noon, ship A is 100 km west of ship B. Ship A is sailing south at 35 km/h and ship B is sailing north at 25 km/h. How fast is the distance between the ships changing at 4:00 PM?

MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 16 SOLUTIONS

1. If
$$x^2 + y^2 + z^2 = 9, \qquad \frac{dx}{dt} = 5, \qquad \frac{dy}{dt} = 4,$$
find dz/dt when $(x, y, z) = (2, 2, 1)$.

   Differentiate with respect to t:
$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} + 2z\frac{dz}{dt} = 0.$$
   Substitute the given values:
$$2(2)(5) + 2(2)(4) + 2(1)\frac{dz}{dt} = 0.$$
   Thus
$$20 + 16 + 2\frac{dz}{dt} = 0,$$
   so
$$\boxed{\frac{dz}{dt} = -18}.$$

2. Each side of a square is increasing at a rate of 6 cm/s. At what rate is the area of the square increasing when the area of the square is 16 cm^2?

   Let s be the side length and A the area. Then
$$A = s^2.$$
   Differentiate with respect to t:
$$\frac{dA}{dt} = 2s\frac{ds}{dt}.$$
   When $A = 16$, we have $s = 4$ cm. Since $ds/dt = 6$ cm/s,
$$\frac{dA}{dt} = 2(4)(6) = 48.$$
   Therefore
$$\boxed{\frac{dA}{dt} = 48 \text{ cm}^2/\text{s}}.$$

3. The radius of a sphere is increasing at a rate of 4 mm/s. How fast is the volume increasing when the diameter is 80 mm?

   The volume of a sphere is
$$V = \frac{4}{3}\pi r^3.$$

Differentiate with respect to t:

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

A diameter of 80 mm means $r = 40$ mm. Since $dr/dt = 4$ mm/s,

$$\frac{dV}{dt} = 4\pi(40)^2(4) = 25600\pi.$$

Thus

$$\boxed{\frac{dV}{dt} = 25600\pi \text{ mm}^3/\text{s}}.$$

4. At noon, ship A is 100 km west of ship B. Ship A is sailing south at 35 km/h and ship B is sailing north at 25 km/h. How fast is the distance between the ships changing at 4:00 PM?

   Let y be the north-south distance between the ships and let d be the distance between them. The east-west distance remains 100 km, so

   $$d^2 = 100^2 + y^2.$$

   Since the ships are moving in opposite north-south directions,

   $$\frac{dy}{dt} = 35 + 25 = 60 \text{ km/h}.$$

   At 4:00 PM, four hours have passed, so

   $$y = 4(60) = 240 \text{ km}.$$

   Thus

   $$d = \sqrt{100^2 + 240^2} = 260 \text{ km}.$$

   Differentiate $d^2 = 100^2 + y^2$:

   $$2d\frac{dd}{dt} = 2y\frac{dy}{dt}.$$

   Therefore

   $$\frac{dd}{dt} = \frac{y}{d}\frac{dy}{dt} = \frac{240}{260}(60) = \frac{720}{13}.$$

   Hence the distance between the ships is increasing at

   $$\boxed{\frac{dd}{dt} = \frac{720}{13} \text{ km/h} \approx 55.4 \text{ km/h}}.$$

MATH 105 Calculus with Algebra Fall 2026

HOMEWORK 17

Due **Friday, November 6** at 11:55pm, on Gradescope

1. A particle moves according to a law of motion $s = f(t)$, $t \geq 0$, where t is measured in seconds and s in feet. Here
$$f(t) = \frac{9t}{t^2 + 9}.$$
   (a) Find the velocity at time t.
   (b) What is the velocity after 1 second?
   (c) When is the particle at rest?
   (d) When is the particle moving in the positive direction?

2. If V is the volume of a cube with edge length x and the cube expands as time passes, find dV/dt in terms of dx/dt.

3. A plane flying horizontally at an altitude of 1 mi and a speed of 500 mi/h passes directly over a radar station. Find the rate at which the distance from the plane to the station is increasing when it is 2 mi away from the station.

4. A baseball diamond is a square with side 90 ft. A batter hits the ball and runs toward first base with a speed of 24 ft/s.
   (a) At what rate is his distance from second base decreasing when he is halfway to first base?
   (b) At what rate is his distance from third base increasing at the same moment?

MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 17 SOLUTIONS

1. A particle moves according to a law of motion $s = f(t)$, $t \geq 0$, where t is measured in seconds and s in feet. Here

$$f(t) = \frac{9t}{t^2 + 9}.$$

(a) The velocity is the derivative of the position:

$$v(t) = f'(t).$$

Using the quotient rule,

$$\begin{aligned} v(t) &= \frac{9(t^2+9) - 9t(2t)}{(t^2+9)^2} \\ &= \frac{81 - 9t^2}{(t^2+9)^2}. \end{aligned}$$

Thus

$$\boxed{v(t) = \frac{9(9-t^2)}{(t^2+9)^2}}.$$

(b) At $t = 1$,

$$v(1) = \frac{9(9-1)}{(1+9)^2} = \frac{72}{100} = \boxed{\frac{18}{25} \text{ ft/s}}.$$

(c) The particle is at rest when $v(t) = 0$. Since the denominator of $v(t)$ is always positive, we need

$$9 - t^2 = 0.$$

Thus $t = \pm 3$. Since $t \geq 0$,

$$\boxed{t = 3 \text{ seconds}}.$$

(d) The particle moves in the positive direction when $v(t) > 0$. Again, the denominator is positive, so this occurs when

$$9 - t^2 > 0.$$

For $t \geq 0$, this means

$$\boxed{0 \leq t < 3}.$$

2. If V is the volume of a cube with edge length x, then

$$V = x^3.$$

Differentiate with respect to t:

$$\boxed{\frac{dV}{dt} = 3x^2 \frac{dx}{dt}}.$$

3. A plane flying horizontally at an altitude of 1 mi and a speed of 500 mi/h passes directly over a radar station. Find the rate at which the distance from the plane to the station is increasing when it is 2 mi away from the station.

   Let x be the horizontal distance from the station to the point directly below the plane, and let s be the distance from the plane to the station. Since the altitude is always 1 mi,

$$x^2 + 1 = s^2.$$

   We are given

$$\frac{dx}{dt} = 500 \text{ mi/h}.$$

   When $s = 2$,

$$x^2 + 1 = 4,$$

   so

$$x = \sqrt{3}.$$

   Differentiate the equation with respect to t:

$$2x\frac{dx}{dt} = 2s\frac{ds}{dt}.$$

   Therefore

$$\frac{ds}{dt} = \frac{x}{s}\frac{dx}{dt} = \frac{\sqrt{3}}{2}(500) = 250\sqrt{3}.$$

   Thus the distance from the plane to the station is increasing at

$$\boxed{250\sqrt{3} \text{ mi/h} \approx 433 \text{ mi/h}}.$$

4. A baseball diamond is a square with side 90 ft. A batter hits the ball and runs toward first base with a speed of 24 ft/s.

   Let x be the runner's distance from home plate. Then

$$\frac{dx}{dt} = 24 \text{ ft/s}.$$

   When the runner is halfway to first base, $x = 45$ ft.

   (a) Let d be the runner's distance from second base. The horizontal distance from the runner to second base is $90 - x$, while the vertical distance is 90, so

$$d^2 = (90 - x)^2 + 90^2.$$

   When $x = 45$,

$$d = \sqrt{45^2 + 90^2} = 45\sqrt{5}.$$

Differentiate:

$$2d\frac{dd}{dt} = 2(90 - x)(-1)\frac{dx}{dt}.$$

Hence

$$\frac{dd}{dt} = -\frac{90 - x}{d}\frac{dx}{dt}.$$

At $x = 45$,

$$\frac{dd}{dt} = -\frac{45}{45\sqrt{5}}(24) = -\frac{24}{\sqrt{5}}.$$

Thus the distance from second base is decreasing at

$$\boxed{\frac{24}{\sqrt{5}} \text{ ft/s} \approx 10.7 \text{ ft/s}}.$$

(b) Let r be the runner's distance from third base. Then

$$r^2 = x^2 + 90^2.$$

When $x = 45$,

$$r = \sqrt{45^2 + 90^2} = 45\sqrt{5}.$$

Differentiating,

$$2r\frac{dr}{dt} = 2x\frac{dx}{dt}.$$

Therefore

$$\frac{dr}{dt} = \frac{x}{r}\frac{dx}{dt} = \frac{45}{45\sqrt{5}}(24) = \frac{24}{\sqrt{5}}.$$

Thus the distance from third base is increasing at

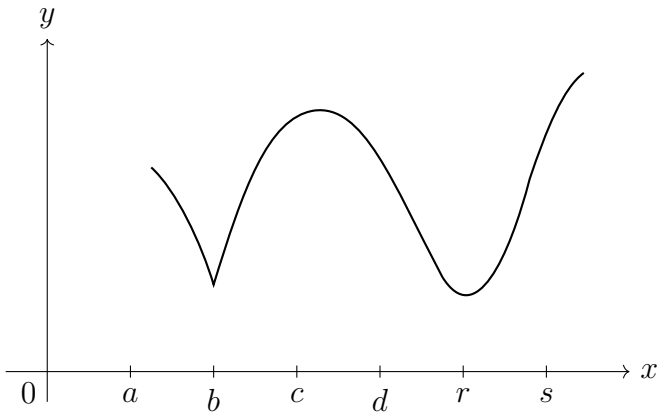
$$\boxed{\frac{24}{\sqrt{5}} \text{ ft/s} \approx 10.7 \text{ ft/s}}.$$

MATH 105 Calculus with Algebra Fall 2026

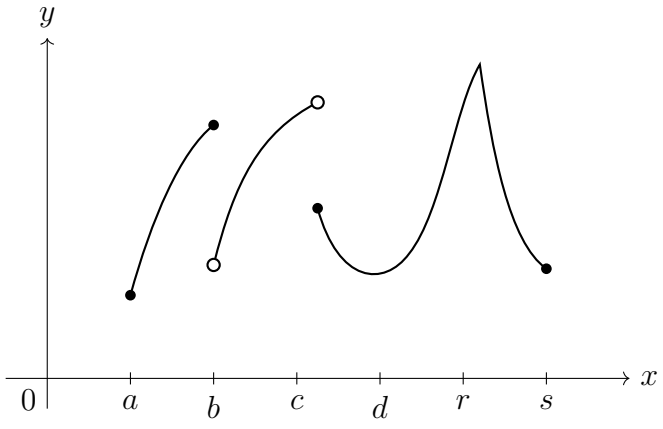
HOMEWORK 18

Due **Monday, November 9** at 11:55pm, on Gradescope

1. For each of the numbers a, b, c, d, r, and s, state whether the function whose graph is shown has an absolute maximum or minimum, a local maximum or minimum, or neither a maximum nor a minimum.



2. For each of the numbers a, b, c, d, r, and s, state whether the function whose graph is shown has an absolute maximum or minimum, a local maximum or minimum, or neither a maximum nor a minimum.



3. Find the critical numbers of the function.

$$f(x) = 4 + \frac{1}{3}x - \frac{1}{2}x^2.$$

4. Find the critical numbers of the function.

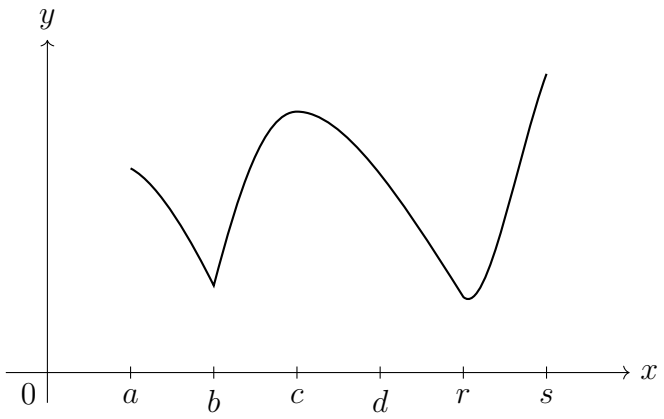
$$g(t) = t^4 + t^3 + t^2 + 1.$$

5. Find the absolute maximum and absolute minimum values of f on the given interval.

$$f(x) = 12 + 4x - x^2, \qquad [0, 5].$$

MATH 105 Calculus with Algebra Fall 2026
HOMEWORK 18 SOLUTIONS

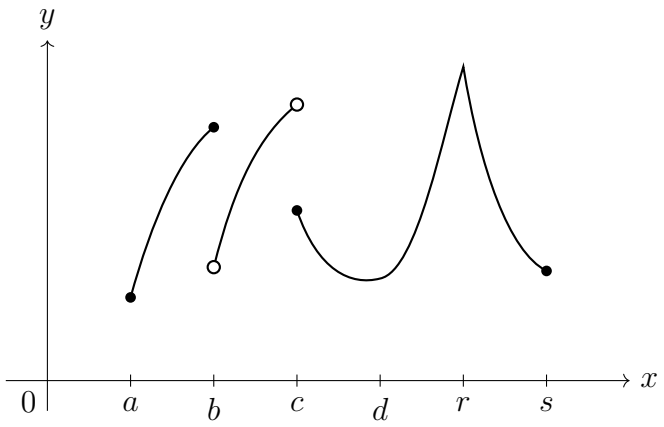
1. For each of the numbers a, b, c, d, r, and s, classify the corresponding point.



Using the definition of local extrema relative to the domain:

- At $x = a$, f has a **local maximum**.
- At $x = b$, f has a **local minimum**.
- At $x = c$, f has a **local maximum**.
- At $x = d$, f has **neither** a maximum nor a minimum.
- At $x = r$, f has an **absolute minimum** (and also a local minimum).
- At $x = s$, f has an **absolute maximum** (and also a local maximum).

2. For each of the numbers a, b, c, d, r, and s, classify the corresponding point.



Therefore:

- At $x = a$, f has an **absolute minimum** (and also a local minimum).
- At $x = b$, f has a **local maximum**.
- At $x = c$, f has **neither** a maximum nor a minimum.
- At $x = d$, f has a **local minimum**.
- At $x = r$, f has an **absolute maximum** (and also a local maximum).
- At $x = s$, f has a **local minimum**.

3. Find the critical numbers of
$$f(x) = 4 + \frac{1}{3}x - \frac{1}{2}x^2.$$
Differentiate:
$$f'(x) = \frac{1}{3} - x.$$
The derivative exists for every x, so the only critical numbers occur where $f'(x) = 0$:
$$\frac{1}{3} - x = 0.$$
Thus
$$\boxed{x = \frac{1}{3}}.$$

4. Find the critical numbers of
$$g(t) = t^4 + t^3 + t^2 + 1.$$
Differentiate:
$$g'(t) = 4t^3 + 3t^2 + 2t = t(4t^2 + 3t + 2).$$
Thus $g'(t) = 0$ when
$$t = 0$$
or
$$4t^2 + 3t + 2 = 0.$$
The quadratic has discriminant
$$3^2 - 4(4)(2) = 9 - 32 = -23 < 0,$$
so it has no real solutions. Therefore the only critical number is
$$\boxed{t = 0}.$$

5. Find the absolute maximum and absolute minimum values of

$$f(x) = 12 + 4x - x^2$$

on $[0, 5]$.

Differentiate:

$$f'(x) = 4 - 2x.$$

The critical number in $[0, 5]$ is found by solving

$$4 - 2x = 0,$$

so

$$x = 2.$$

For the Closed Interval Method, evaluate f at the critical number and at both endpoints:

$$f(0) = 12,$$

$$f(2) = 12 + 8 - 4 = 16,$$

$$f(5) = 12 + 20 - 25 = 7.$$

Therefore the absolute maximum value is

$$\boxed{16 \text{ at } x = 2},$$

and the absolute minimum value is

$$\boxed{7 \text{ at } x = 5}.$$